

Honeywell

TotalPlant Solution (TPS) System

Fieldbus Implementation Guide

FB12-611a

R611

06/06

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About This Document

This document describes,

- TPS Builder/Fieldbus fundamentals
- Fieldbus OLE controls
- Fieldbus connections
- Configuring a Fieldbus link
- Configuring CM strategies

Release Information

Document Name	Document ID	Release Number	Publication Date
Fieldbus Implementation Guide - FB12	FB12-611a	611	06/06

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE –EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE – INTERNAL: Identifies an additional source of information within the bookset.

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1. System Requirements

1.1 Before You Begin

Prerequisites

You should have installed:

- TPS Builder (see “TPS Builder Configuration Tool” and “Fieldbus Libraries” in this document).
- The required hardware and software (see “Platforms” in this document).
- Get the 32-character manufacturer’s device identification numbers for all devices you intend to configure for the links.

TPS Builder Configuration Tool



ATTENTION

Fieldbus point configuration cannot be built using the Data Entity Builder (DEB).

If you are installing the Fieldbus IOP in an existing HPM that is already configured in the TPS system, you can backbuild the HPM configuration using TPS Builder (see “Configuration Example 1” in this document). TPS Builder can backbuild all your existing process units.

Configuration data for CM (the new point type for fieldbus points) are saved in the History Module checkpoint files. This is the configuration data used to restore the FBIOP.

Field device configuration data is saved in the TPS Builder Repository. The Device Save/Restore is a separate function.

1. System Requirements

1.2. Platforms

1.2 Platforms

Workstation Configuration

TPS Builder can be used to configure Fieldbus using a workstation with NT, Service Pack 4 with a minimum of 128Mb memory. TPS Builder R130 is required. Refer to the Software Change Notice (SCN) for the correct release of TPS Builder.

The project database must be transported to the Global User Station (GUS) for downloading. Data can be transferred using the network facilities; otherwise the project data can be copied from its directory, which is on the drive on which the SQL server is installed, in the folder:

\HWIAC\databases\TPS Builder\.

TPS Builder Implementation of Fieldbus

Refer to the *TPS Builder User's Guide* and the latest SCN accompanying the guide for requirements.

Although a workstation can be used for configuration, **downloading** of data to links and devices requires a Global User Station (GUS), a Primary Domain Controller (PDC), and a TPN Server.

Primary Domain Controller (PDC)

In addition, the TPS system requires the installation of an NT TPS Primary Domain Controller. Fieldbus devices can be implemented only on a GUS with 128mb RAM with TPN Server software to communicate the special records to the LCN/HPM/Fieldbus. To use the TPN server, an NT domain and TPS domain are required in that the TPN server can be configured only on a GUS within a TPS domain. This means that a GUS node must be connected to a PDC, since the PDC holds the NT and TPS domain configuration. Currently the GUS or APP nodes are not recommended for use as a PDC. Refer to the *TPS Domain Node Types* in the *TPS System Implementation Guide* for installation and administration.



ATTENTION

The TPN server must be configured as a **HCI TPN Server**, and the default access level must be set at **Point Builder**.

NI Configurator

The NI Configurator card must be installed on a PC running Windows 95 (or later) and it must not reside on the same station that is used to download the configurations from TPS Builder. NI Tool software (refer to the SCN for the correct software version) must be present to download new revisions of the firmware to devices.

Libraries



ATTENTION

GUS users should ensure that they install the Microsoft Redistributable Libraries option when installing TPS. If you are installing TPS Builder on a workstation other than a GUS node, you must install the Microsoft Redistributable Libraries option.

The TPS Builder installation option must include the UCN Library option for the fieldbus configuration to be present.

TPS Builder Release Setting

The TPS Builder release must be set to R600. See the *TPS Builder User's Guide* or "Configuration Example 1" in this document if you need instructions for setting this release.

1. System Requirements

1.2. Platforms

2. TPS Builder/Fieldbus Fundamentals

2.1 Introduction to Fieldbus Implementation

The Fieldbus Configuration Tool

TPS Builder is the configuration tool for fieldbus implementation.

Fieldbus functionality is available as one of the optional subsystems for TPS Builder.

On-line or off-line Configuration Support

The fieldbus subsystem provides the ability to fully configure fieldbus networks off-line, and to download the configuration to the fieldbus network.

Downloading to Networks

TPS Builder also provides the ability to configure the TPN strategies required to support a fieldbus network, and to download the configuration to the TPN. The off-line configuration of the fieldbus network includes building of Link Schedules and Function Block schedules.

Template-based Configuration

The configuration of the fieldbus network is based upon the extensive use of templates. Templates for Fieldbus Foundation standard entities are provided as part of TPS Builder. Templates for manufacturer devices are created from the Manufacturer supplied Device Description and Capabilities files.

Checker and Error Logs

The configuration can be checked for errors, using the Checker functionality, which is extended to provide fieldbus specific checks. Errors are added to the TPS Builder Error Log for later viewing and correction.

Data Checkpoint/Save

After the configuration of fieldbus is complete, downloaded to the fieldbus network and TPN, this configuration can be checkpointed and the device data can be saved. Operational data such as CM point data or IOP checkpoint data is saved in the TPN history module; device and TPS Builder configuration data is saved in the TPS Repository.

2.2 Fieldbus Libraries

Introduction to Fieldbus Libraries

Three standard fieldbus libraries are supplied as part of the fieldbus subsystem.

- The **Function Block** Library contains templates for the standard fieldbus Function Blocks, Transducer Blocks and other fieldbus entities. It also contains templates for any manufacturer-specific function blocks required to support the fieldbus device templates in the device library.
- The Fieldbus Device Library contains templates for fieldbus devices. The device library as supplied contains a template for a generic device, and templates for the devices supported in the R610 Fieldbus release.
- The Fieldbus Link Library contains templates for a Fieldbus Link.

Function Block Libraries

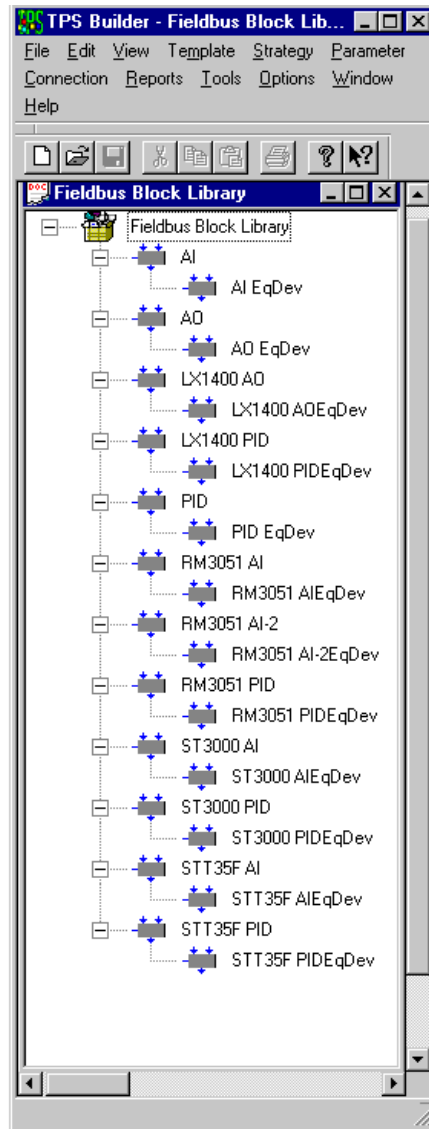
The Function Block Library contains templates for the following fieldbus entities.

Standard Function Block Templates

Block Class	Block Name
Input	Analog Input (AI)
Output	Analog Output (AO)
Control	PID, PI, P Controller (PID)

The following figure shows an expanded view of the Fieldbus Block Library. The AI, AO, and PID blocks are standard function blocks. The other blocks are manufacturer-specific blocks.

Fieldbus Block Library



Standard Transducer Blocks

The following are the first seven standard transducer blocks.

Block Name
Standard Pressure with Calibration Basic Access
Standard Temperature with Calibration Basic Device Access
Standard Radar Level with Calibration Basic Device Access
Standard Flow with Calibration Basic Device Access
Standard Basic Positioner Valve Basic Device Access
Standard Advanced Positioner Valve Basic Device Access
Standard Discrete Positioner Valve Basic Device Access

Transducer Block templates have Fieldbus Link view. The transducer block is included in the device template. Releases 610 and 611 support only one transducer block per device.

Manufacturer Specific Function Blocks and Transducer Blocks

The library also contains templates for any manufacturer-specific blocks, function and transducer, required for the fieldbus device templates present in the Device Library. Although the manufacturer-specific blocks have a minimum functionality of a standard block of the same type, there is no relationship, in TPS Builder terms, between a standard block and a manufacturer-specific block of the same type. As shown in above, the blocks that have the names ST3000 and LX1400 are manufacturer-specific blocks.

Transducer Block Views

A manufacturer-specific Function Block has Equipment and Fieldbus Link view. A manufacturer-specific Transducer Block has Fieldbus Link view.

Supported devices are listed in the Fieldbus Device Library.

Blocks with Manufacturer-Specific Parameters

A manufacturer-specific block may have manufacturer-specific parameters added. The specification for these parameters is added to the Fieldbus Block Library as part of the process of creating the Fieldbus Device template.

Fieldbus Device Library

Templates

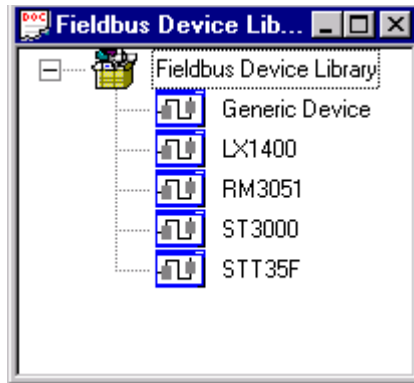
The Device Library contains templates for fieldbus devices. The Device Library is supplied with templates for the devices supported by the current release of fieldbus. Device and block templates are added by processing device descriptions and importing them into the template libraries. The Device Library also contains templates for fieldbus VFDs.

A device template comprises an identification block. The template also contains a number of references to the entities, function blocks and transducer blocks that make up the device.

The Fieldbus Device Library also contains a template for a generic device. The standard device, including the VFD, forms the basis of all manufacturer Fieldbus Foundation compliant devices.

The following figure shows the Fieldbus Device Library.

Fieldbus Device Library



A device template has Fieldbus Link view. In the following figure, the project is expanded to show the link, link scheduler, sequences, and devices.

Fieldbus Link Library

Templates

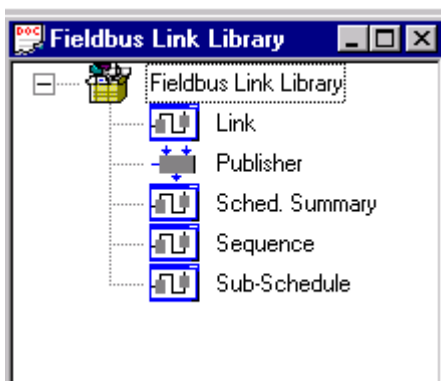
The Fieldbus Link Library contains:

- a standard template for a Fieldbus Link, and
- a standard LAS template contained in the Fieldbus Link module

Both templates have Fieldbus Link view membership.

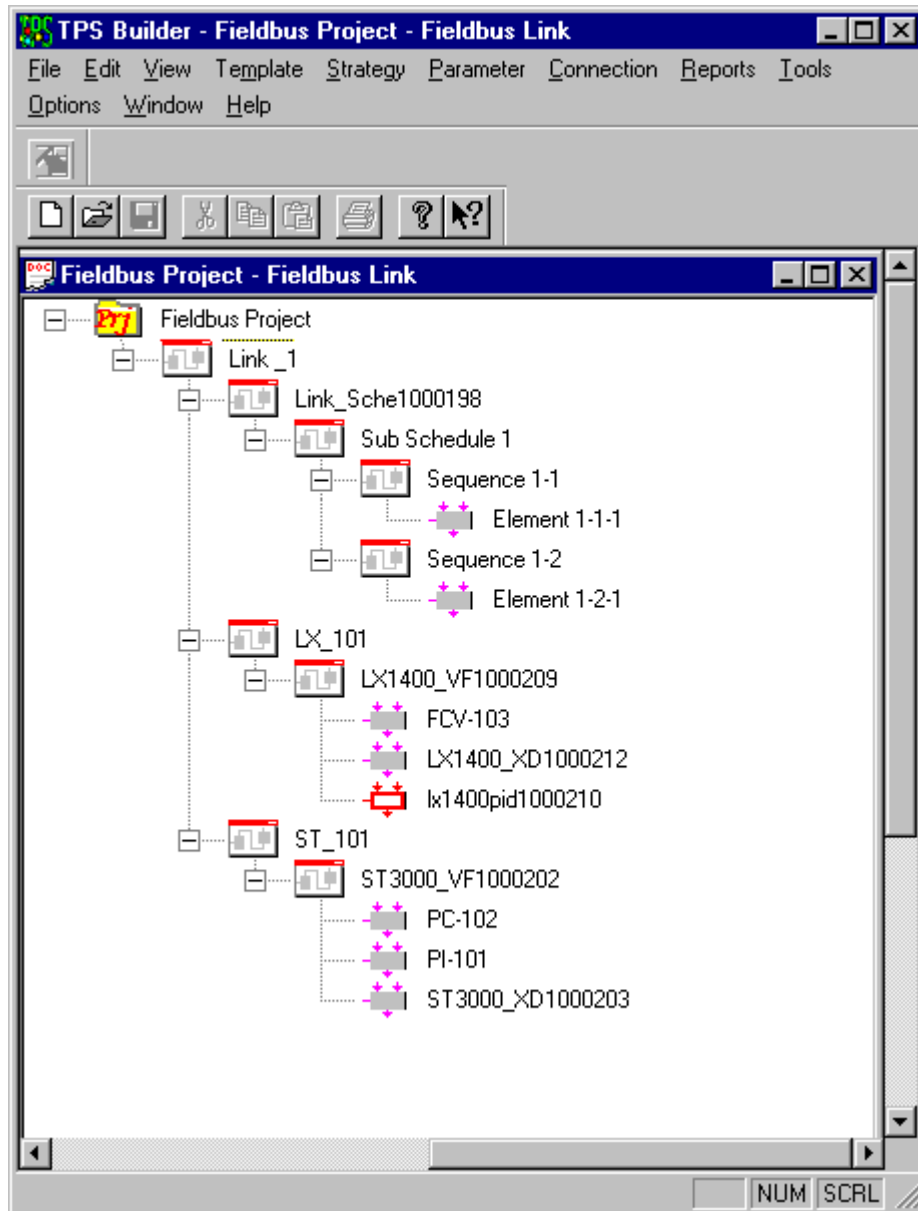
The following figure shows the content of the Fieldbus Link Library:

Fieldbus Link Library



The following figure shows a Link view of a project expanded to show the devices, VFDs, function blocks, and schedule sequences after the schedule has been created.

Link View of a Project Expanded

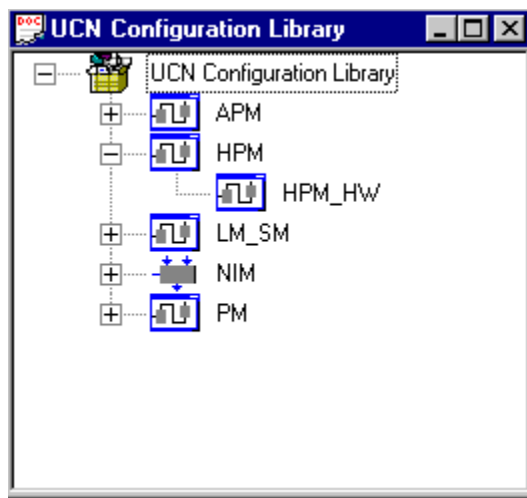


2.3 UCN Configuration Library

HPM Hardware

The HPM hardware is located in the UCN Configuration Library.

UCN Configuration Library



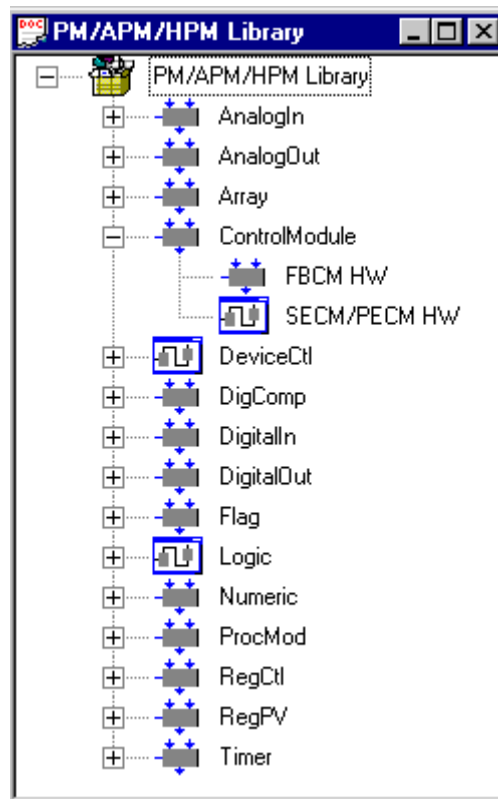
2.4 PM/APM/HPM Library

CM Point Type Template

The template for the CM point type (SECM, PECM and FBCM variants) is included in the PM/APM/HPM Library. This template is present in the library only if the fieldbus subsystem is installed. Ordinarily, these templates are not used in building fieldbus, but the SECM may be used for certain conditions, such as for links which have already been configured without using TPS Builder, as described in "Downloading only a SECM with No Devices" in this document.

The following figure shows the CM point types in the library.

CM Points Types PM/APM/HPM Library

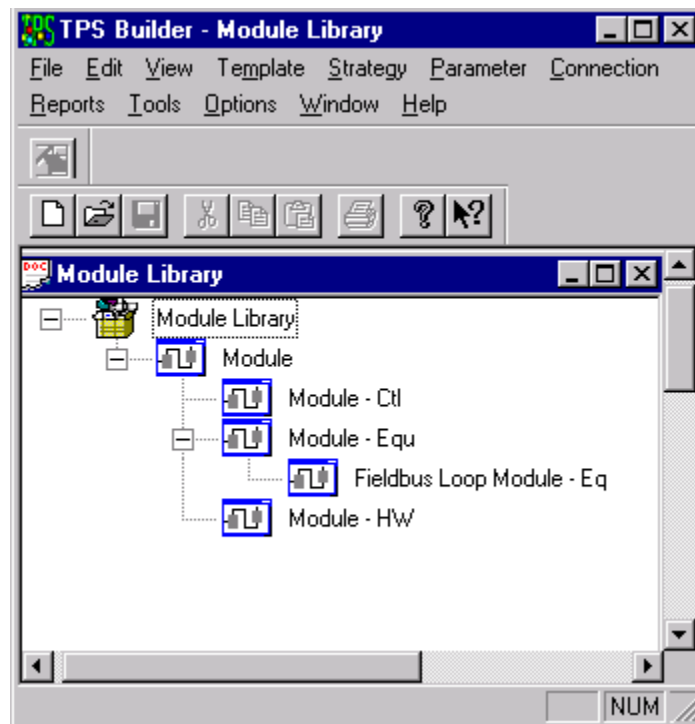


2.5 Module Library

Use of Module Control Template

The Module control is used in TPS Builder to draw loops. A specialized Module template, which has Equipment and Hardware views, is available to draw fieldbus Control Loops. This module contains Loop Timing parameters already present. The values in these parameters will define the default timing behavior of all loops contained in the module. If a module contains a loop that requires different timing values, these can be set on the individual function blocks in the loop.

Module Library



Specializing a Template

Templates normally would be specialized from device templates, which create specialized VFD, function blocks and transducer block templates as well. This is done from the Blocks' dialog using Device/Right Mouse Button.

The specialized Fieldbus Module is added to the Module Library during installation of the fieldbus subsystem.

2.6 Overview of the Fieldbus Configuration Process

Introduction

This section gives you an overview of the configuration process. The details for configuration are provided in “Configuration Example 1” in this document.

The Link View Configuration

Configuration of a fieldbus network is performed mainly in the project’s Fieldbus Link view, which shows the configuration of the project in physical fieldbus terms. The Fieldbus Link view shows each Fieldbus Link and the devices assigned to the link. Each device shows the VFDs associated with the device, and the Function and Transducer Blocks appearing in each VFD. The Link may contain multiple Link Schedules.

2. TPS Builder/Fieldbus Fundamentals

2.6. Overview of the Fieldbus Configuration Process

Two Approaches to the Configuration

One of two approaches can be adopted for fieldbus configuration:

Approach	Description
1	Base TPS Builder functionality is first used to create control drawings, using fieldbus standard and manufacturer specific function blocks and transducer blocks. Function blocks and transducer blocks created in control drawings also appear in the Fieldbus Link view of the project, initially at the project level, until they are assigned to devices. Connections between function blocks are made, to represent data flow from one function block to another. At this point in the configuration process, complete connections data, some of which is derived from the device description, is not available, as the devices to be used are not yet defined. Connections are defined as the CM Mapping and Link Scheduling are performed. After control drawings have been produced, the Fieldbus Links are configured. Using the Fieldbus Link view, the user defines the link and the devices to appear on the link. When first created, devices have 'placeholders' that represent the function blocks. The user then associates function blocks with devices using drag and drop.
2	Alternatively, a Fieldbus Link can be configured by assigning devices directly to the link. In this case actual function blocks and transducer blocks are created in the Fieldbus Link view and in the Equipment view. The Equipment view is then used to create control drawings and define connections between blocks, if required. If control drawings are not required, connections are defined using the parameter configuration form for the blocks.

Fieldbus Entities Mapped to TPS

All physical fieldbus entities are mapped to corresponding TPS Network (TPN) CM strategies.

- a Fieldbus Link is mapped to a SECM
- a Fieldbus Device to a PECM
- each Fieldbus Function Block to a FBCM. A fieldbus transducer block shares the same FBCM as its related function block.

It is not necessary for the user to perform this mapping. The TPS Builder tool performs this mapping during the CM mapping process. See “Fieldbus Parameters Mapping to TPS Builder Parameters” in this document.

Schedule Creation

After a Link has been fully configured, the Link schedule and associated FB schedules can be created. TPS Builder calculates a suggested schedule for a link, using the FB connectivity and FB execution times and execution periods. The user is able to make changes to the suggested schedule by changing the execution sequence and the execution start time of FBs, using a graphical interface. The user can maintain several schedules for a Link. During creation of a Link Schedule, TPS Builder creates Link Objects and VCRs to represent the connections.

Checker Facility

At any time during the configuration of a project, the configuration may be checked using the checker functionality. CM strategies undergo the same checking as existing TPS Network strategies. A set of specific checks are performed on a Fieldbus Link, such as ensuring that no duplicate identifiers are used, that all device addresses are in the correct range and are not duplicated, and that configuration is complete.

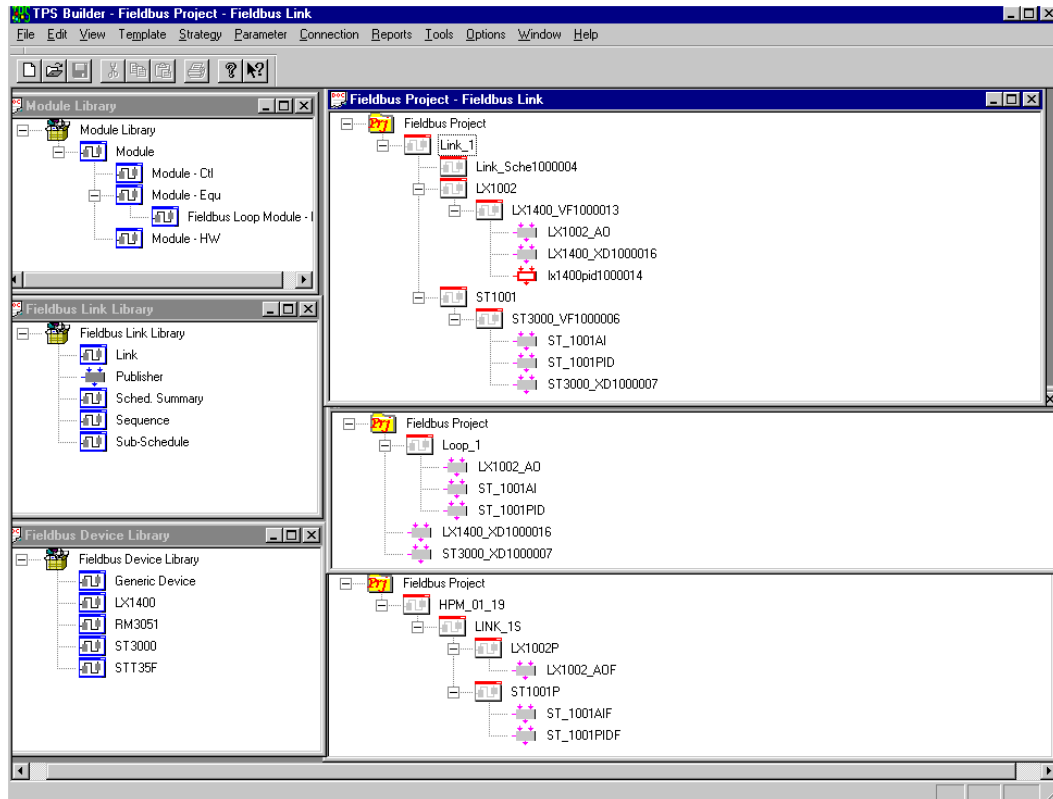
2. TPS Builder/Fieldbus Fundamentals

2.6. Overview of the Fieldbus Configuration Process

Managing the Views and Libraries

Some users find it helpful when configuring fieldbus to keep the libraries open on the left side of the windows, and to keep the views on the right hand side as shown in the following figure.

Managing Libraries and Views



3. Fieldbus OLE Controls

3.1 Functional Description

Terminology Clarification

TPS Builder contains references to the acronym LAS, referring to the currently active link schedule. This should not be confused with the term “LAS” as used in the FOUNDATION Fieldbus specification, which refers instead to the link device that is current function as the active scheduler for the link.

OLE Controls added to TPS Builder for Fieldbus

A number of new controls are added to TPS Builder to support the fieldbus entities.

Type of Control	Purpose
Link	This control represents a Fieldbus Link. It contains Assignment parameters that identify the Link and Link Active Schedule parameters for the link. It is a module control that has Link Active Schedule and Device controls associated with it.
Link Active Schedule (LAS)	This control represents the Link Active Schedule for a Fieldbus Link. It contains parameters that define the schedule and further LAS controls to define the sub-schedules, elements and sequences that make up the schedule.
Device	This control represents a Fieldbus Device. It contains Assignment parameters that identify the device being used. It also contains parameters that model the SMIB and the NMIB, and the directories for the SMIB and the NMIB. These parameters are define the template and therefore cannot be changed. The values for these parameters are present in the Capabilities File for a device. It is a module control that has VFD controls associated with it.

3. Fieldbus OLE Controls

3.1. Functional Description

Type of Control	Purpose
Virtual Field Device (VFD)	This control represents a Virtual Field Device within a fieldbus device. A device may have more than one VFD. It contains parameters that model the Resource Block appearing in a Function Block Application Process (FBAP). It also contains parameters that model the FBAP directory and the Block Link Objects. The VFD is a module control and can therefore contain other controls. It contains 'placeholders' to the Function Blocks and Transducer Blocks that can be part of the device.
Function Block	This control represents a Fieldbus Function Block. Function Blocks may be of two main types: Fieldbus Standard or Manufacturer-Specific.
Transducer Block	This control represents a Fieldbus Transducer Block. Transducer Blocks can be of two main types: Fieldbus Standard or Manufacturer Specific. Manufacturer-specific blocks may be derived from a Standard block or a generic block.

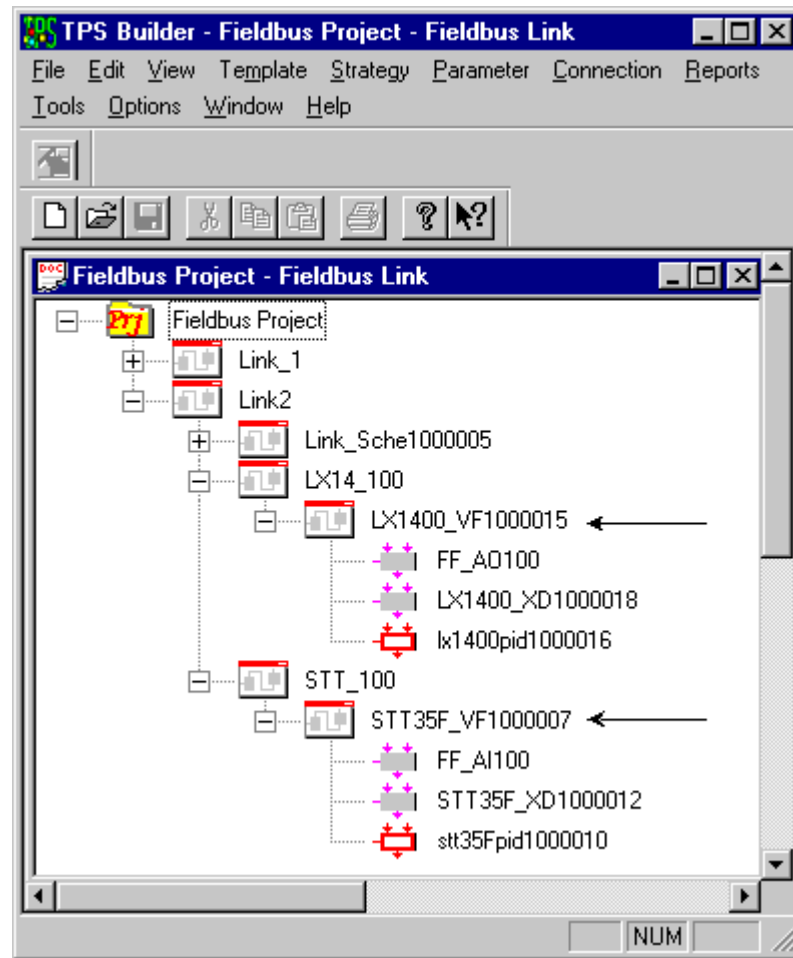
Device Modeling

VFDs and VCRs

A device is modeled by its Virtual Field Device (VFD). The VFD provides access to the Network Management Information Base and the System Management Information Base, the Link Active Scheduler (if the device is a Link Master), and Virtual Communications Relationships (VCRs).

TPS Builder creates the VFD based on the device and block templates chosen by the user. The following figure shows the graphic representation of the VFD in a TPS Builder project.

VFD in TPS Builder Project



3. Fieldbus OLE Controls

3.1. Functional Description

Four Default Views for Fieldbus

TPS Builder Views allow a project to be looked at from different perspectives. Views organize strategies into different hierarchies. The four default views in TPS Builder are:

- **Equipment View** — This view uses an equipment association as a basis for organizing strategies.
- **Hardware View** — This view uses TPN hardware addressing as a basis for organizing strategies (that is, Network, Node and the points assigned to them).
- **Fieldbus Link View** — This view, which is specific to the fieldbus subsystem, uses the layout of a Fieldbus Link as a basis for organizing strategies (that is, Link, Devices, and Function Blocks/Transducer Blocks).
- **Control View** — This view uses control functionality (for example, Optimization) as a basis for organizing strategies.

Default View Membership

The controls have the following default views:

Type Of Control	View Membership
Link	Fieldbus Link View
LAS	Fieldbus Link View
Device	Fieldbus Link View
VFD	Fieldbus Link View
Function Block	Fieldbus Link View, Equipment View
Transducer Block	Fieldbus Link View, Equipment View
CM	Hardware View

3.2 Fieldbus Parameter Types

Introduction

Many of the parameter types used by fieldbus are invisible to the user in TPS Builder. Those that are important in the implementation of fieldbus are discussed here. Many of the fieldbus parameters are projected to TPS parameters. Other parameters may be decomposed and written to TPS parameters, as shown in “Structured Parameters” in this document. For complete information on fieldbus parameters or a specific parameter, refer to the *Fieldbus Parameter Reference Dictionary*.

Structured Parameters

Definition of Structured Parameter

A structured parameter is a parameter that is composed of multiple elements or variables. These elements or variables are modeled as sub-parameters and must be of the FF basic variable types. These element parameters are usually read or written as one unit.

TPS Builder handles the following new parameter types to support FF parameter types not currently supported.

Data Type	Attribute
Date	The Date parameter is 7 octets of data with structured fields that represent months, the day of the month, a two-digit year, day of the week, hours, minutes, milliseconds, and a daylight savings or standard time indicator. TPS Builder holds this string for the display of the fields. A 64-bit unsigned integer, which is the time in 1/32s of a millisecond.
Time Value	Eight octets of data, which is time to 1/32 of a millisecond.
Bitstring	One, two, three, or four octets of data, with each bit representing the selection of an option.
Integers	The different types of FF Integers are mapped into the one TPS Builder integer data type. The actual FF parameter type and size is also maintained in the database.

3. Fieldbus OLE Controls

3.2. Fieldbus Parameter Types

Fieldbus Parameters Mapping to TPS Builder Parameters

Fieldbus data types are mapped to existing TPS Builder data types as follows:

Mapping Fieldbus to TPS Parameters

Fieldbus		TPS Builder	
Data Type	Index	Data Type	Parameter Type
Boolean	1	Logical	4
Integer8	2	Integer	1
Integer16	3	Integer	1
Integer32	4	Integer	1 or 5 (See Note)
Unsigned8	5	Integer	1
Unsigned16	6	Integer	1 or 5 (See Note)
Unsigned32	7	Integer	1 or 5 (See Note)
Floating Point	8	Real	2
Visible String	9	String	3
Octet String	10	String (Integer)	3 (1)
Date	11	Date	9
Time of Day	12	Date	9
Time Difference	13	Date	9
Bitstring	14	Integer	1
Time Value	21	Date	9

Note: If the parameter is an enumeration parameter, the parameter type is 5. The fieldbus data type and the number of octets used by the data type are also held as part of the parameter definition. These are used to convert data when downloading data to a device.

Identifying Fieldbus Structured Parameters

Structured parameters are identified on the parameter configuration form with a control showing an ellipsis. The detailed parameter values are displayed separated by a comma.

Selecting this control opens another Parameter Configuration Form showing the detailed parameters that make up the structured parameter. The individual parameters can then be amended. The figure below shows the Parameter form for an Analog Input transducer (XD) scale. When opened, the attributes are exposed.

Parameter Configuration Form

The image shows two overlapping windows from a software interface. The top window is titled "Parameters" and has three tabs: "Block Assign - AI", "Block Header - AI", and "Oper. Parameters - AI". The "Oper. Parameters - AI" tab is active, showing a list of parameters on the left and their values on the right. The parameters include ST_Rev, Tag_Desc (AI Input Block), Strategy (0), Alert_Key (0), Block_Err, PV, Out, and XD_Scale (100.000000,0.000000,K,0). The bottom window is titled "Attributes of XD_Scale" and shows a list of attributes on the left and their values on the right. The attributes include EU_At_100 (Range Upper Limit) with value 100.000000, EU_At_0 (Range Lower Limit) with value 0.000000, Units_Index with value K, and Decimal_Point with value 0. Both windows have OK, Cancel, and Help buttons at the bottom.

Parameter	Value
ST_Rev	
Tag_Desc	AI Input Block
Strategy	0
Alert_Key	0
Block_Err	
PV	
Out	
XD_Scale	100.000000,0.000000,K,0

Attribute	Value
EU_At_100 (Range Upper Limit)	100.000000
EU_At_0 (Range Lower Limit)	0.000000
Units_Index	K
Decimal_Point	0

17521

Date and Time

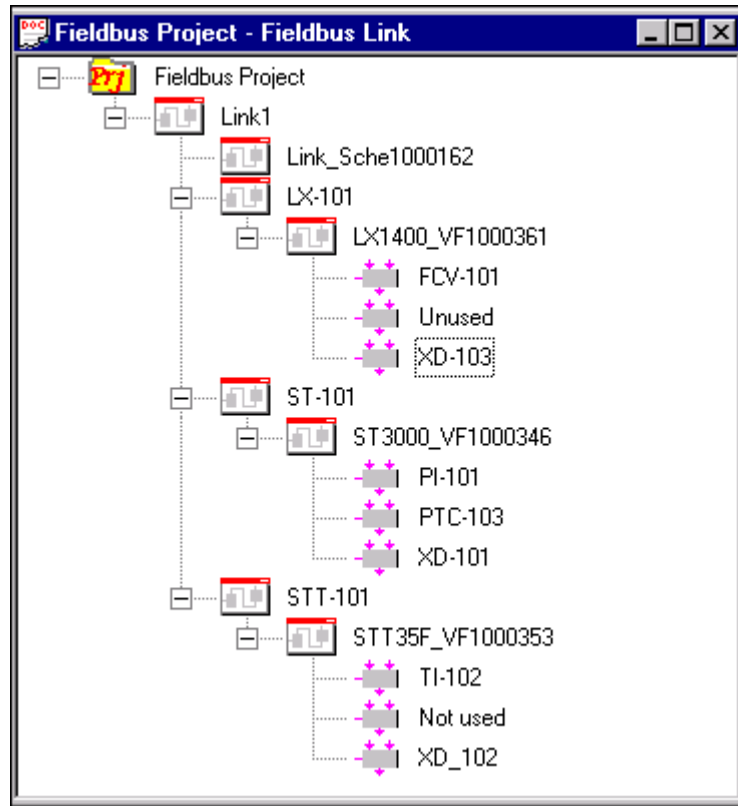
Date and time must be synchronized in TPS Builder with TPS time.

The Date and Time parameter type is displayed in the format mm/dd/yy hh:mm:ss + nnn nn/32. The first part of the display, date and time to number of seconds is displayed in the local date and time format as selected in the Windows NT Regional Settings.

A Date and Time parameter is displayed in a control showing an ellipsis. Selecting this control opens the Date and Time dialog. Date and Time values are entered using this formatted dialog that shows and accepts input for each part of the date and time separately.

To call up the Date and Time Dialog Box:

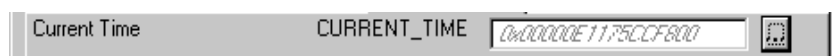
Step	Action
1	In the Fieldbus Link View, position the cursor on a device (LX_101 ST_101, and STT-101) in the following figure are device templates:



3. Fieldbus OLE Controls

3.2. Fieldbus Parameter Types

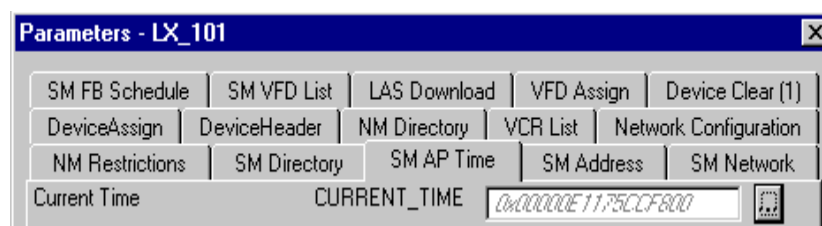
Step	Action
2	Two methods can be used to call up the Date and Time from the Device's Parameter Configuration Form: Setting the Parameter Configuration Form to alphabetical listing of parameters, and then selecting the device, right mouse button click, and choosing parameters. In this format, the parameter Current Time is easily visible. Click on the ellipsis, and the Date and Time Dialog Box opens as shown:



CURRENT_TIME Dialog Box

A screenshot of a dialog box titled 'CURRENT_TIME'. It contains fields for setting the date and time. The 'Month' field is set to 11, 'Day' to 1, and 'Year' to 1999. The 'Hour' field is set to 10, 'Minute' to 26, 'Second' to 0, 'Millisec' to 0, and '1/32 of ms' to 0. There are 'OK' and 'Cancel' buttons at the bottom.

- Positioning the Cursor on the Device, right mouse button click and the Parameters are shown. Choose the SM AP Time tab, as shown in the following figure, to click on the CURRENT_TIME parameter.



Bitstring

Each bit in the string represents an option. The value of the parameter is displayed as a hex value, with a control showing an ellipsis. When this control is selected a dialog showing a check box for each of the options is displayed. The label for each check box is taken from the enumeration values held for the parameter. A checked box represents 1; an unchecked box represents 0 in the bit string.

3. Fieldbus OLE Controls

3.2. Fieldbus Parameter Types

4. Fieldbus Connections

4.1 Connectivity Actions

Connections between Strategies

TPS Builder provides the functionality to make connections between TPN strategies, either graphically in a control drawing, or by entering a *tag.parameter* value using the parameter configuration form.

For fieldbus, the communication relationship between a block parameter and an interface device or a block parameter and another block parameter are defined using Function Block Link objects.

Purpose of the Link Object

A link object provides mapping between resources and information exchanged through network communications.

Link objects define the following:

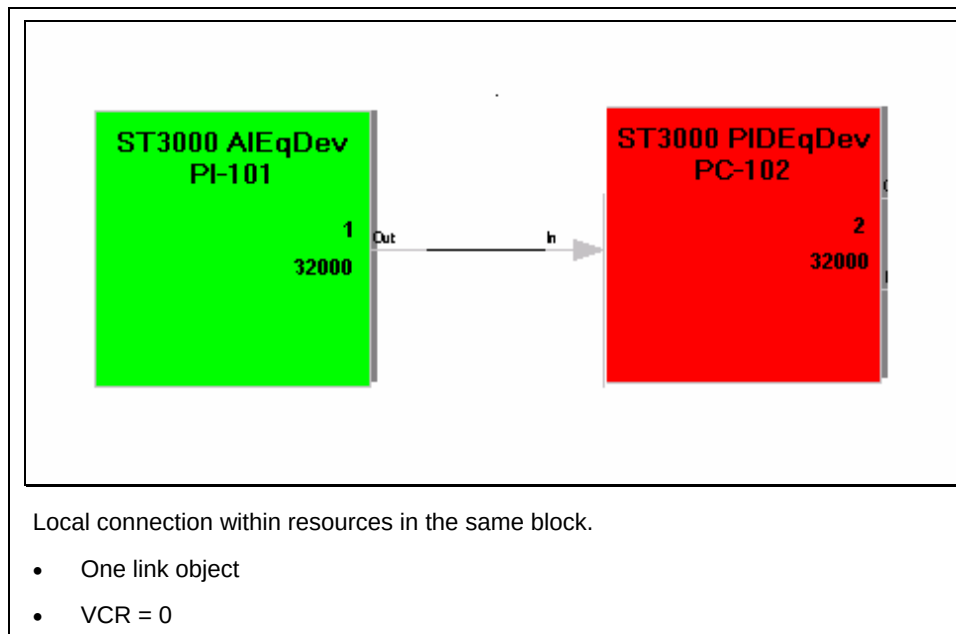
- An association between an output parameter on one block and the input parameter on another block.
- Alert notifications for interface devices, automatically configured to the Fieldbus IOP.
- Trend reports for interface devices. Foundation Fieldbus Trends are not supported; however, trends are available through the TPS Network in the usual manner, by trending the *tag.parameter*. The point, however, should be historized if the user is trending other than current data. (See the *Process Operations Manual*.)

4.2 Block to Block Connection

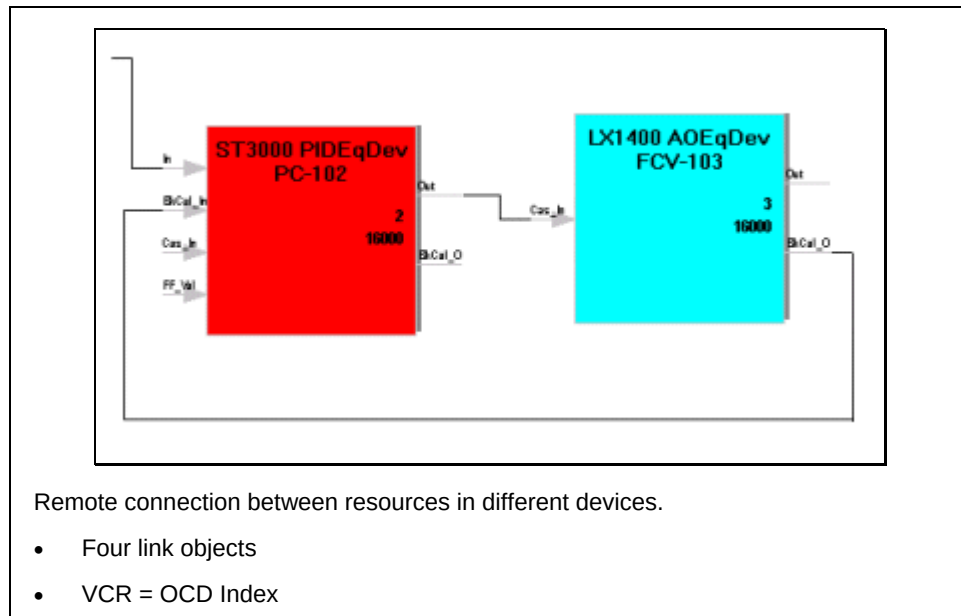
Local and Remote Connections

Blocks may be connected locally or they may be connected remotely. A remote connection is specified by a VCR data transport mechanism. Different types of connection require different numbers of link objects and use different methods to create VCR Numbers.

Link Object Local Connection



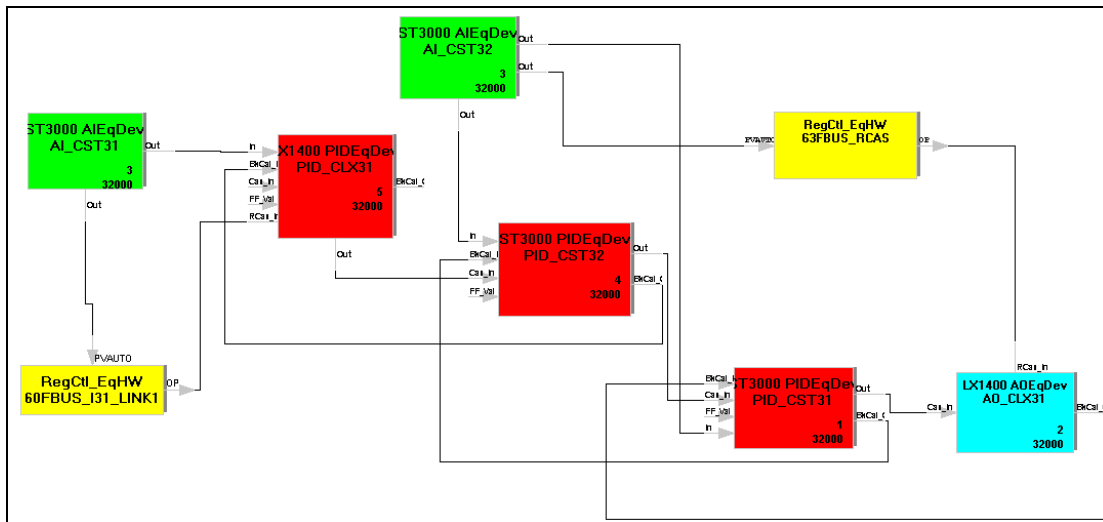
Link Object Block to Block Connections



4. Fieldbus Connections

4.2. Block to Block Connection

Complex Strategy with Blocks in Fieldbus and Points in HPM



- 19 Link Objects — Multiple inputs and outputs
- VCR = OCD Index

4.3 Link Object Relationship to Block Input-Output Parameters and VCRs

Definition

The linkage of parameters together is the first step in defining a connection relationship between function blocks and between function blocks and interface devices on a fieldbus link. The link object contains five fields that describe the linkage:

No	Field Name	Description								
1	Local Index	For a block that produces data, this is the OD Index of the output parameter. For a block that consumes data this is the OD Index of the input parameter.								
2	VCR Number remote communication	For a block that produces data this is the OD Index of the producing VCR. For a block that consumes data this is the OD Index of the consuming VCR.								
3	Remote Index	For the following link objects: <table><tr><td>Subscriber</td><td>OD index of Published Output parameter</td></tr><tr><td>Client</td><td>OD index of Servers Input Parameter</td></tr><tr><td>Local-Consumer</td><td>OD index of producer Input Parameter</td></tr><tr><td>Others</td><td>0</td></tr></table>	Subscriber	OD index of Published Output parameter	Client	OD index of Servers Input Parameter	Local-Consumer	OD index of producer Input Parameter	Others	0
Subscriber	OD index of Published Output parameter									
Client	OD index of Servers Input Parameter									
Local-Consumer	OD index of producer Input Parameter									
Others	0									
4	Service Operation	This defines the type of service operation defined for this block parameter: local, publisher, subscriber, client, server, alert notify, trend.								
5	Stale Count Limit	This is the maximum number of retries before the input parameter status is set bad. This number is used only on the subscriber, server, and the local-consumer (resource to resource) link objects.								

The linkages in the foregoing are managed by TPS Builder during the building of the link, schedules, and blocks.

4. Fieldbus Connections

4.3. Link Object Relationship to Block Input-Output Parameters and VCRs

The Communication Profile of a device restricts the type of I/O linkage that may be defined for a device. The most prevalent function block linkage for a control loop is the publisher/subscriber relationship rather than the client/server relationship.

Device Class	Class Name	FB Link Service Supported
001	Passive Device Class	N/A
101	Intelligent I/O Device	Publisher/Subscriber, Server, AlertNotify-Trend (source)
102	Display-Control Device	Publisher/Subscriber, Client/Server, AlertNotify-Trend (source/sink—see following note)
103	Temporary Device	Subscriber
104	Interface Device	Publisher/Subscriber (when AP is present), AlertNotify-Trend (source/sink—see following note)

Note: A **source** is a *sender* of a message, alert, or data. A **sink** is a *receiver* of the message, alert, or data.

4.4 Valid I/O Connections

Rules for Connecting Inputs and Outputs

The table shown below shows the valid input and output parameters on each of the standard FF function blocks. The general rule is that only output parameters can be linked to input parameters through link objects. Contained parameters may not be specified in link objects. One exception to the general rule is that an input parameter may be linked to another input parameter in the same resource.

Parameters that are linked must be of the same type. I/O parameters can be of the types:

- DS-65 Value & Status — Floating Point,
- DS-66 Value & Status —Discrete,
- or DS-67 Value & Status — Bit String.

Valid Input/Out Parameters on Function Blocks

<i>Parameter</i>	<i>Use</i>	<i>Algorithms</i>	<i>Data Type</i>
BkCal_In	IN	PD, PID	DS-65
Cas_In	IN	PD, PID, AO	DS-65
FF_Val	IN	PD, PID	DS-65
In	IN	PD, PID	DS-65
Trk_In_D	IN	PD, PID	DS-66
Trk_Val	IN	PD, PID	DS-65
BkCal_Out	OUT	PD, PID	DS-65
Out	OUT	AI, PD, PID, AO	DS-65

4.5 I/O Parameter Connections

Two step Connections

Fieldbus FB connections are made in two steps:

Step	Description
1	When a graphical connection is made between an input parameter and an output parameter, the connection is created. This is the connection made by the user.
2	When the Fieldbus Link has been configured and FBs have been assigned to the actual devices present on the link, the actual connections are created automatically by TPS Builder. This process creates the Link Objects and VCR Objects required to support the particular type of connection.

The control parameter connection is derived from the TPN's I/O parameter connection methodology. For instance, TPN control algorithms have a CISRC(n) (Control Input Connection Source) parameter that specifies the "Tagname.Parameter" of the control input value(s) of the control algorithm. FF function blocks do not have the concept of an *input source* parameter that resides on the block and specifies the block name and parameter from which an input is retrieved.

HPM Points Connected to Fieldbus Points

If an HPM control point or logic point is used in the strategy, the point must specify its source when it is an input from the fieldbus block; if it outputs to a fieldbus output block, it must specify the output. An FF block cannot *pull* a value from the HPM point nor can it *push* a value to an HPM point.

A fieldbus block can read from or write to any function block on the same segment.

Restrictions on the Use of Certain Parameters

Refer to the *Parameter References, CL Access (Fieldbus Parameters)* section in the *PM Family Parameter Reference Dictionary* for certain restrictions for read/write parameters when using CL/AM and CL/HPM.

TPS Builder User Support Parameters for Function Block I/O

To specify the control-input connection four *User Support FF* parameters reside on each standard function block template. The *User Support FF* parameters *FB_NOCIN*, *FB_CISRC*, *FB_CIDST*, and *FB_CITYPE* exist to support the block's input parameters.

Parameter Name	Purpose
<i>FB_NOCIN</i>	The number of connected inputs on the block. As each input parameter is connected, <i>FB_NOCIN</i> is incremented. As an input parameter connection is deleted, <i>FB_NOCIN</i> is decremented.
<i>FB_CISRC</i>	An array parameter, with a low index of 1 and a high index equal to the number of input parameters on the FB. <i>FB_CISRC(n)</i> references the BlockTag and parameter (Tag.Param) of the input source. The input source is normally an output parameter. The <i>FB_CISRC(n)</i> value of an unconnected parameter is represented as a null (---- ----).
<i>FB_CITYPE</i>	An array parameter, with a low index of 1 and a high index equal to the number of input/output parameters on the FB. <i>FB_CITYPE(n)</i> specifies the data transport type of the connection: NotConf, Local, Subscriber, or Client. The default is NotConf. When a connection is made, TPS Builder assumes that the connection is a Publish — Subscribe, and the <i>FB_CITYPE</i> is changed to Subscriber. If the connection is actually a client-server type, the connection type must be changed using the Parameter Configuration Form. If the function blocks are in the same device, TPS Builder assumes that it is a local connection; otherwise it is assumed to be a Publish — Subscribe.

The following table shows the assignment of the *FB_CIDST* parameter to function block input parameters for each algorithm type that supports input parameters.

The above User Support FF parameters provide a framework for the building of link objects and VCR connections when actual devices are associated with the function blocks and configured.

4. Fieldbus Connections

4.5. I/O Parameter Connections

The following table assigns the FB_CIDSTN input parameters per algorithm type that support input parameters:

Input Parameter Assignments

Algorithm	FB_CIDSTN index	Input Parameter
PD	1	IN
	2	BkCal_In
	3	Cas_In
	4	Trk_In_D
	5	Trk_Val
	6	FF_Val
PID	1	IN
	2	BkCal_In
	3	Cas_In
	4	Trk_In_D
	5	Trk_Val
	6	FF_Val
AO	1	Cas_In

4.6 Connecting Fieldbus Blocks and TPN Strategies

Mixing Fieldbus and HPM Block Types

TPS Builder allows control schemes to be created which mix fieldbus blocks and TPN strategies. The general rule is that the Fieldbus Blocks may be connected only to TPN HPM RegCtl, RegPV and logic point types. The allowable connection types are maintained in the TPS Builder Repository.

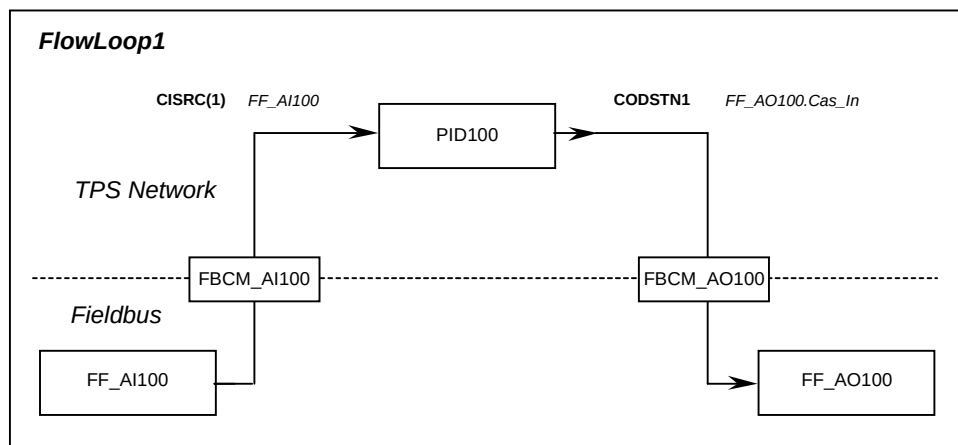
When a connection is made between an HPM control point and a Fieldbus function block, the function block *BlockId* and the function block *ParamName* are held in the **CISRC(n)** or **CIDSTN(n)** parameters in a *tag.parameter* format. However, the true connection attributes of this connection include using a FBCM strategy. Therefore, when the TPN control strategy is Forward Built, the connection parameter is converted to the CM parameter to which the FB parameter is mapped.



TIP

Remember that fieldbus points can neither “pull” their inputs nor “push” their outputs to HPM points. In a configuration in which a HPM Regulatory, Regulatory PV, or Logic point is used, the HPM points must get their inputs from and write their outputs to the fieldbus points.

Flow Loop with PID in HPM RegCtl, Input and Output in Fieldbus



4. Fieldbus Connections

4.6. Connecting Fieldbus Blocks and TPN Strategies

Parameter Name	Parameter Configuration Form	EB File
PID1.CISRC1	FF_AI100Out	FBCM_AI100.PV
PID1.CODSTN1	FF_AO100.Cas_In	FBCM_AO1.\$CASIN

Control Attributes Parameters

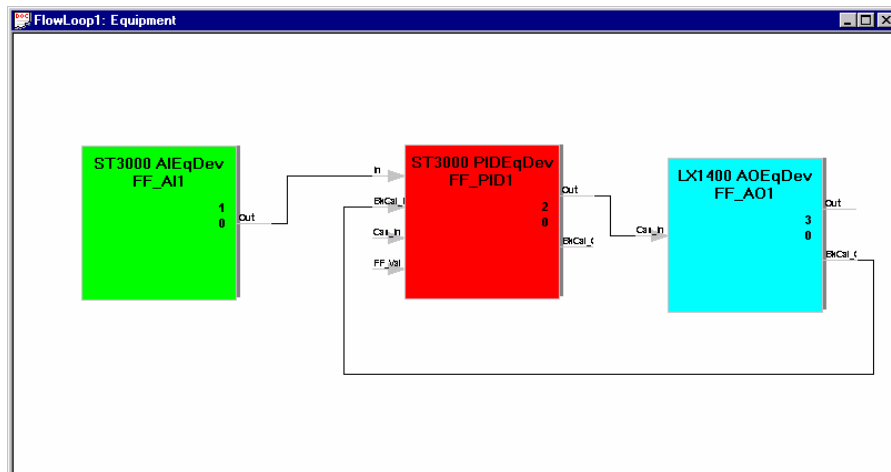
The parameters for the HPM RegCtl (as shown in preceding figure) are accessed by selecting the function block (selection is indicated by the six black “handles”) and selecting *<Control Attributes>*.

4.7 The BkCal_Out to BkCal_In Connection

When (1) a connection is made from the *Out* parameter of a fieldbus control block to the *Cas_In* of another fieldbus block, or (2) a connection is made from a TPN Control point **CODSTN(n)** parameter to the *Cas_In* parameter fieldbus block, then the supporting *BkCal_Out* to *BkCal_In* connection will be automatically made by TPS Builder, if it is required.

The following figure illustrates the *BkCal_Out* parameter connection in a strategy using three fieldbus function blocks, AI, PID, and AO.

BkCal_Out to BkCal_In



Normal connection usage parameters are exposed. If the back-calculation (*BkCAL_Out*) is not required, the connectors should be deleted.

4. Fieldbus Connections

4.8. Storing Constants in an Input Parameter

4.8 Storing Constants in an Input Parameter

Introduction

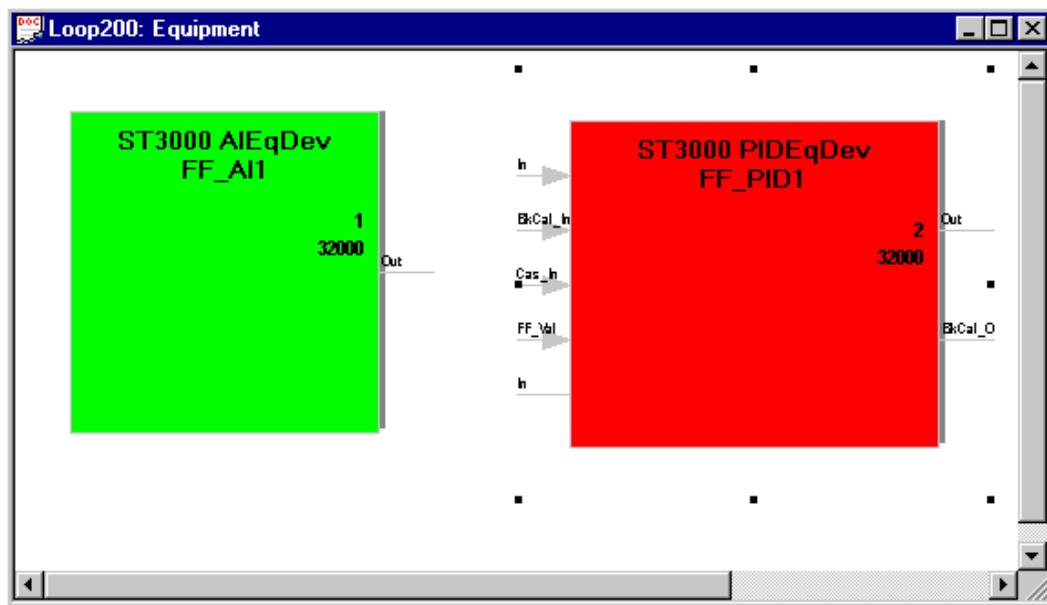
TPS Builder allows a constant value to be entered into an Input parameter. If a constant value is required, this is entered into the input parameter using the Parameter Configuration Form.

Store a Constant in an Input Parameter

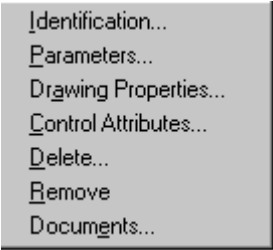
The procedure for storing a constant into an input parameter follows. The block selected is a PID block on an ST3000 device, as shown in the following illustration.

Prior to this procedure, the engineer has dragged the Fieldbus Control Module Loop onto Project in the Project's Equipment View. Onto the loop he has dragged the ST3000 device, and named the AI Block FF_AI1 and the PID block FF_PID1. Step-by-step instructions for building the strategy are in "Configuration Example 1" in this document.

Storing a Constant in an Input Parameter

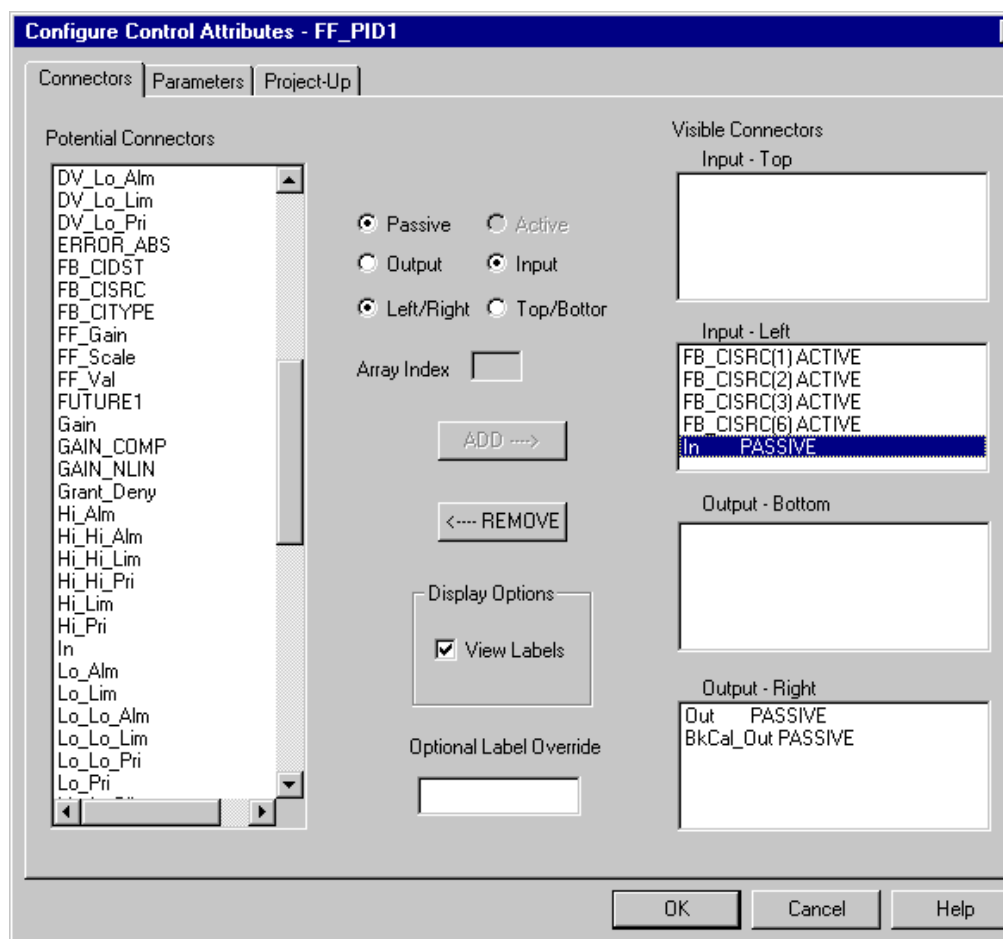


Procedure: Store Constant in Input Parameter

Step	Action	Result
1	Place the cursor on the block FF_PID1 as shown in the illustration. Right click.	The selection list appears. 
2	Select <Control Attributes>.	The Configure Control Attributes dialog box appears.

4. Fieldbus Connections

4.8. Storing Constants in an Input Parameter

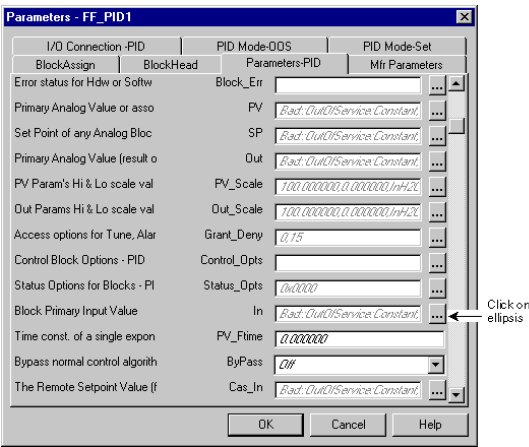


- 3 Select the Connectors tab and click on Input option button. In the Potential Connectors column (left), choose *In*, click on the ADD →. The *In* parameter is added to the Input Left parameters.
- 4 Click OK and close the form.

4. Fieldbus Connections
4.8. Storing Constants in an Input Parameter

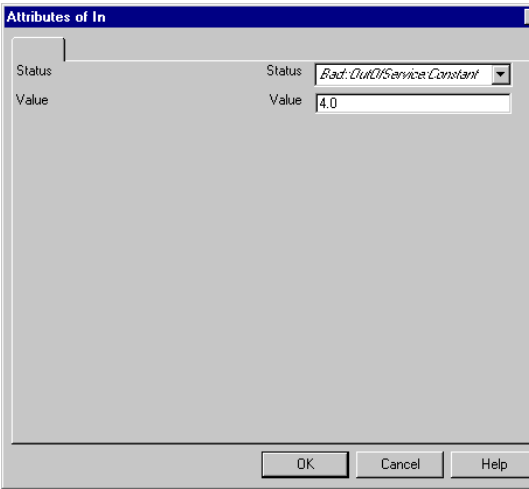
- 5 Select the PID block in the drawing. Handles appear (as in the illustration above this procedure). Right click and select <Parameters> from the selection menu.

The **Parameters** form appears.

The image shows a dialog box titled "Parameters - FF_PID1". It has a tabbed interface with four tabs: "I/O Connection -PID", "PID Mode-D05", "Parameters-PID", and "PID Mode-Set". The "Parameters-PID" tab is selected. It contains a list of parameters with their current values and a column of ellipses (...) for editing. The parameters include: Error status for Hdw or Softw (Block_Err), Primary Analog Value or asso (PV), Set Point of any Analog Bloc (SP), Primary Analog Value (result o (Out), PV Param's Hi & Lo scale val (PV_Scale), Out Params Hi & Lo scale val (Out_Scale), Access options for Tune, Alar (Grant_Deny), Control Block Options - PID (Control_Opts), Status Options for Blocks - PI (Status_Opts), Block Primary Input Value (In), Time const. of a single expon (PV_Ftime), Bypass normal control algorithm (ByPass), and The Remote Setpoint Value (f (Cas_In). An arrow points to the ellipsis next to the "In" parameter with the text "Click on ellipsis". At the bottom are "OK", "Cancel", and "Help" buttons.

- 6 Click on the ellipsis.

The **Attributes of In** form appears.

The image shows a dialog box titled "Attributes of In". It has a tabbed interface with two tabs: "Status" and "Value". The "Value" tab is selected. It contains a "Value" parameter with a text box showing the value "4.0". At the bottom are "OK", "Cancel", and "Help" buttons.

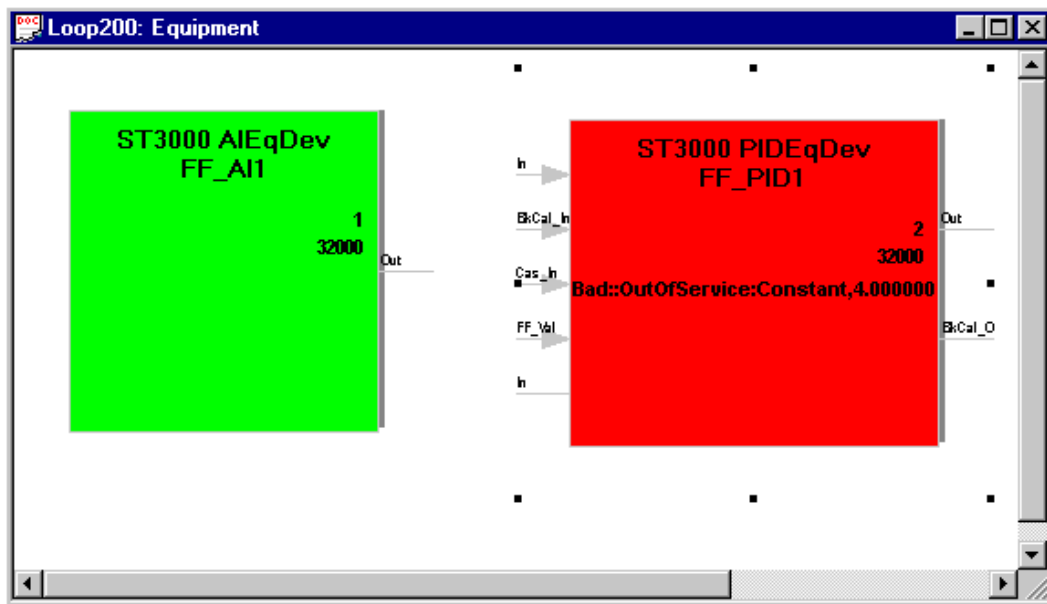
- 7 In the Value parameter, enter the desired constant (4.0 is shown in this illustration). Click OK and the form closes.

The constant is stored in the parameter.

4. Fieldbus Connections

4.8. Storing Constants in an Input Parameter

- 8 The constant, as stored appears in the PID block, as shown in the following figure



Rules

The rules for connecting fieldbus output parameters to fieldbus input parameters are more rigid than the parameter connections between TPN points. Fieldbus Foundation specifications name the output and input parameters by block type. You do not need to define connectors as input/output or passive/active. When a function block is inserted into a drawing, all of the specified output connectors and specified input connectors are graphically displayed.

Not all connection parameters are exposed in drawings of standard blocks; normally, not all will be used.

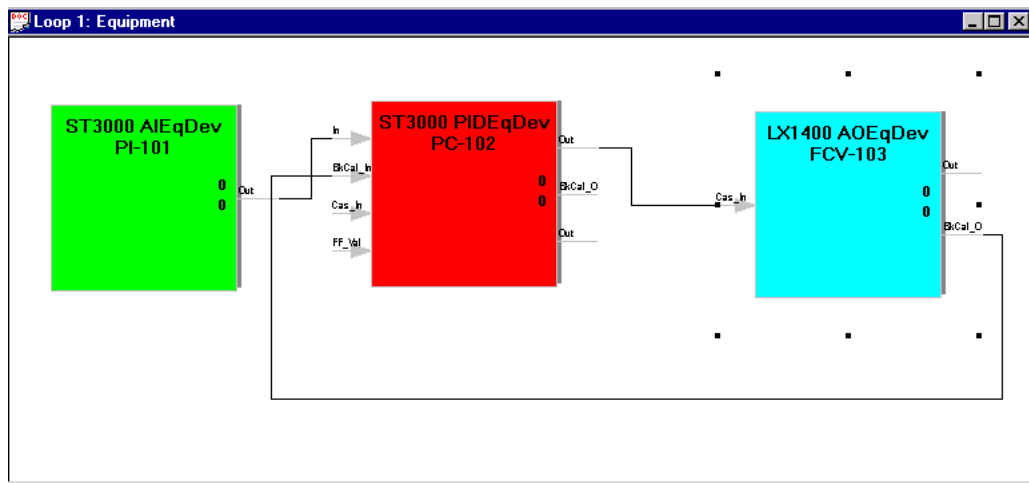
Location of Input and Output Connectors

Input connectors are shown on either the left or top of the block. Output connectors are shown on either the right side or bottom of the block. Output connectors may be connected only to input connectors. Under certain conditions, input connectors may be connected to other input connectors. Connection parameters can be of only two data types: floating point with status (DS65) and discrete with status (DS66). **Parameters of different data types cannot be connected.**

Example of Simple Loop

The following scenario describes the connection of a control loop consisting of an AI block, a PID block and an AO block. Previously the control engineer has created a control module in the project and has selected a standard AI, PID, and AO template from the Block Library to drop into the control module drawing.

Loop with AI, PID and AO Blocks



4. Fieldbus Connections

4.8. Storing Constants in an Input Parameter

Table of Actions for Scenario in Preceding Figure

Step	Action	Response or Comment
1	The control engineer opens the control loop drawing.	The AI, PID, and AO block are graphically displayed, along with all of their exposed input and output connectors.
2	The control engineer selects Connection/connect from the main menu.	The ability to provide graphical connections between control components (blocks) is enabled.
3	The control engineer selects (left-mouse-click) the <i>Out</i> output connection of the AI block and then selects the <i>In</i> input connection of the PID block. The control engineer selects the <i>Out</i> output connection of the PID block and then the <i>Cas_In</i> input connection of the AO block.	A line is drawn between the <i>Out</i> parameter of the AI block and the <i>In</i> parameter of the PID block. A line is drawn between the <i>Out</i> parameter of the PID block and the <i>Cas_In</i> parameter of the AO block. A second connection is shown on the drawing, representing the connection between the <i>BkCal_In</i> parameter of the PID block. If this connection is not required, the engineer uses <i><Connection/Delete Connection></i> on the toolbar to remove the connection.
4	Result of a successful connection	The <i>FB_CISRC</i> input array element tag.param value is changed from null (---) to the tag.param reference of the source parameter. The <i>FB_CITTYPE</i> element of the connection is changed from NotConf to Subscriber or local. The <i>FB_NOCIN</i> parameter of the destination block is incremented by 1 to indicate the number of link objects associated with the block.

5. Configuring a Fieldbus Link

5.1 Introduction to Link Configuration

Project Views

A fieldbus link is configured in the context of a Project. The Fieldbus Link view shows the fieldbus representation of the link, the Hardware view shows the TPN representation of the link, and the Equipment view shows the control strategies.

5.2 Fieldbus Configuration

Two Approaches for Configuration

One of two general approaches can be used for fieldbus configuration:

Approach 1 (Control-based Design)

No.	Step
1	Use the Equipment view to create the required control drawings.
2	Use the Fieldbus Link view to define the physical layout of links and devices. This includes: • Create fieldbus links and select the devices appearing on each link. • Assign the blocks created when producing the control drawings to the devices.
3	Use CM Mapping to create the necessary CM points in the Hardware view.
4	Use the Fieldbus Link view to create the schedule for the link.

5. Configuring a Fieldbus Link

5.2. Fieldbus Configuration

Approach 2 (Device-based Design)

No.	Step
1	Use the Fieldbus Link view to define the physical layout of links and devices. This includes: • Create fieldbus links and select the devices appearing on each link. • Identify any blocks in the selected devices that are not going to be used.
2	Use the Equipment view to create the required control drawings and establish block connectivity. If control drawings are not required, connectivity is established using the parameter configuration form for the blocks.
3	Use CM Mapping to create the necessary CM points in the Hardware view.
4	Use the Fieldbus Link view to create the schedule for the link.

5.3 Fieldbus IOP

The Interface to the HPM

Fieldbus interfaces to the TotalPlant system through a special IOP card that resides in the HPM. Each Fieldbus IOP has a corresponding Field Termination Assembly (FTA) to which up to four Fieldbus H1 segments may be connected.

Configuration of the IOP

The Fieldbus IOP is one of the enumeration values displayed for the IOMTYPE(#) parameter. This parameter can be updated either by using the parameter configuration form or the IOM Assignment dialog. When the Fieldbus IOP is assigned to a card, a module number is also assigned. See “Step 4 — Defining the Hardware” in this document for the procedure to define the IOP.

Link Master Attributes of the IOP

The Fieldbus IOP is the *primary* Link Master for each of the four segments that can be configured for it. It is given the first **device** address on the link. The Link Master contains the Link Active Scheduler (LAS) and therefore controls the scheduling of execution on a block.



TIP

A **software** address, shown below, is not the same as a **device** address on the link.

5. Configuring a Fieldbus Link

5.4. Setting the Release Level

IOP Slot Configuration

CM strategies, SECMs, PECMs and FBCMs, are assigned to slots on the Fieldbus IOP by the CM Mapping function. A procedure for slot assignment is shown in “Step 5 — Adding Fieldbus Link” in Configuration Example 1 in this document.

The Fieldbus IOP has 128 slots. These may be assigned as follows:

Slot No	Use
0	Unused by assignment
1 – 119	Available for PECM and FBCM points. FBCM points will use slots 1 and up. PECM points will use slots 119 and down. The first 32 FBCM slots will be assigned to points mapping to AI and PID function blocks.
120 – 123	Available for SECM points. Slots are assigned to the SECM mapped to Links in the following way: Link at position 1 on IOP assigned to Slot 120 Link at position 2 on IOP assigned to Slot 121 Link at position 3 on IOP assigned to Slot 122 Link at position 4 on IOP assigned to Slot 123
124-127	Reserved to support redundancy.

The CM strategies assigned to each slot in a Fieldbus IOP can be viewed in the same way as for other IO Module types.

5.4 Setting the Release Level

For Fieldbus the Release is 600

Before you can build a link in a project you must set the release to R600. Prior releases of fieldbus do not include the fieldbus libraries and devices. The complete procedure for setting the release level is shown in “Step 1 — Creating a Project” in Configuration Example 1 in this document.

5.5 Creating a Fieldbus Link in a Project

Fieldbus Links

Fieldbus links are also known as fieldbus segments.

A fieldbus link strategy is created in the Fieldbus Link view of a project. To create a fieldbus link:

Step	Action
1	Drag the Link template from the Fieldbus Link library into the project.
2	Use the parameter configuration form to assign a Link Address, in the range 120 to 123, to the link, where 120 represents the first FTA position. When a Fieldbus Link strategy is created, a Fieldbus LAS strategy is also created.

The Link — LAS relationship is modeled as a Containment Relationship in TPS Builder. **Controls that represent sub-schedules, sequences and elements cannot be created by dragging from the Fieldbus Link library.** These are created by the schedule function.

The creation of fieldbus links and schedules is shown step-by-step in “Step 5 — Adding Fieldbus Link” in Configuration Example 1 in this document.

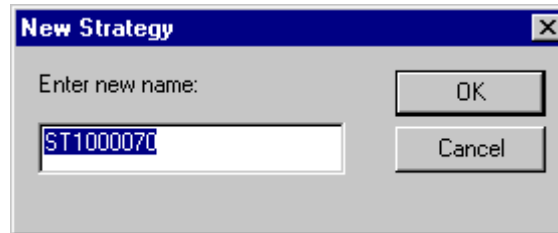
5. Configuring a Fieldbus Link

5.5. Creating a Fieldbus Link in a Project

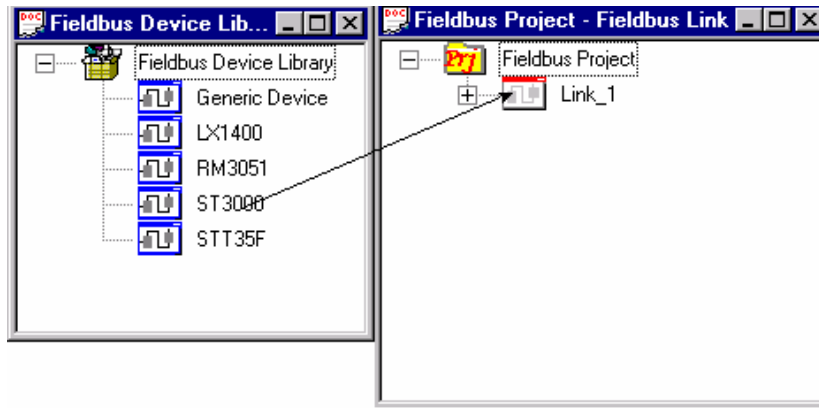
Creating Fieldbus Devices in a Project

A fieldbus device strategy is created in the Fieldbus Link view of a project. To create a Fieldbus Device in a project:

Step	Action	Result
1	Drag the Device template from the Fieldbus Device library into the project.	A New Strategy dialog box appears with a default name in the port. If you want to give the device a name, name it here.



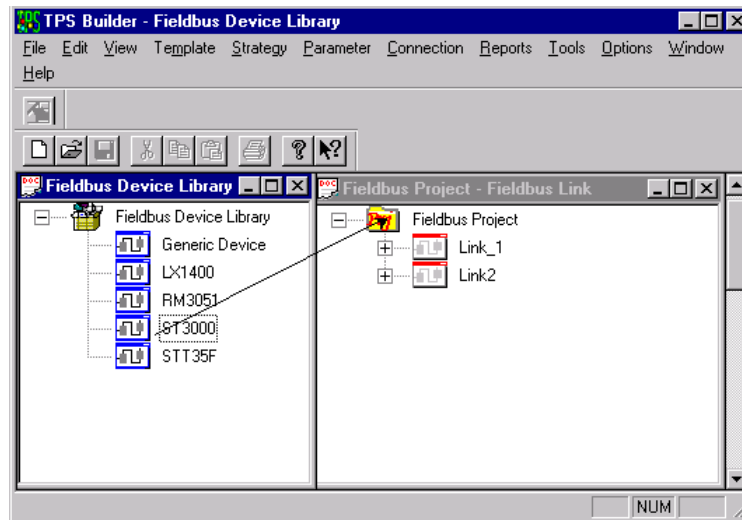
- 2 The device can be dragged onto a fieldbus link, in which case the Link Name assigned to the fieldbus link is assigned; that is, it is 'propagated down' to the device.



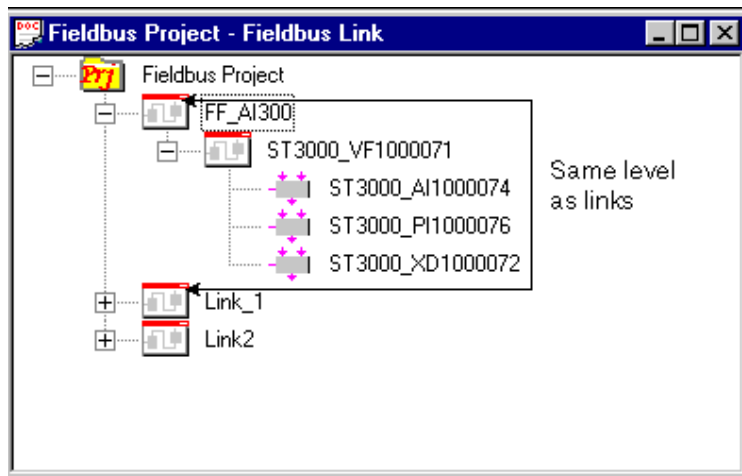
5. Configuring a Fieldbus Link

5.5. Creating a Fieldbus Link in a Project

Step	Action	Result
3	Alternatively, it can be dragged onto the project level, in which case it does not have a Link Name assigned, and must be moved manually to a Link subsequently, when a Link Name is assigned.	



When assigned directly to the project, the device and its accompanying blocks appear on the same level as the links.



5. Configuring a Fieldbus Link

5.6. Associating a Fieldbus Link with a Fieldbus IOP

The Link-Device relationship is modeled as a Containment Relationship in TPS Builder.

Note that the device has function blocks and transducer blocks assigned.

Control-based Design with Placeholders

If Control based design is being used, the device does not have any specific function blocks assigned, but has placeholders for future assignment. Any function blocks that appear in the Fieldbus Link view as a result of creating Control Strategies appear at the Project level. The assignment of function blocks to placeholders is performed by dragging function blocks onto the correct device or automatically by the function block Assignment function.

Device-based Design

If Device based design is being used, all function blocks and transducer blocks defined as part of the device are created and automatically assigned to the device. The type of design being used is defined in the TPS Builder Settings dialog. Two options are available:

Option	Action
Control	Creates block placeholders in devices for subsequent assignment of actual blocks.
Device	Creates actual blocks and assigns them immediately.

5.6 Associating a Fieldbus Link with a Fieldbus IOP

Using the Parameter Configuration Form

A fieldbus link is associated with its Fieldbus IOP FTA by software slot assignment using the parameter configuration form. Configuration assignments, including Network Number, Node Number and Module Number, identify the IOP; the Link Number, range 120 – 123, identifies the SECM's position in the IOP. That is, the FTA position 1 corresponds to address 120, FTA position 2 corresponds to 121, etc.

The Parameter Configuration Form is reached by positioning the cursor on the icon for the link in the project's link view, right mouse button click, and choosing *<Parameters>*. A procedure for configuring the link is in "Step 5 — Adding Fieldbus Link" in Configuration Example 1 in this document.

5.7 Assigning Function Blocks

FB Assignment is the function that matches function blocks to the physical devices present on a fieldbus link. A function block is assigned to a device by dragging the block strategy onto the placeholder it is to replace.

Before completing the assignment, TPS Builder ensures that the device contains an unassigned function block placeholder of the required function block type.

The device – function block relationship is modeled as a Containment Relationship in TPS Builder.

Note: If a Block was previously assigned to another device; that is, the function block is moved from one device to another, the other device is also updated so that an empty block placeholder is now present in the device.

5.8 Relationship of Fieldbus Entities to TPN Strategies

CM Mapping

To enable users to have access to fieldbus entities present on a Fieldbus Link, the fieldbus entities are mapped to corresponding TPN strategies. These TPN strategies are assigned to software slots on the Fieldbus IOP. TPN CM strategies are created by the CM Mapping function in TPS Builder. CM Mapping is performed from the Fieldbus Link view. The CM Mapping function is available from the mouse right-button menu of a Link control.



ATTENTION

Since CM Mapping maps the strategies to the correct CM point type, SECM, PECM or FBCM, the PED should not be used to configure the CM points.

Note: CM Mapping can be performed at any time, but if function blocks have not been assigned to devices, then no FBCM strategies are created.

5. Configuring a Fieldbus Link

5.8. Relationship of Fieldbus Entities to TPN Strategies

The CM Mapping function examines each entity on the Fieldbus Link, and creates a related TPN CM point, of the required variant (SECM, PECM or FBCM). For the selected Fieldbus Link, CM Mapping performs as follows:

Step	Action
1	For the Fieldbus Link strategy, create a SECM. Assign this to the address designated for the Fieldbus Link, which comprises Network Number, Node Number, Module Number and Slot Number.
2	For each Fieldbus Device on the Fieldbus Link, create a PECM. Assign this to the next available slot in the FB IOP, in the range 119, 118, 117 . . . 1.
3	For each Function Block in each Device on the Link, create a FBCM. Assign this to the next available slot in the FB IOP, in the range 1, 2, 3, 4, 5 . . . 119.
4	For each Transducer Block in each Device on the Link that has been assigned its own CM strategy, create a FBCM. Assign this to the next available slot in the FB IOP, in the range 119 – 1.

Error Log

CM mapping errors are written to the TPS Builder Error Log. The most likely reason for errors is that no slots are available on the Fieldbus IOP, so an entity cannot be mapped to a TPN CM strategy.

Note: CM Mapping does not attempt to map fieldbus entities to existing CM strategies. If the user has created CM strategies and assigned these to a slot in the Fieldbus IOP, these CM strategies may prevent the creation of CM strategies required during CM Mapping.

Naming Conventions

The CM strategies created during CM Mapping are assigned a unique name, the TPN *NAME* parameter. The name is constructed as follows:

Character	Description
1 – 15	The first 15 characters of the PD_TAG assigned to the fieldbus entity.
16	S — Strategy is a SECM P — Strategy is a PECM F — Strategy is a FBCM

Note: If a strategy with a duplicate NAME is about to be created, characters 13 to 15 are replaced by _nn, where nn is a number between 01 and 99 which makes the NAME unique. If _01 to _99 already exist, a error message is written to the error log and the CM mapping is not be created for the entity.

5.9 Changing the Configuration of a Link

Configuration Changes

If the configuration of a link is changed after FB Assignment and CM Mapping have been performed, these operations must be performed again, as FB Assignment and CM Mapping are not performed automatically as the configuration is changed.

Adding a Device to the Link

When a device is added to the Link, function blocks may or may not be automatically assigned to the device. If they are not assigned, the user must drag the required blocks into the device. The user must also perform CM Mapping again to create the related CM strategies and create the schedule again to update it.

Removing a Device from the Link

When a device is removed from the Link, the user is prompted to remove the function blocks and transducer blocks assigned to the device. If the user chooses to retain the function blocks, the TBs and FBs are kept at the project level. The CM strategies that are mapped to the device being removed are automatically deleted.

Connections to/from the removed device are also removed.

Replacing an Existing Device

If a device is replaced on a link, scheduling **must** be performed again. The scheduling function configures the device ID in the SECM.

5. Configuring a Fieldbus Link

5.9. Changing the Configuration of a Link

An existing device may be replaced under the following conditions:

By an Identical Device with a Different Device ID

To make this change the user must:

- Create a new schedule.
- Load the device configuration, and associated TPN strategies, to the device.
- Match devices (on the Forward Build/Download).

By a Different Device

This change requires a change to the link schedule. To make this change the user must:

- put the PECM and FBCM points assigned to this device in an INACTIVE state
- delete the PECM and FBCM points associated with this device (from the TPN network as well as TPS Builder database)
- remove the existing device from the link
- add the new device to the link physically and through TPS Builder
- assign new function blocks to the device
- perform CM Mapping to create new PECM and FBCM strategies
- create a new schedule
- forward build/download the new devices and schedule

Deleting a Fieldbus Link

When a user deletes a fieldbus link, the devices on the link are also deleted. The CM strategies mapped to the fieldbus entities are automatically deleted.

If you wish to retain the devices, drag them to the project level. If associated CM points exist, the address parameters (NTWKNUM, NODENUM, MODNUM, and SLOTNUM) are set to zero.

If the device is subsequently assigned to another link, a new address is assigned when CM Mapping is performed.

Deleting TPN Strategies

All TPN points (SECMs, PECMs, and FBCMs) must be deleted using the TPS Builder (delete entity function). The TPN database should be checkpointed after the deletion.

5.10 Device Addressing

Address Range

The fieldbus node address range for the assignment of devices is in the range of 0 – 247. FF recommends that devices that have Link Master capability be assigned addresses from 20 upward. Basic devices (those without link master capability) are assigned addresses 247 downward.

It is recommended that users let TPS Builder assign the addresses, unless the devices have an address assigned (that is, **node addresses** have been previously downloaded); in such case the *<Options/Fieldbus/Match Devices>* should be selected and the device address should be defaulted.

- The FBIOP will be assigned the address 20.
- Devices with Link Master capability will be assigned in the range of 21-35; Basic devices have the range 247-233. If the engineer has assigned addresses to the device using the Parameter Configuration Form, these must also be in this range.



ATTENTION

If an address is assigned outside the range of those listed above, it will not be recognized by the segment until the first unpolled address is reset and the link is downloaded.

5.11 Alert Notifications

Definition

An alert is defined as a system event which or alarm which may or may not return to normal. See the *Alarms in User Applications* in the *TPS Fieldbus Overview* for a detailed explanation of alerts.

Alert objects support the reporting of events to interface devices and other field devices.

The Link Objects and VCR entries for Alert Notifications are automatically created by TPS Builder, during the creation of other Link Objects and VCR entries prior to creation of a Link Schedule.

An Alert Notification Link Object and VCR Entry are created for each VFD (that is, FBAP), in each Device assigned to a Link.

Alarms and the reporting of alarms must be configured in the Parameter Configuration Form. See “ Step 13 — Configuring the ACK_Options for the Blocks” and “Step 14-A — Configure the Resource Block's Features and Features Select” in Configuration Example 1 in this document.

Process Alarm Configuration

The configuration of values for process alarms can be delayed until after strategies have been forward built/downloaded to devices. At that time, alarm parameters can be selected and configured on the detail displays screens. If alarms are configured on the VIEW pages, they must be correlated as shown in “Configuration of Alarms” in this document.

6. Fieldbus Link Schedules

6.1 Introduction to the Scheduling Function

Link Active Scheduler definition

The FF Link Active Scheduler (LAS) is that node currently operating on the link, which is:

- 1) functioning as a Link Master or Bridge class device
- 2) currently administering a schedule to the link.

Scheduled and Unscheduled Communications

The transfer of data on the network is both scheduled and unscheduled. The system management function provides for the scheduling of function blocks. Blocks are scheduled to execute in a manner such that the block that is to receive the data will execute after the block that provides the data. In some cases, the manner in which function blocks are scheduled is manufacturer-specific.

Other operations are unscheduled, which prevents background information from interfering with block execution time.

6.2 Link Schedules

Overview of the Schedule Building Process

In TPS Builder, a fieldbus link may have a number of link schedules. When a Link strategy is created in a project, a single link schedule is created for the Link. However, the user may create additional schedules for a Link if he chooses. All link schedules are initially empty. The LAS control has a build method, which invokes the creation of the link schedule. Before a link schedule can be built, the 'soft' links created when the control drawing was drawn must be translated into true connections. This process creates the real connections and the link object and VCR objects.

6. Fieldbus Link Schedules

6.2. Link Schedules

Building of a schedule is in three parts:

No	Description
1	TPS Builder generates a recommended link schedule for the device, using the connectivity of the devices and their blocks assigned to the link.
2	TPS Builder displays the recommended link schedule in a graphical format. The schedule is displayed with each loop assigned to a sub-schedule.
3	The user is able to make changes to the link schedule using the graphical display. When the scheduled is completed, it is saved. At this time TPS Builder creates additional controls to represent the sub-schedules, sequences, and elements included in the schedule.

If a link schedule has already been built, the current link schedule is displayed and can be changed. The user has the ability to revert to the recommended link schedule while changing the schedule.

Commands

The following commands apply to schedules:

- **Load** — allows a link schedule that has been previously created and saved to be loaded and displayed.
- **Save** — allows a link schedule to be written as a collection of controls, and as a file that defines the graphical view of the schedule.
- **New** — allows an additional, Supplementary, link schedule to be created for the Link. When created a Schedule Name is assigned. Initially the new schedule is the same as the schedule currently being displayed.
- **Delete** — allows a Supplementary link schedule to be removed from the Link.
- **Download** — the link schedule(s) are downloaded to a device.

In addition to the commands available from the link schedule dialog, the user is able to select the element in a schedule and activate a menu using the right mouse button. This menu will have the following functions available:

- **Change** — allows the execution time of the selected block to be changed. (This dialog can also be accessed by double clicking on the required element.)
- **Deactivate/Activate** — allows the selected block to be activated (or deactivated) in this version of the link schedule. Deactivating a block will also cause the associated Publish elements to be deactivated. Deactivated blocks will appear differently when the schedule is displayed.



TIP

Blocks that are not included in the master link schedule cannot be added to a supplementary link schedule. The addition of blocks will require a change to the VCRs, Link Objects and Function Block Schedules configured in the IOP and/or Devices configured on the Link. This would mean that the resulting link schedule could not be used as a backup, as it would require a change to the configuration of the link to support it. Therefore, if blocks and/or devices are intended to be used as backup, they must be configured on the link, then deactivated in the master link schedule.

6.3 Creating a Link Schedule

Link Schedule Types

TPS Builder supports two types of link schedules: **Master** and **Supplementary**.

Master Link Schedule

Explanation

The link schedule is created when the Link is first created. This link schedule dictates the creation/configuration of VCRs, Link Objects and Function Block Schedules in the IOP and devices on the Link. The master link schedule is always the schedule that is downloaded to the first schedule parameter in the IOP. The master link schedule may also be downloaded to any number of devices present on the link if required.



ATTENTION

Note: Do not create a link schedule for a link until all the devices are in the correct strategies and CM Mapping has been performed for the link. See "Configuration Example 1" in this document for the correct sequences for building, mapping, and downloading strategies.

Supplementary Link Schedule

Explanation

Additional link schedules may be created as required. These link schedules do not cause any configuration of VCRs, Link Objects and Function Block Schedules; they must use those created by the master link schedule. Supplementary link schedules can be loaded to any number of devices, but cannot be loaded to the first schedule parameter in the IOP.

For example, a supplementary link schedule that omits all IOP communications might be created for downloading to memory-limited Link Master field devices. Such a link schedule would maintain function block execution after a loss of power to the IOP, in which case communication with the unpowered IOP has no value.

Create Master Link Schedule

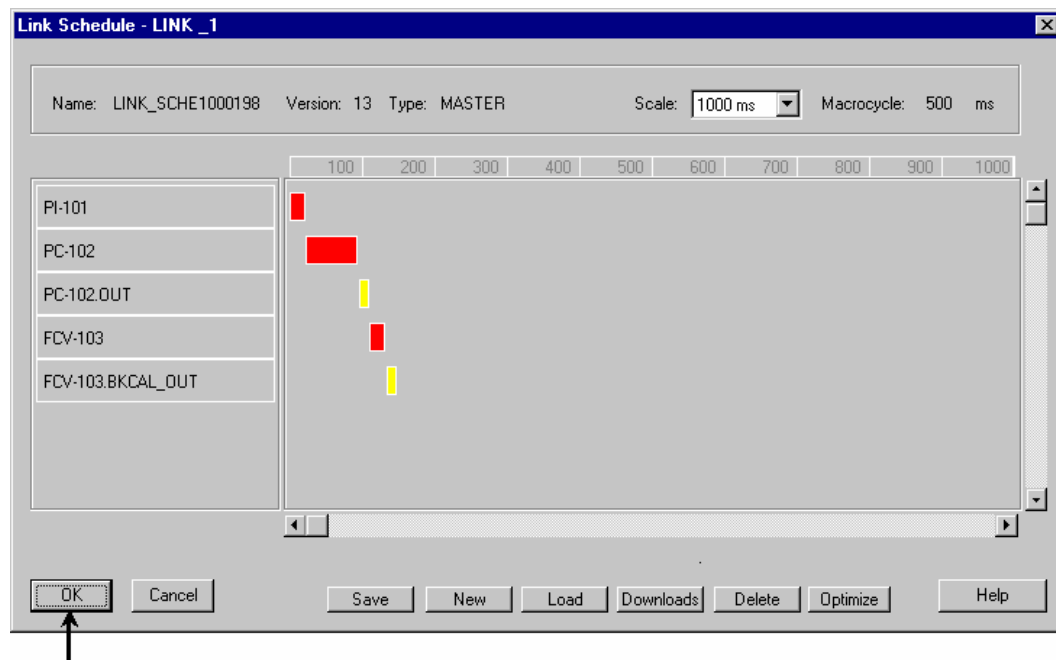
To create the Master link schedule:

Step	Action	Result/Comments
1	Check to see if CM Mapping has been performed for the link.	If it has, the SECM point for the link is shown in the hardware view. (Example: if the link is named Link_1, LINK_1S appears in the Hardware View .)
2	All blocks must have execution sequences and execution periods before the schedule will build correctly.	The execution sequence is found on the BlockAssign Parameter tab. The execution period is found on the BlockHead tab.
3	Position the cursor on the Link and right click.	The menu appears.
4	Select the <Schedule> option.	TPS Builder constructs and displays the recommended link schedule. See the example following this discussion.
5	At this point changes can be made to the link schedule.	Blocks can be activated and deactivated by selecting the block and right clicking. A selection menu for activation/deactivation appears.
6	Click the <Save> option or click OK.	The schedule must be saved or no schedule can be downloaded.

6. Fieldbus Link Schedules

6.3. Creating a Link Schedule

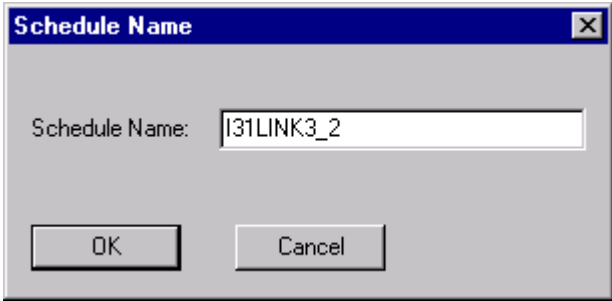
Example of Master Link Schedule



After the Master link schedule has been saved, it is now possible to create Supplementary link schedules and to define the devices to which a link schedule should be downloaded. See the following figure.

Creating a Supplementary Link Schedule

To create a supplementary link schedule:

Step	Action	Result
1	Select the New command from the schedule dialog.	The Create Supplementary Schedule dialog is displayed.
		
2	Enter the name of the new link schedule and click on OK to create the schedule	The new link schedule is created and displayed. At this time it is <i>exactly the same</i> as the schedule it was created from, apart from the name, schedule revision and schedule type.

To change the schedule, blocks can be activated/deactivated, or the execution time of a block can be changed.

Deactivate/Activate a Block

To deactivate/activate a block:

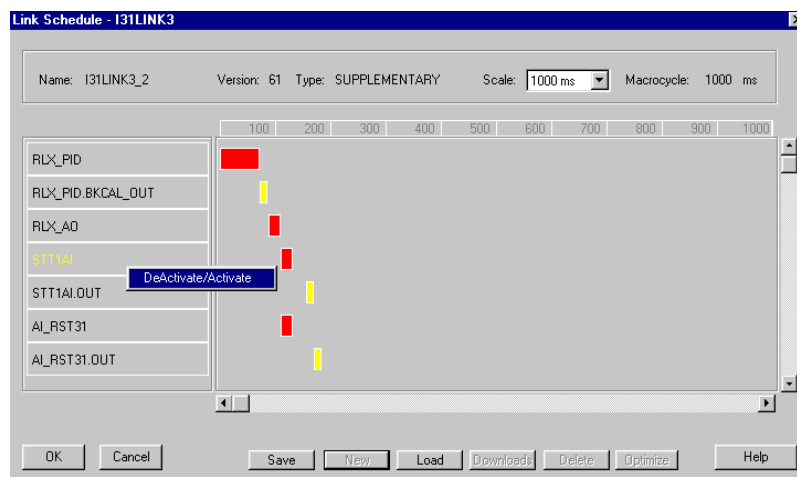
Step	Action	Result
1	Select the block by pressing the left mouse button on the block tag displayed on the left hand of the schedule dialog.	The currently selected block is shown in yellow font.
	Right click the mouse.	A popup menu appears.

6. Fieldbus Link Schedules

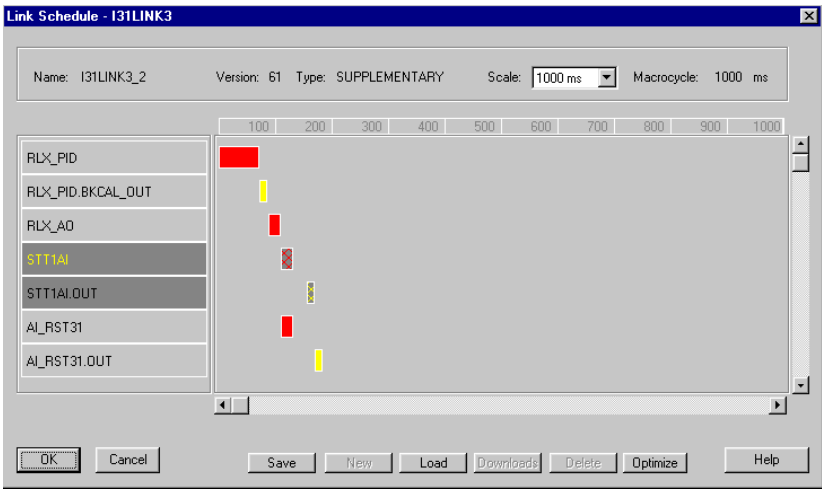
6.3. Creating a Link Schedule

Step	Action	Result
2	Select Deactivate/Activate .	The block, and any associated parameters being published, is deactivated. The deactivated elements in the schedule are displayed in a darker color and are hatched. See the example in the following figure.
3	To reactivate the block, make it the current selection, and select Deactivate/Activate from right mouse button menu.	The block and any associated parameters being published are once again included in the schedule.

Example of Deactivating a Block



Example of a Supplementary Link Schedule



Change Execution Time of a Block

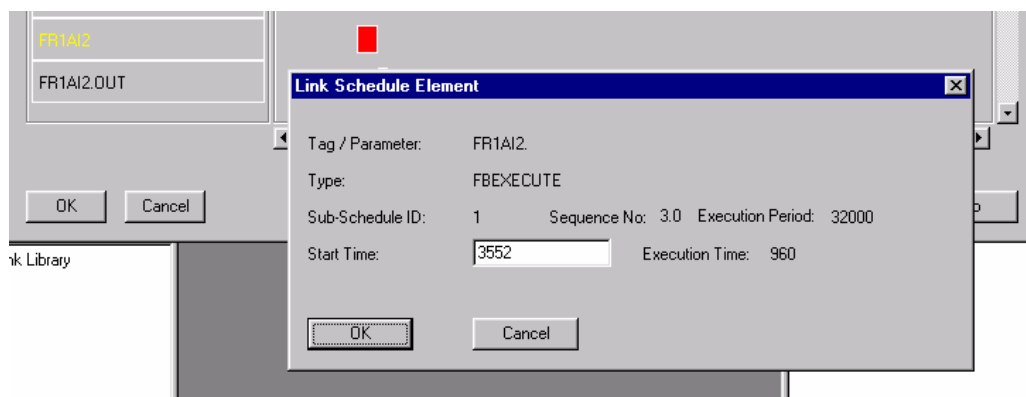
To change the time at which a block will execute in the macrocycle:

Step	Action	Result
1	Double click on the block tag display	The Link Schedule Element dialog is displayed, which shows the current details for the block
2	Update the start time (in 1/32 ms from the start of the macrocycle)	All other blocks executing after this block are rescheduled, and the new schedule is displayed when the dialog is closed.

6. Fieldbus Link Schedules

6.3. Creating a Link Schedule

Changing Execution Time for Block



Save Link Schedule

To save the details of a Supplementary link schedule:

Select the <Save> command.

This function creates a text file of the link schedule details in a folder on which the SQL server is located:

HWIAC\databases\TPS Builder\ProjectName\linkunit

using a .las file extension, which is used when supplementary link schedules are subsequently loaded.

Parameters Tab — LAS Download

Parameters - RLX		
NM Restrictions	SM Directory	SM AP Time
DeviceAssign	DeviceHeader	NM Directory
SM FB Schedule	SM VFD List	VCR List
LAS Download		Network Configuration
VFD Assign		Device Clear (1)
Schedule Names to Downloa	FB_LasDownload(1)	I31LINK3_2
Schedule Names to Downloa	FB_LasDownload(2)	
Schedule Names to Downloa	FB_LasDownload(3)	
Schedule Names to Downloa	FB_LasDownload(4)	
Schedule ID to Download	FB_LasDownloadID(	1003808
Schedule ID to Download	FB_LasDownloadID(	0
Schedule ID to Download	FB_LasDownloadID(	0
Schedule ID to Download	FB_LasDownloadID(	0
Schedule Name to Activate	FB_LasScheduleActi	I31LINK3_2
Schedule Version Number to	ScheduleActivationV	63
OK Cancel Help		

Although these parameters can be configured using the Parameter Configuration Form, this should not be necessary as they are configured from the LAS Download dialog.

6. Fieldbus Link Schedules

6.3. Creating a Link Schedule

Loading a Supplementary Link Schedule

When a supplementary link schedule is saved, a new group of schedule controls are created as part of the Link. These controls look the same as those of the master link schedule. However, a supplementary link schedule does not have the capability of the ‘Schedule’ command, and a message is displayed if this is selected from the Right Mouse Button Menu. To subsequently load and display a supplementary schedule that has previously been saved, select the <**Load**> command. A list of existing link schedules is displayed. Select the link schedule to be opened. The schedule is opened and displayed in the schedule dialog.

Recreating the Original Recommended Link Schedule

To revert to the original recommended link schedule, select the <**Optimize**> command. The original recommended link schedule is recalculated and displayed. Any previously deactivated blocks are now included in the link schedule and all start times are recalculated.

Deleting a Supplementary Link Schedule

To delete a no longer required supplementary link schedule, select the <**Delete**> command while the Master schedule is displayed. Select the link schedule to be deleted from the available supplementary link schedules list, and select <**Delete**>. The selected link schedule is deleted.

6.4 Downloading a Link Schedule to a Fieldbus Device

Overview

TPS Builder provides the functionality to download a link schedule to the schedule domain of a device, and to configure the parameter in the device that defines the link schedule to be activated.

This functionality consists of two sub-functions:

- the creation of multiple link schedules for a fieldbus link, and the definition of the link schedule(s) which should be downloaded to the selected device(s).
- the downloading of the link schedules to the assigned devices configured on the link.

This capability is determined from the following device parameters:

DeviceClass	= LINKMASTER
NumOfSubSchedulesPerSchedule	>= No of Sub Schedules in Current schedule
MaximumSingleLasScheduleSize	>= Size of current schedule

Device Assuming the Role of Link Active Scheduler

In the event of loss of power to the IOP, the IOP loses its role as Link Active Scheduler. Should this occur, if the segment still has power and more than one device has the link schedule loaded, the active device with the lowest address will assume the role of Link Active Scheduler.

Creation of Multiple Link Schedules for a Fieldbus Link

Master Link Schedule

When a link is first established in a project, a single link schedule control is also created. This link schedule, designated MASTER, is created using the **Schedule** option from the right mouse button menu of the Link Schedule. The schedule option also configures the devices on the link and the associated CM points. The entities configured include VCR Static Entries, FB Start Entries and Link Objects for devices and VCR Static Entries, FB Start Entries, Link Objects, Device List and other parameters for CM points. Therefore, the 'Schedule' function can be accessed only from the MASTER schedule.

6. Fieldbus Link Schedules

6.4. Downloading a Link Schedule to a Fieldbus Device

Additional, Supplementary Link Schedule(s)

Additional link schedules, designated SUPPLEMENTARY, can be created only once the MASTER schedule has been created and saved. Also, any devices and/or blocks required for any of the schedules must be included in the master schedule, to ensure that the necessary VCRs, FB Start Entries and Link Objects are configured and downloaded to the device.

Restrictions on Link Schedule Downloads

Once a link schedule has been created, either the MASTER or a SUPPLEMENTARY, this can be designated to be downloaded to any or all of the devices on the link which are defined as having Link Master capability. **A link schedule cannot be downloaded to a Basic-class device.**

Blocks that are not included in the master link schedule cannot be added to a supplementary link schedule. The addition of blocks requires a change to the VCRs, Link Objects and Function Block Schedules configured in the IOP and/or Devices configured on the link. This means that the resulting link schedule could not be used as a backup, as it would require a change to the configuration of the link to support it. Therefore if blocks and/or devices are intended to be used as backup, they must be configured on the link, then deactivated in the master link schedule.

Downloading a Link Schedule to a Device

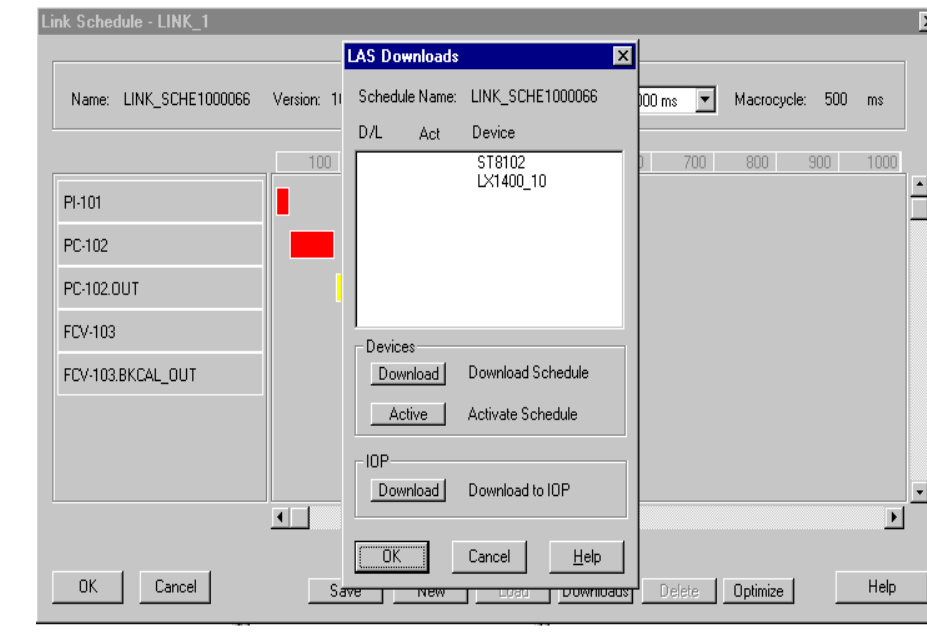
Downloading of a link schedule to the FBIOP is part of the normal download process. Only devices designated as having link master capability can accept a link schedule.



ATTENTION

As failures are unpredictable, it is deemed advisable to download the link schedule to all devices that have the capability of assuming the role of Link Active Scheduler.

Devices have a limited amount of memory, and large link schedules may not download without error. In such cases, it is advisable to determine the critical elements, make a supplementary link schedule, and download this supplementary link schedule to the devices.

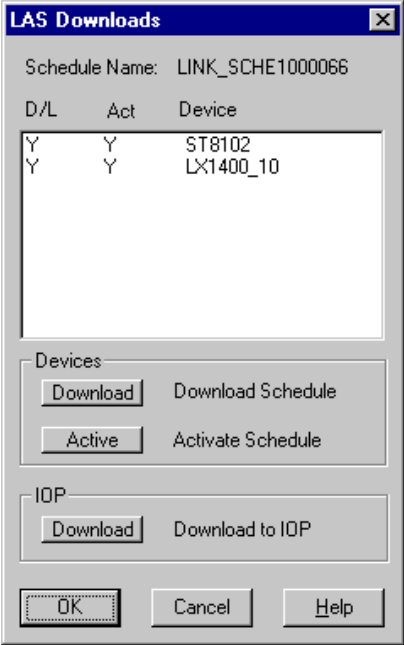


6. Fieldbus Link Schedules

6.4. Downloading a Link Schedule to a Fieldbus Device

Procedure for Downloading a Schedule to a Device with LinkMaster Capability

Step	Action	Result
1	Open the schedule by right clicking on the schedule element.	The schedule appears.
2	Select the Download Schedule target.	The devices with LinkMaster Capability are displayed. (In the example shown, schedules have been downloaded and activated.)



D/L	Act	Device
Y	Y	ST8102
Y	Y	LX1400_10

Devices

Download Download Schedule

Active Activate Schedule

IOP

Download Download to IOP

OK Cancel Help

3	Select the device to which you wish to download the schedule, and select the Download target.	The schedule is downloaded and the letter "Y" appears in the D/L area for the device.
4	To activate the schedule, select the device and select the Active target.	The letter Y appears in the Act area for the device.

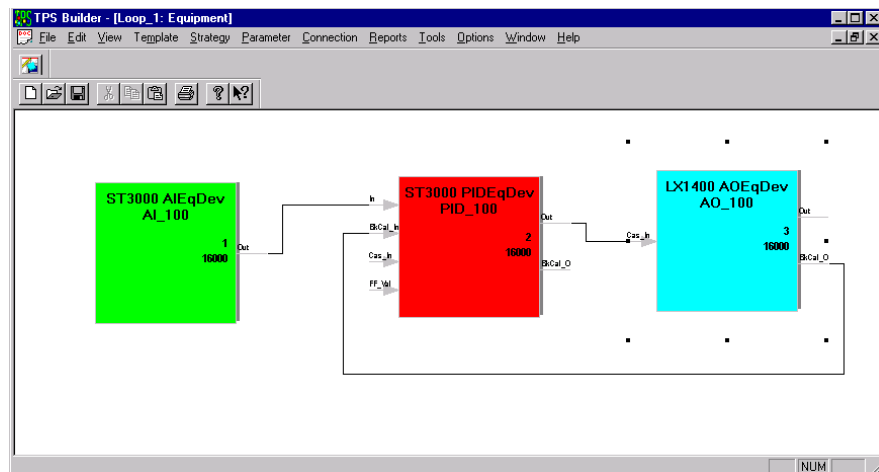
6.5 Loop Execution

Execution Cycles with PID in Input Device

Loop execution period is a multiple of $\frac{1}{4}$ second. The loop execution contains two distinct actions: the execution of the function blocks, and the publishing of data after each block executes. Therefore, a gap exists in the schedule from the time one block executes until the next block in the sequence executes. The macrocycle is the least common multiple of the of all the periods of the loops in the sequences. The macrocycle is created by unrolling each loop as many times as required to fill the macrocycle period.

As illustrated below, a simple loop with three function blocks, Analog In, PID, and Analog out are configured.

Example of Simple Loop with Three Blocks. PID in Input Device



6. Fieldbus Link Schedules

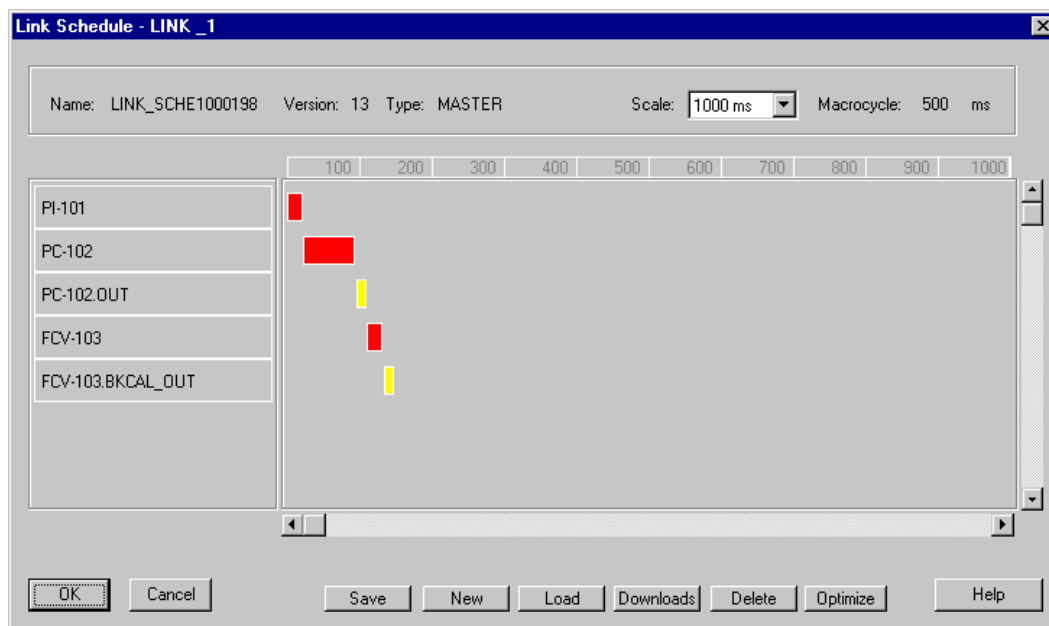
6.5. Loop Execution

The block execution times for Loop1 are:

Loop execution period is 500 ms (the execution period for the loop = 16000 clock units)

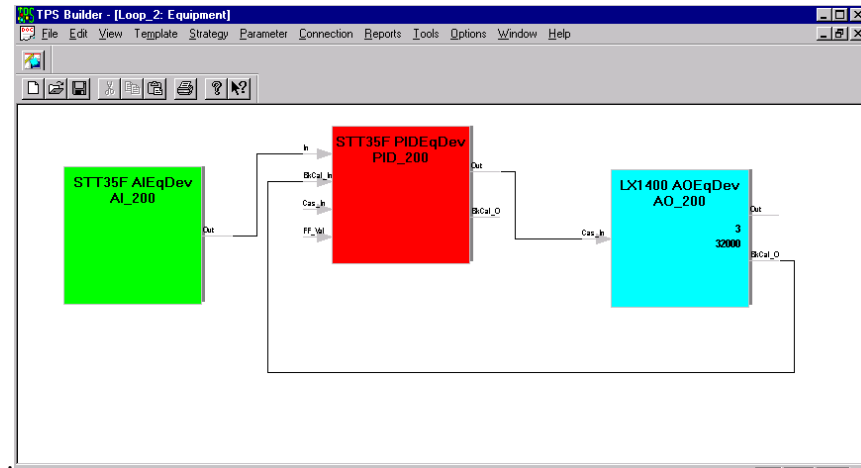
Block	Clock Units	Milliseconds
AI_100 execution time	736	23
PID_100	2336	73
PID_100.OUT	480	15
AO_100	736	23
AO_100.BKCALOUT	<u>480</u>	<u>15</u>
Total Execution Loop 1	4768	149

Link Schedule for Loop1



If we add another simple loop with three blocks with an execution period of one second, we see that the macrocycle for this loop is 1000 (one-second = 32000 ms) and the execution times for the blocks are the same as Loop1. The link schedule displays two cycles of Loop1 and 1 cycle of Loop2, as shown in the following figure.

Example of Simple Loop2, PID in Input Device

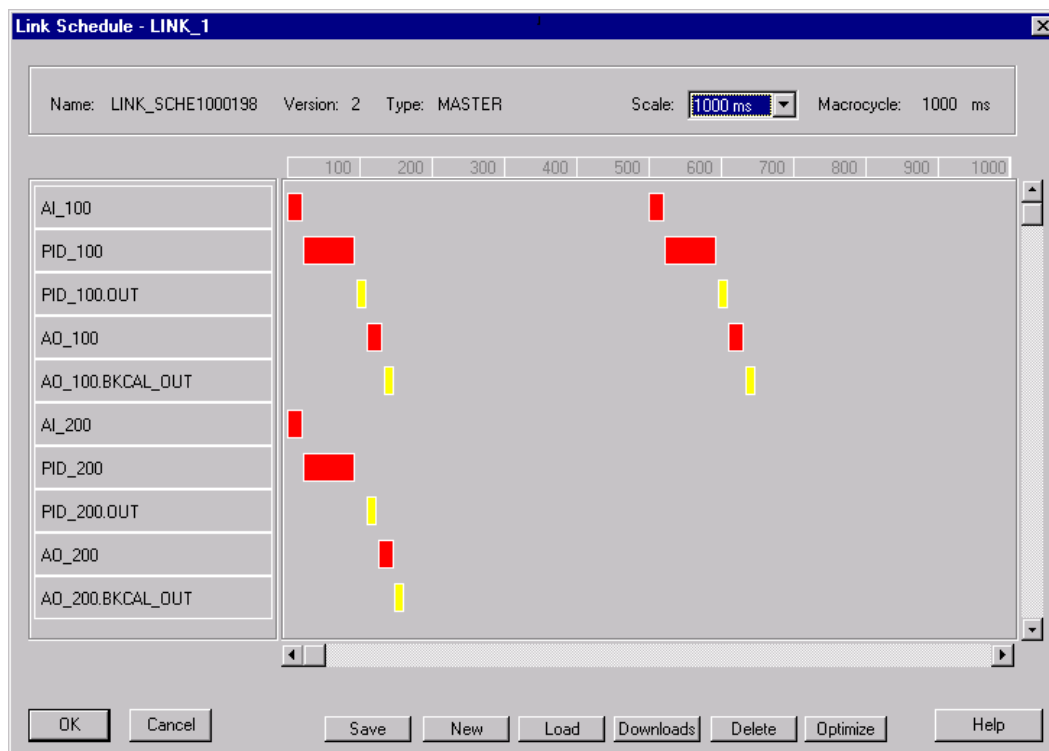


6. Fieldbus Link Schedules

6.5. Loop Execution

The link schedule for both loops follows.

Link Schedule for Loop1 and Loop2, PID in Input Devices



(We have concatenated the display to show the two loops in one picture, but in actual practice the vertical scroll bar must be used to see both cycles.) The block execution times for both loops are the same; only the cycle is changed. Note that Loop1 is displayed twice, but the longer cycle, Loop2 is displayed only once.

The block execution times for both loops are:

Macrocycle 1000 ms (the execution period for the loop = 32000 clock units)

Block/element	Execution Time	
	Clock Units	Milliseconds
AI_100	736	23
PID_100	2336	73
PID_100.OUT	480	15
AO_100	736	23
AO_100.BKCAL_OUT	480	15
AI_200	736	23
PID_200	2336	73
PID200.OUT	480	15
AO_200	736	23
AO_200.BKCAL_OUT	480	15



ATTENTION

It may not be wise to extend the link MacroCycle Duration past one hour because switching to an alternate Link Schedule is only at the end of MacroCycle Duration. These periods or loop times would be the visible configuration parameters for the user. During implementation, it may be prudent to reduce the number of sub-schedules to the first six periods internally. For example, a 10-second loop sequence could be created using a 5-second sub-schedule that executed every other MacroCycle.

6.6 Performance Considerations in Configuring Loops

PID Blocks within the Loop

In general, when fieldbus input and output devices both contain PID blocks of acceptable capability, the PID Block in the output device should be used as the primary PID, rather than the PID in the input device.

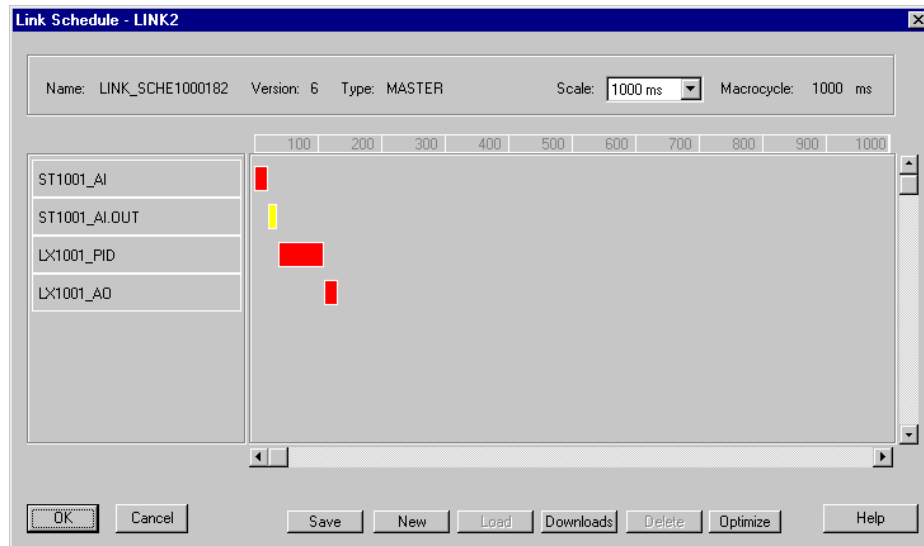
This choice:

1. minimizes the deadtime of the loop
2. minimizes the fieldbus communications load of the loop
3. provides potentially better fallback control of the output should fieldbus communications start to fail, for example from a nearby impulse noise source.

This same rationale applies to the choice of PID in the output device versus PID in the HPM.

See the following schedule for an example of a PID in the output device.

Schedule for Loop with PID in Output device



Compare the execution with that for the corresponding schedule when the PID is in the input device, shown previously in this document as “Execution Cycles with PID in Input Device” in this document. This schedule completes earlier than the other, and all PID-AO handshaking is internal to the device.

Block	Start Time	Execution Time	Elapsed Time
AO block	0	736	736
AI.out	736	480	1216
PID execution	1216	2336	3552
PID out	3552	0*	3552
AO execution	3552	736	4288
AO Back Calc out	4288	0*	4288

*data transfer internal to device

6. Fieldbus Link Schedules

6.7. Scheduling Fieldbus and TPN Strategies

6.7 Scheduling Fieldbus and TPN Strategies

HPM Strategies

HPM Regulatory Control, Regulatory PV, and Logic Points can read from [certain] Fieldbus parameters. Regulatory Control and Logic points can write to [certain] fieldbus parameters. Always consult the *Fieldbus Parameter Reference Dictionary* when writing to and reading from fieldbus points.



ATTENTION

When a TPN control strategy is included in a loop, TPS Builder is not able to schedule the TPN control strategy, either when it will run, or when it will pass the output value to the FBCM strategy. TPS Builder schedules TPN strategies at the end of the loop sequence.

6.8 Recreating the Original Recommended Link Schedule

Optimize

To revert to the original recommended link schedule, select the **Optimize** command. The original recommended link schedule is recalculated and displayed. Any previously deactivated blocks are now included in the link schedule and all start times are recalculated.

7. Configuring CM Strategies

7.1 Introduction to Configuring Strategies

Configuring CM Strategies in TPS Builder

Only minimal configuration of CM strategies is required, such as the Name, Point Descriptor, EU Descriptor and Keyword. Once a CM strategy has been mapped to a fieldbus entity, only the TPN specific parameters on the CM may be updated from the CM. Parameters that appear on the fieldbus entity which are projected or mapped to TPN parameters may be updated only on the fieldbus entity.

The relationship between TPS parameters and Foundation Fieldbus (FF) parameters is discussed in *TPS Fieldbus Overview* section of the *TPS Fieldbus Overview*.

7.2 Creating CM Strategies

CM Mapping

CM strategies are created by the CM Mapping function, when fieldbus entities are matched to a corresponding TPN entity. The user is then able to configure the strategy using the Parameter Configuration Form.

7.3 Parameter Projection and Mapping

Introduction

Parameter projection and parameter mapping concern the ability to view the value of a fieldbus parameter in the corresponding TPN CM strategy.

Parameter Projection

Parameter projection is used when there is a one-to-one relationship between a fieldbus parameter and a TPN parameter.

The fieldbus parameter *Dv_Hi_Lim* is projected to the TPN parameter **FFDVHILM**.

Parameter Mapping

Parameter mapping is used when there is a one-to-many relationship between a fieldbus parameter and a TPN parameter. This is normally used to support the fieldbus structured parameter.

The fieldbus parameter *Bk_Cal_Out* creates the following mapping:

- The value variable *Bk_Cal_Out.Value* is mapped to the TPN parameter **FFBCOPVL**.
- The status variable *Bk_Cal_Out.Status* is mapped to the TPN parameter **FFBCOPST**.

Manufacturer-Specific Parameters

Manufacturer-specific parameters may be added to the TPS Builder repository during processing of a Device Description to create a manufacturer-enhanced or a manufacturer-custom block template. If required, these parameters may be mapped so that they can be accessed in TPS. Mapping is performed to an array of generic parameters. These arrays exist for:

- IDXUS8
- IDXUS16
- IDXUS3
- IDXVS
- US8
- US16
- US32
- FLOAT

data types. For any block, the user is able to select the parameters to be mapped (this could be any unmapped parameter, not just manufacturer-specific parameters) and then the position in the relevant array to where it should be mapped. The display will show any parameters already mapped into the relevant array. The mapping will be held and written to the FBCM point.

7.4 Control Module Generic Parameter Mapping

Purpose

The purpose of mapping fieldbus parameters to TPN parameters is to allow the user to map a parameter to a TPN parameter, thereby making it visible in the TPS system (where otherwise it would not be).

Description

This section describes the user interface in the TPS Builder used to map the parameters of fieldbus function blocks, transducer blocks and resource blocks to CM Generic parameters.

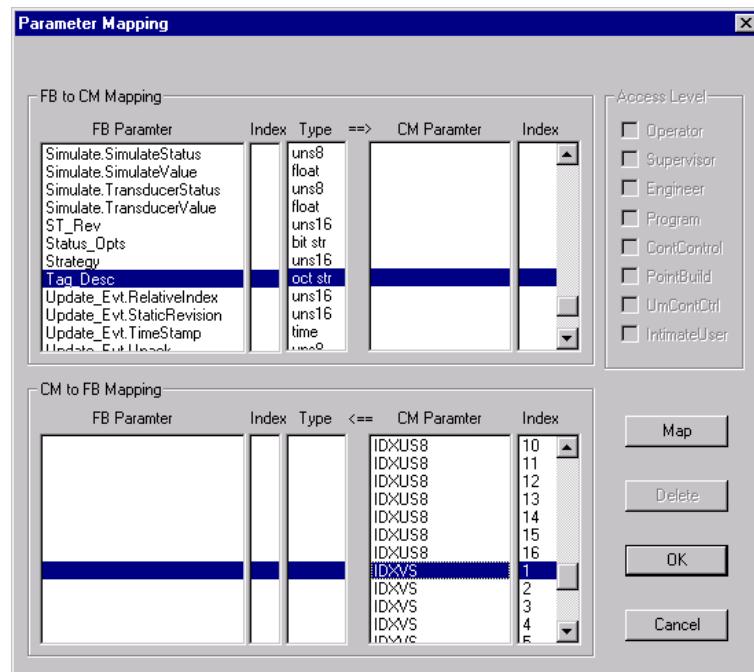
Parameters are mapped at the device and function block level. This means that each different manufacturer's function block can have different mapping applied.

Invoking the Parameter Mapping Dialog

The Parameter Mapping functionality is invoked from the right mouse button menu of a device or a function block template.

Selecting this menu item displays the Parameter Mapping dialog.

Generic Parameter Mapping Dialog Box



The dialog is divided into two groups of list boxes. The top group shows the mapping of fieldbus Parameters to TPN Parameters; the bottom group shows TPN Parameters to Fieldbus Parameters.

Note: While the dialog box shows two ways of looking at the same mapping; two levels of mapping are not included.

A group of checkboxes allows the access level to be defined for the CM Parameter being mapped.

List of Block Parameters

The list of **Block Parameters** includes all parameters present on the block being mapped, Resource or Function Block/Transducer Block.

7. Configuring CM Strategies

7.4. Control Module Generic Parameter Mapping

List of Parameters

The list of parameters includes all the CM Type, PECM or FBCM parameters.

- Device resource block parameters
- Block either function or associated transducer block parameters

List of CM Parameters

The list of CM Parameters includes all the CM Type, PECM or FBCM parameters.

Mapping Parameters

To map a fieldbus parameter to a CM parameter, first select the block parameter/index to be mapped in the top left control. Then select the CM parameter/index that is to be mapped to using the bottom left control. The mapping appears in both views.

Note that a parameter/index cannot be used in more than one mapping. The preceding example (see “Invoking the Parameter Mapping Dialog” in this document) shows Tag_Desc being mapped to IDXVS(1), and an Access Level of Operator being assigned.

After the block and FBCM parameters are selected, confirm the mapping by selecting the “Map” control.

Setting the Parameter Access Level

The Access Level of the parameter is defined by selecting the parameter and selecting the level(s) from the boxes in the Access Level group.

Deleting a Mapping

To delete an existing mapping, select the parameter/index mapping to be deleted using the top left control. Select the “Delete” command to remove the mapping.

Saving the Parameter Mapping

To save the changes made to the mapping, select the OK command. The mapping is saved and the dialog is closed.

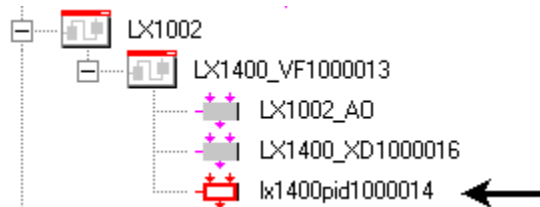
To leave the dialog without saving changes, select the Cancel command. The dialog is closed.

7.5 Deleting CM Strategies

If a CM strategy that is mapped to a fieldbus entity is deleted, the mapping to the fieldbus entity is deleted, but the fieldbus entity remains.

Delete Unused Function Blocks

If a device with function blocks is part of a strategy, such as a loop, and it is not needed, delete it before downloading. The following picture illustrates a deleted function block.



7.6 Configuration of Alarms

Standard TPN Process Alarms

Standard process alarm parameters, such as priorities, high and low PV, deviation alarms, etc. are available for the AI, AO and PID function blocks, as they would be for any other TPN point.

To access the alarm parameters after CM mapping has been performed, position the cursor on the point in the Hardware View and select *<Parameters>*. Select the FBCM Alarm tab; the process alarms available for this point type can be configured from this tab.

7. Configuring CM Strategies

7.6. Configuration of Alarms

Alarm configuration in FBCM points requires the use of priority numbers, as shown in the following table. This table correlates the number used by FF priorities with the TPN alarm priorities. After the points and the MetaData based point detail display are built, the TPN Alarm Priorities are displayed on the point's detail page.

Correlation of FF Alarm Priorities /TPN Alarm Priorities

FF Prior.No.	<i>FF Description</i>	<i>TPN Alarm Priority</i>
12-15	Critical	Emergency
8-11	Critical	High
5-7	Advisory	Low
4	Advisory	Printer
3	Advisory	JnlPrint
2	Low fixed	Journal
1	No Notification	NoAction
0	No Indication	NoAction

Configuring the ACK_Option

The ACK_Option (bitstring parameter) must be set to the value that enables auto acknowledgment for all alarms that are valid for the block being configured. See the *Fieldbus Parameter Reference* for a definition of this bitstring.

This option may be configured by default in the template, but some manufacturers may not define the ACK_Option. The TPS Builder ACK_Option configuration for the AI Block, PID, and Resource blocks is shown in detail in “Step 13 — Configuring the ACK_Options for the Blocks” section in this document.

Features and Features Sel

The manufacturer of a device has a select group of features that are part of the FOUNDATION Fieldbus specifications; the features provided are defined in the device descriptions. Identification of these features, and the user's options to choose from the provided features are configured in TPS Builder. These Features and the Feature_Sel are found in the Resource Block Parameters in the Link View of a project (position the cursor on the VFD, prefix VF, and select *<Parameters/Resource Block>*).

This option may be configured by default in the template, but some manufacturers may not define it. A procedure for configuring the features can be found the Configuration examples, "Step 13 — Configuring the ACK_Options for the Blocks" in this document.

7. Configuring CM Strategies

7.6. Configuration of Alarms

8. Building and Loading CM Strategies

8.1 Introduction to Building/Downloading Strategies

Downloading and Uploading Functions of TPS Builder

The download and upload functions are discussed fully in “Download/Upload Functional Description” section of this document.

Functional Description

CM Strategies automatically have the ability to Forward Build because they are TPN strategies. Forward Build creates an EB file for the selected entities. The Load function is added to all TPN strategies. The Load function automatically loads TPS Builder configuration data to TPN, creating an EB file for selected entities and then using Command EC files and the EC Trigger to activate the load on TPN. The Forward Build and Load options are accessed either by right-clicking for the control, or by selecting item from the <Tools> menu. Choosing the Download by right-clicking (after selecting the control) automatically creates both the forward build and load.

The process of Downloading creates all the necessary VCRs, loads the schedules, and the data to the device(s). In addition, the TPN points are created and the MetaData builds the accompanying TPN displays and VIEWS.

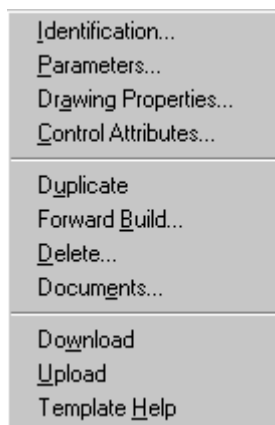
Strategies can be downloaded by link or by selected entities. In all cases, the user must set the path for the online operations.

8. Building and Loading CM Strategies

8.1. Introduction to Building/Downloading Strategies

Forward Build

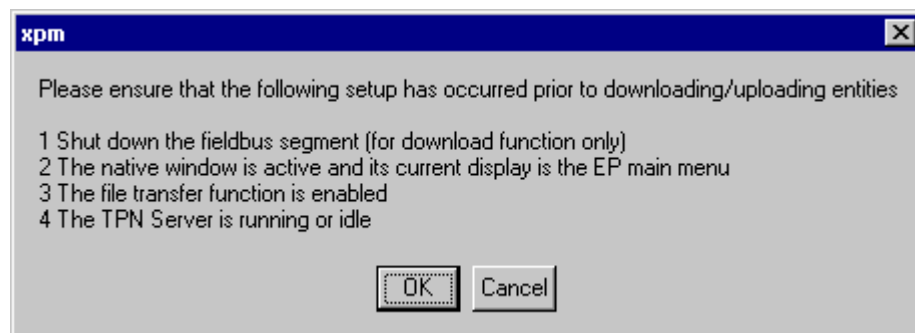
Menu Selections from Control Right-Mouse Button



Position the cursor on the control and select Download from the menu.

Download/Upload Requirements

Before a download or upload can proceed, the following requirements must be met. A warning message is displayed before each download/upload action.



Downloading Fieldbus Links

When the fieldbus link has been configured with the devices, it is customary to forward build and download the strategy in one operation, as shown in “Step 16 — Downloading” in Configuration Example 1 in this document. To accomplish this, the strategies must be on a TPS Builder that is running on a GUS station with a TPN Server.

The Download option creates an EB file for the selected entities, creating an EC trigger, which downloads the files to the configured TPN and fieldbus devices. A bar displays the operation’s progress.

Setting the Path

Set the path as shown in “Setting the Default Paths for Download” in the “Download/Upload” section of this document.

If you are downloading a SECM only for the purpose of identifying the devices, their identification numbers, and/or their addresses, this is customarily done by downloading from the control of the device, right-clicking and selecting Download from the menu. See the procedure in “Downloading only a SECM with No Devices” in this document.

8. Building and Loading CM Strategies

8.1. Introduction to Building/Downloading Strategies

9. Checking the Configuration

9.1 Introduction to the Checker Function

The TPS Builder Checker function allows the configuration of any part of a Library or a Project to be checked in a batch operation. This operation is in addition to checks that are made as data is entered using the parameter configuration form and other user dialogs. Checker performs both individual parameter value checks and specific checks, such as for duplicate addresses, depending on whether or not a Library or Project is selected and the type of strategy being checked.

9.2 Checking CM Strategies

Checker automatically performs the standard TPN checks on a CM strategy, as it is a standard TPN strategy. See information on additional checking performed in “Checking a Fieldbus Link” in this document, following this topic.

9.3 Checking a Fieldbus Link

Standard Parameter Checks

Checker performs standard checks on all parameters on all fieldbus entities. The scope of checking is defined in the Checker parameter dialog.

Fieldbus Specific Checking

Checker also performs a number of fieldbus specific checks when the Checker is executed.

Step	Action
1	The <i>tag.parameter</i> of all fieldbus entities is checked to ensure that it is unique in the project or library. This entails checking the <i>PdTag</i> , <i>BlockTag</i> and other parameters that identify fieldbus entities.
2	Device addresses are checked to ensure that none are duplicated, and that they are in the ranges permitted.
3	The <i>ManufacturerId</i> , <i>ModelName</i> , <i>DeviceRevision</i> and <i>DeviceId</i> are checked to ensure that none are duplicated.
4	All associated Function Blocks and Transducer Blocks are checked to ensure they are assigned to the same Device.

9. Checking the Configuration

9.3. Checking a Fieldbus Link

Additional Checking

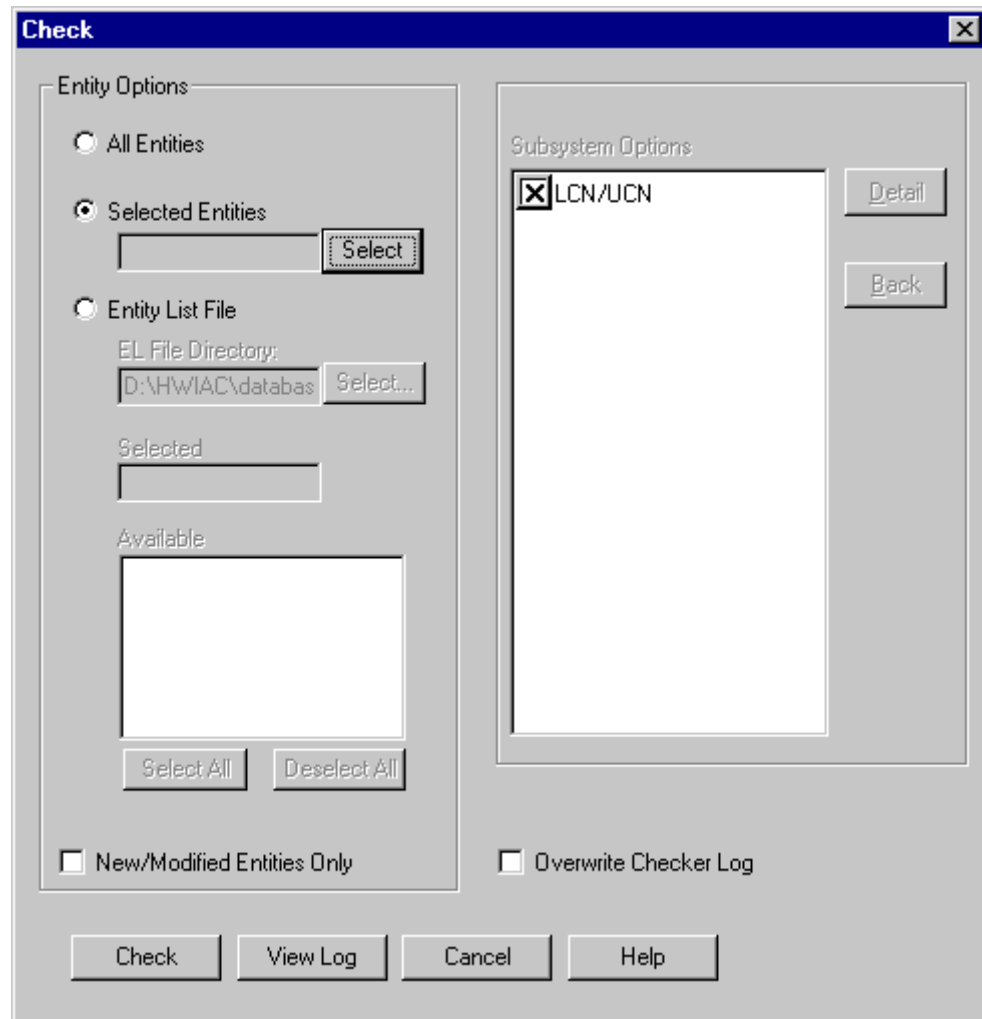
The following limits are checked, mainly in the CM Mapping and Scheduling functions.

No of Devices (PECM) per Link	10
No of Function Blocks (FBCM) per Link	Up to 119 The total of PECM & FBCM cannot exceed 119. The positioning of PECM and FBCM slots is interchangeable.
No of Function Blocks per FBCM	1
No of Transducer Blocks per FBCM	1 [An FBCM must contain at least 1 Function Block or 1 Transducer Block]
No of FBAP VFDs per device	1
Any mix of Function Block types is supported	
Any mix of Transducer Block types is supported	
No of VCRs in IOP	63
No of FB Schedule Objects in IOP	20
Maximum sub-schedules per schedule	4
Maximum number of Sequences per sub-schedule	10
Maximum number of Sequences per schedule in IOP.	29
Maximum size of a schedule	380 bytes.
Minimum size of a schedule	55 bytes (must have at least 1 sequence)

9.4 Accessing the Checker

The Checker function is available from the <File> menu. The name of the menu item is *Check*.

Checker Access

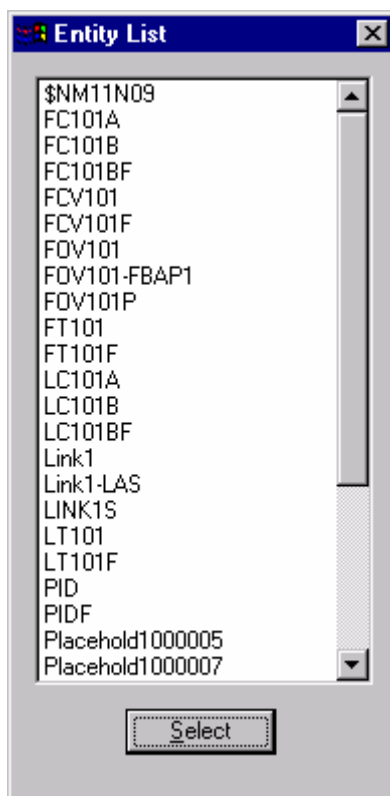


9. Checking the Configuration

9.4. Accessing the Checker

Depending upon the item(s) selected to be checked, you might get a secondary menu from which to select the entities to be checked.

Entity List



10. Detail Display MetaData Files

10.1 Introduction to MetaData

The Detail Display MetaData file describes the structure of the fieldbus device data to a NIM. Each NIM receives a complete definition of the fieldbus Function Block Library. The MetaData file is created from TPS Builder, using the Libraries installed for building the MetaData.

10.2 Creating the MetaData File

Procedure for Creating MetaData

Before you are able to view data from fieldbus devices using the detailed displays you must create a MetaData file on TPS Builder and transfer the file to the NIM. With GUS 341.X or later, File Transfer does not require a remote GUS machine to map the HM. File Transfer maps HM as a namespace and not as a Network drive. Hence the path of the HM is preceded by “!\ “. For Example: !\<volume name>

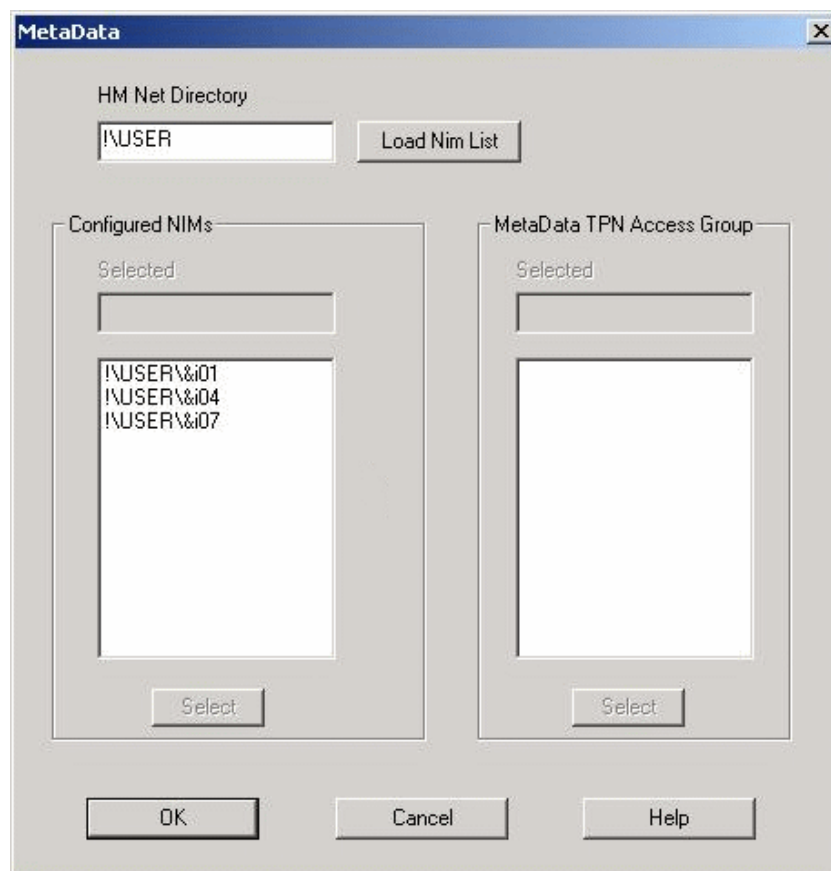
To create a MetaData file, follow these steps:

Step	Action	Result
1	Select <Tools/MetaData> from the TPS Builder main menu.	The MetaData dialog is displayed.

10. Detail Display MetaData Files

10.2. Creating the MetaData File

Step	Action	Result
2	Click the Load NIM List button alongside the input area to display the following	Configured NIMs are displayed.



10. Detail Display MetaData Files

10.2. Creating the MetaData File

Step	Action	Result
3	From the list, select the HM Net Directory. For example, !\<volume name> Click Select and OK .	TPS Builder performs a number of steps to create a MetaData file. The file is then distributed to the NIMs on the LCN, and the parameter \$NTWRKnn.METADATA is set TRUE (where nn is the UCN Number). The NIM will immediately read the new MetaData file and use this file to interpret data read from the device and displayed using the detailed displays. The MetaData file must be in the same directory as the checkpoints.

10. Detail Display MetaData Files

10.2. Creating the MetaData File

11. Creating a Value File

11.1 Functional Description of Value File

Purpose of a Value File

A Value File is a text file, in FF Common File Format, that contains the configuration of a particular device, or device type. A Value file can be used to update a Device template in a library, or a Device strategy in a Project.

The TPS Builder configuration for a fieldbus device can be updated from a Value File using the Device Backbuild function.

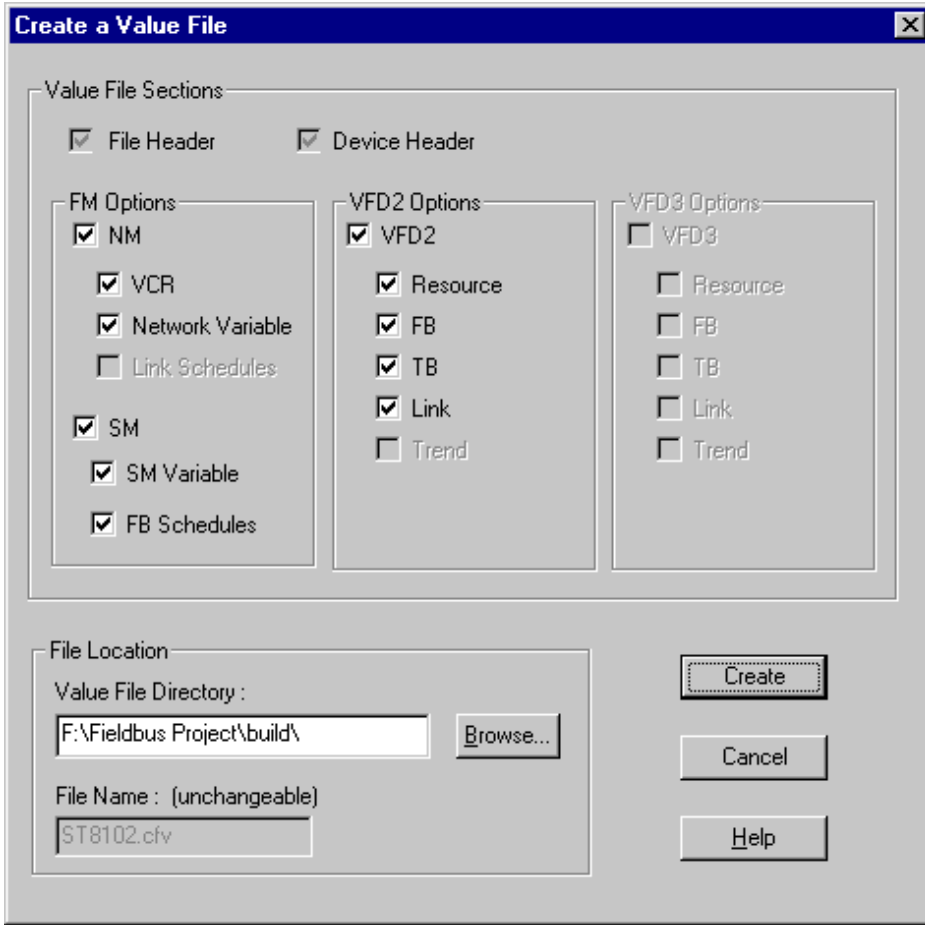
The Device Backbuild function updates only existing strategies or templates; a template or strategy cannot be created. Any errors encountered during the backbuild process are recorded in the TPS Builder Error Log.

Device Backbuild does NOT update any parameter on a strategy that is identified as “Template Defining.” If the Value file contains an entry for such a parameter, and it is different from the “Template Defined” value, an error message is created. When changes are made to a “Template Defining” parameter on a Template, all strategies created from the template are also updated.

11. Creating a Value File
11.1. Functional Description of Value File

Procedure for Creating a Value File

Step	Action	Result
1	In the Fieldbus Link View , position the cursor on the device and right-click.	The menu appears. Save Restore Backbuild ValueFile Identification... Parameters... Drawing Properties... Control Attributes... Duplicate Delete... Documents... Template Help
2	Select Value File .	The Create Value File dialog box appears.

Step	Action	Result
		
3	Set the Value File Directory path, using the Browser and click Create .	The value file is placed in the named directory with the name of the device and an extension .cfv, for example LX1001.cfv .

11. Creating a Value File

11.2. Backbuilding a Device from a Value File

11.2 Backbuilding a Device from a Value File

Accessing the Backbuild Function

The Device Backbuild function is available from the right mouse button menu of the device control. When selected, a standard file open dialog is displayed, which defaults to the 'build' directory for the project and a file extension of ".cfv." The file open dialog allows any file in any directory to be selected as input to the function.

Before processing commences, TPS Builder ensures that the file is a valid file to use to update the selected template or device.

You should also ensure that the Value File is in the correct format and has the correct entries. This involves examining the file for a File Header section, and ensuring that it contains a FileType=Download File.

Example of Value File

The following is a very small excerpt from a Value File [text file] that contains the pertinent information.

```
//=====
```

```
// File Header
```

```
//=====
```

```
[File Header]
```

```
Description=This is a downloaded Value File for device LX1001
```

```
FileType=Download File
```

```
FileDate=2000,08,08
```

```
[Device Header]
```

```
VendorName=FlowServe
```

```
ModelName=LX1400FF
```

```
Revision=Rev 0.67
```

```
DeviceClass=LINKMASTER
```

DeviceName=Valve Controller

CommClass=0x0070

CommSubClass=0xE4

DEV_ID=464C530201-VAL-LX1400-00DEF5FFB0

[Tag Assignment]

[FM OD]

PDTag=LX1001

NodeAddress=22

Vfd=0

VersionOD=1

1]

VcrListOdIndex=291

DlmeLinkMasterOdIndex=332

LinkScheduleListOdIndex=347

DlmeBridgeOdIndex=0

PlmeBasicOdIndex=340

11. Creating a Value File
11.2. Backbuilding a Device from a Value File

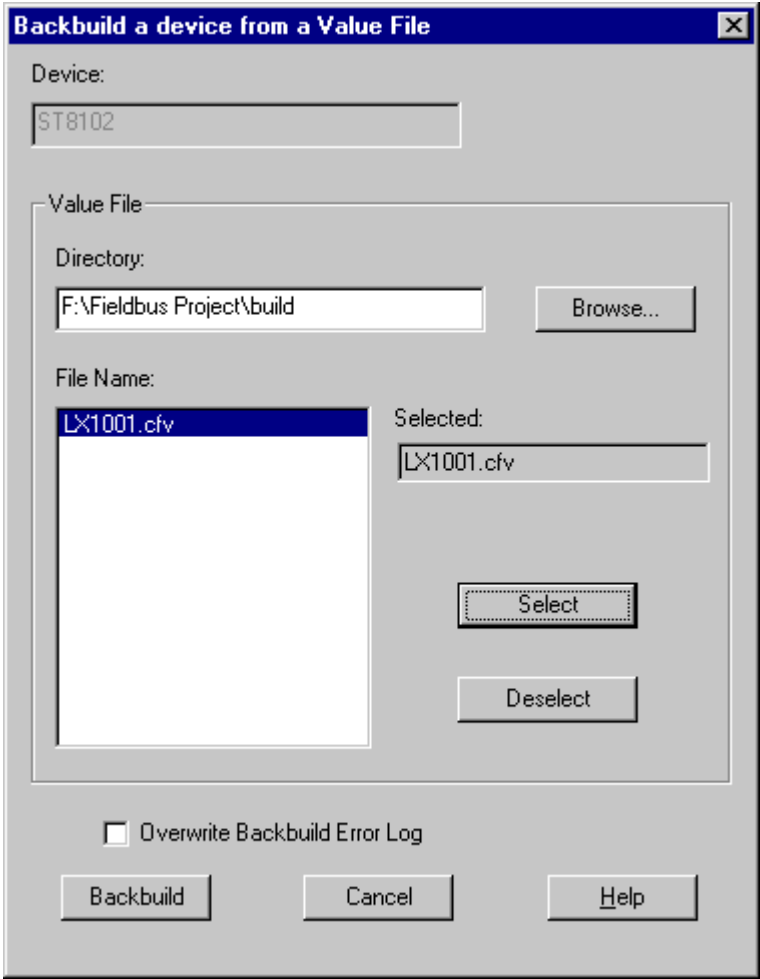
Procedure to Backbuild Device from Value File

Use the following procedure to backbuild a device from a Value File. The Value File must be created prior to backbuilding the device. See “Procedure for Creating a Value File” in this document.

Step	Action	Result
1	In the Fieldbus Link View of the project, position the cursor on the Device control and right click.	The menu appears: 
2	Select Backbuild .	The Backbuild Device from Value File dialog box appears.

11. Creating a Value File

11.2. Backbuilding a Device from a Value File

Step	Action	Result
		
3	Use the browse function of the dialog box to navigate to the directory in which the Value File is stored.	The directory is placed in the port.
4	Select the Value File.	It is placed in the Selected port.
5	Click Backbuild .	The device is backbuilt from the Value File.

11. Creating a Value File

11.2. Backbuilding a Device from a Value File

12. Download/Upload

12.1 Functional Description

Purpose of the Download/Upload Functions

Download

When the Download function is used, EB files are created in the Work Station/GUS environment of TPS Builder and are copied to the LCN environment, using file transfer. During the download operation, addresses, VCRs, and system management functions may be assigned. The EC trigger is used to invoke the processing of the file(s) to be copied to the TPS environment.

Upload

The Upload Link configuration manages the transfer of configuration data from the TPS environment and the fieldbus link to the TPS Builder database. The operation comprises uploading of CM strategies associated with the Link and the devices configured on the link. The EB file building, file transfer, and EC trigger functions are used to accomplish the transfer of data.

12.2 Download Link Configuration

Functionality

The Download Link Configuration transfers the complete configuration data for all entities configured on a fieldbus link:

Entity Type	Notes
SECM strategy	One per link
PECM strategies	One per device FBAP
FBCM strategies	One per function block used in a control loop
Devices	All devices configured on the link

12. Download/Upload

12.2. Download Link Configuration

Downloading a Link

Note: before attempting a download, ensure that all items in the checklist, as shown in “Checklist before Download Strategies” in this section (of this document) are compliant.



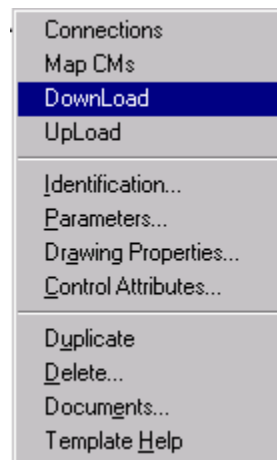
ATTENTION

All points that have been previously built and are to be overwritten must be set **INACTIVE**. All SECM points previously built must be set to the **STK LDED** state.

Downloading the Link Strategy

Set the path for the download, and in the **Fieldbus Link View**, select the link, right-click and select **Download** from the menu. This loads the entire link strategy.

Right-Mouse Click Menu



Sequence of Download

Step	Action	Result
1	Clear Devices	All devices have their VCRs and FB Schedules cleared (set to NOTCONFIG). Additionally, the ScheduleActivationVariable is set to zero, which deactivates the current schedule in the device. The remainder of the download is then performed on a per-device basis as follows:
2	Configure Management parameters	All devices have their VCRs and FB Schedules downloaded.
3	Schedule domain	Following the download of the management parameters the device Schedule domains are loaded. If the schedule downloads successfully, the final parameter loaded is the ScheduleActivationVariable.
4	Blocks	For each device perform the following:- - Set Resource Block Mode Out Of Service - Load Resource Block parameters For each Function Block in the FBAP: - Set Block Mode Out Of Service - Load Block parameters - Set Block Mode to user configured values Set Resource Block Mode to user configured value.

If errors are encountered during Step 1, the download for all devices is aborted.

If errors are encountered during Step 2 or 3, the download for the current device only is aborted. Other devices continue to be downloaded.

12. Download/Upload

12.2. Download Link Configuration

All errors encountered are added to the error log. These errors, including any found in Step 4 of the sequence shown above, should be evaluated and after correction a further download should be started.

The download is performed in three main steps:

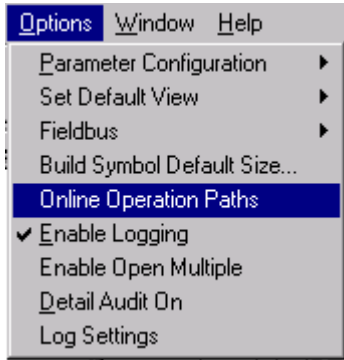
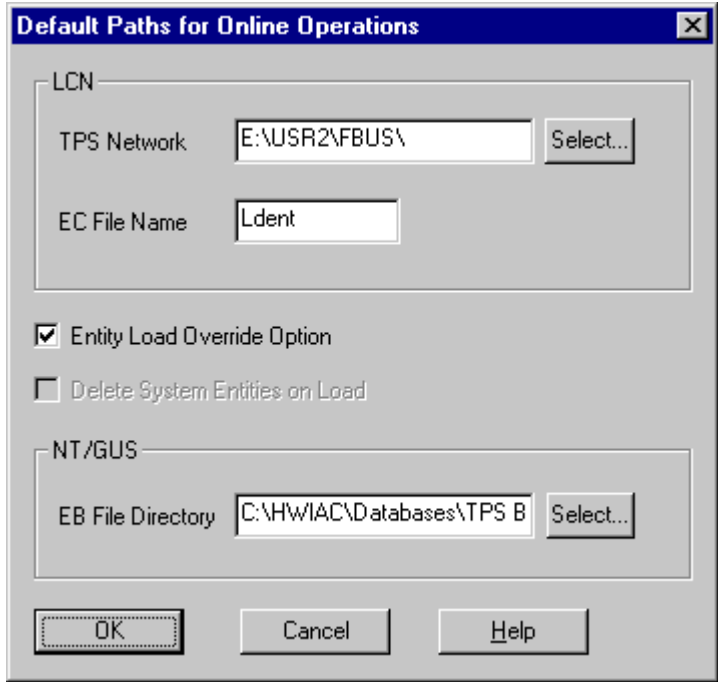
Step	Action
1	The CM strategies associated with the Link are added to an EB file. The EB is copied to the TPN and is used to create the CM strategies on the TPN. The EB file contains only a small number of parameters, those appearing on the Assignment tab of the Parameter Configuration Form.
2	The remainder of the CM point parameters are configured. First, the CM strategy is configured, followed by the PECM strategies, and finally the FBCM strategies.
3	Each device on the Link is configured.

Setting the Default Paths for Download

The default paths for the creation of EB files must be defined for the GUS/Workstation TPS Builder environment and the TPN environment.

Before you set the paths for download/upload, you must have a drive mapped to the History Module, and to the path to the directory in which the database will reside. This is normally the \HWIAC\databases\TPS Builder\download directory.

To set the path for online operations, use the following procedure:

Step	Action	Result
1	From the Options menu, select Online Operations Paths. 	The dialog box for Online Operations appears, as shown in the following figure. 

12. Download/Upload

12.2. Download Link Configuration

Step	Action	Result
2	In a similar manner, set the EB file directory on the Workstation/Gus.	This is usually the directory as shown in the introductory material.
3	Using the Select button for the TPS Network, navigate to the directory on volume on the History Module.	The directory is placed in the port, as shown above.

EC Trigger

The TPN function EC Trigger is used to invoke the processing of the EB file.

Device Mismatch

A device mismatch error may be created by:

- 1) Revision mismatch between the template and firmware.
- 2) Device identification mismatch.
- 3) Device name mismatch.

12.3 Upload Link Configuration

Functional Description

The Upload Link Configuration function is described previously in this section of this document in “Purpose of the Download/Upload Functions.”

Upload Device Configuration

The Upload Link Configuration transfers configuration data for entities configured on a fieldbus link. Note that unlike Download, the complete configuration data is not uploaded, as management parameters and link schedules are not uploaded.

Use the same checklist to ensure that all operations are available, except that you do not have to set the point INACTIVE in order to upload it.

The Upload function uses the EL file selection, EC trigger, and backbuild to upload data from the selected entity.

The follow entities are uploaded, with restrictions on the data uploaded.

Entity Type	Scope of Upload
SECM strategy	EB file parameters
PECM strategies	EB file parameters
FBCM strategies	EB file parameters
Devices	Block parameters (Resource, Function and Transducer)

12. Download/Upload

12.3. Upload Link Configuration

Uploading is performed in two steps:

-
- | | |
|----------|--|
| 1 | CM strategies associated with the Link are added to an EB file on the History Module. The EB is then copied from the LCN to the workstation/GUS. The EB file is then used to update the CM strategies on the link by backbuilding. Note that Backbuild does not create any new CM points; the point must already be present in the configuration. The EB file contains the same parameters as appear on the EB file when the CM points are downloaded. The VCRs, System Management, and Network management configuration are not uploaded. Only TPN parameters are uploaded. |
| 2 | The block parameters from each device on the Link are uploaded. Note that other parameters on the CM point that do not appear on the EB file are NOT uploaded, for example generic parameters. |
-

Device Parameters Uploaded

Parameter Type	Description
Block Parameters	Static and Writeable. Non Volatile and Writeable and I/O.
Management Parameters	NO Management Parameters are uploaded.
CM Parameters	Appearance in EB file.

Checklist before Download Strategies

To make sure that a forward build and download completes successfully, use the checklist below.

Checklist before Forward Build/Download

-
- | | |
|-----------|--|
| 1 | CM Mapping has been completed successfully. |
| 2 | All required parameters have been configured. |
| 3 | Schedule has been created. |
| 4 | Native Window is running with Engineering Main Menu displayed. |
| 5 | The History Module is mapped as a network drive. |
| 6 | GUS station time and LCN time are in synch (at this stage). |
| 7 | TPN Server is running. A warning is given if the server is not configured: |
| 8 | File Transfer is running; Status is "Started." |
| 9 | Online paths in TPS Builder have been configured. |
| 10 | If the CM points to be loaded already exist, they must be set INACTIVE. |
| 11 | The SECM (if it already exists) must be shut down and in STK LDED state. |
-

Error Messages

If errors are encountered during the forward build/download process, you will get a message telling you not to checkpoint the IOP before remedying these errors and forward building/downloading again.

12.4 Checkpointing the IOP

Saving the IOP Data

If the forward build/download completes without errors, you should save the IOP data. In TPS terms, this is Checkpointing. IOP Checkpointing is initiated from the HPM Status Display. See the *Process Operations Manual* if you are not familiar with saving databases.



ATTENTION

The FBIOP Checkpoint Database Save operation may successfully complete, however the FBIOP database then may not be able to be successfully restored. This problem occurs when one or more of the Fieldbus Segments is not properly configured. To assure proper FBIOP Segment configuration three areas of the configuration should be verified prior to performing a Checkpoint database Save.

1. For each Segment (SECM point) go to page 2 of the Device Detail display. The Schedule Version should be displayed.
 2. On the Fieldbus Segment Status display, all the devices should show "DEV OK."
 3. On the Device Detail display for each device on the Fieldbus Segment Status display, Device STAT = ACTIVE and RB STATE = On LINE. If any of these conditions are not met for all segments, the segment should be either reloaded from TPS Builder or activated from the Segment detail displays.
-

12.5 Downloading only a SECM with No Devices

Rationale for Downloading a SECM Only

In the event you have devices on a link and you want to establish the Node Address and Device Ids of any devices installed on a Link, you may create a SECM point for a Link. The details of up to 6 devices on the link will be displayed on the IOP Detail Display.

Creating the SECM

To create only the SECM point for a Link use the following instructions.

Create and Download a Single SECM Point

Step	Action		
1	Create a Link strategy in the Fieldbus Link view of the selected project. Assign the required name to the Link.		
2	Assign a network address to the Link. (NTWKNUM, NODENUM, MODNUM and SLOTNUM parameters)		
3	Create a HPM strategy in the Hardware view of the selected project.		
4	Assign the NTWKNUM and NODENUM values to the HPM point.		
5	Select 'CM Mapping' for the Link. Only a SECM point will be created under the HPM in the Hardware view.		
6	After CM mapping has been performed, in the project's Hardware View of the point, you must have a valid unit before the point will download. An alphabetical listing of the point (in the Hardware View) is a quick way to find the unit number.		
7	Perform the CM Mapping again, to be sure that the unit number propagates to all views.		
8	Create the Schedule for the Link. The schedule will recognize that no devices are connected to the link and prompt if you want to configure the link only for a SECM. If you answer 'Yes', the SECM will be configured as follows:		
	DEVADDR	20	[Appears on the 'IOP Address' tab].
	FirstUnPolledNodeId	248	[Part of the CFLNKSET parameter appearing on the 'IOP Stack' tab].
	NumConsecUnpolledNodeId	0	[Part of CFLNKSET parameter appearing on the 'IOP Stack' Tab].

12. Download/Upload

12.5. Downloading only a SECM with No Devices

Step	Action
9	Select Download from the Right Mouse Button Menu of the Link Control. The download will establish, and then configure, the SECM point in the IOP database. When the SECM is called up, it displays up to 6 devices connected to the Link on the second page of the detailed display.

View Devices on the Link

To view the devices on the link, use the following procedure to call up the Segment Status Display. Get the SECM point's name from the Hardware View of the project; it is the name assigned to the link, with the letter "S" appended.

Step	Action	Result
1	From the GUS (or US), call up the point detail display (DETAIL; enter the point name in the port; ENTER).	The first page of the SECM appears in the Native Window (or on the US).
2	Select the target for Device Detail.	The Device Detail display appears.
3	Select the Fbus Seg Status target.	The Fieldbus Segment Status Display appears. If you see no devices on the link, page forward to the next page to view the devices, their addresses, device ids and tag identifications.

13. Device Save/Restore Application

13.1 Introduction

Device Save/Restore Installation

Device Save/Restore is a separate application and must be installed separately. Refer to the instructions in the "Device Save/Restore Installation", "Software Installation", section of the *TPS Builder Installation, Implementation & Administration Guide*. Follow the instructions carefully, as the security levels are important for data access levels.

Ensure that the TPN Server Priority level is set to 1 for the Device Save/Restore function, as the Device Save/Restore function can restore schedule information.

Use the Device Save/Restore function to save or restore a device.

Purpose

The purpose of the Device Save/Restore function is to ensure that an operator can restore a fieldbus segment or device online without having to download the segment. Device Save/Restore ensures that:

- a device can be restored to its state at the last checkpoint.
- a device can be replaced with an identical device without reconfiguring and downloading the data to the identical device.
- a device can be restored to a prior state from a backup checkpoint.

The Device Save/Restore function is able to show the current version of a Link at all times.

13. Device Save/Restore Application

13.1. Introduction

The Relationship to Download Interface Files

The Device Save/Restore function is directly related to the TPS Builder Download files in the \HWIAC\databases\TPS Builder\download directory. The two applications share this directory. Each time a download is performed, a new set of interface files is created. Any changes made to an individual link or device should be checkpointed at the time the link is determined to be operating correctly. That is, the Device Save/Restore function should be used to save (checkpoint) the devices.

The download interface files consist of the following files:

- A Link Definition file that defines the devices present on the downloaded link, and the details of those devices.
- A Device Schedule File for each Device on the Link that is a LINKMASTER device. The file contains the schedule details that are downloaded. This file is necessary, as the Honeywell devices do not support the FMS Domain Upload service, so Device Save/Restore would not be able to read the schedule from the device. As it is unlikely that any other configuration device will change the schedule, in this case it is safe to take the value downloaded by TPS Builder and use this to Restore to an original or replacement device.
- A file for each device on the link; that is, each device's file can be accessed.

13.2 Main Menu Items

The following options are available from the Main Menu.

Main Menu	Drop-down Menu	Description
File	Save	If a Link is selected, saves ALL devices on the Link. If a device is selected, saves the selected device.
	Restore	If a Link is selected restores ALL devices on the Link. If a Device is selected restores the selected device.
	Find	Find a particular device in the Device Save/Restore tree control. The Find works using the PECM/SECM Name, Device ID, or Device PdTag of a device
	Delete Link	Deletes the selected Link and all associated checkpoint files. A prompt is displayed to confirm this operation before it is performed.
	Retain/Overwrite Checkpoint	Specify whether the Backup checkpoint is to be retained or overwritten when a Save is executed.
	Delete Checkpoint	Deletes the selected checkpoint for the selected device.
	Display	Displays the checkpoint and schedule information for the selected device.
	Update Database	Updates the Device Save/Restore database using the Download Interface Files created by the latest Download.
	Exit	Terminates execution of the Device Save/Restore application.
Logs	Logs	Displays the LOG dialog.
Tools	Refresh	Refresh the current device tree view.

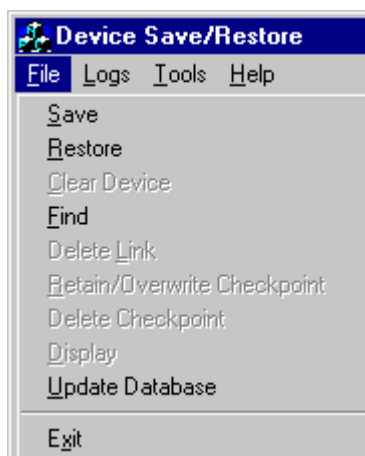
13. Device Save/Restore Application

13.2. Main Menu Items

Main Menu	Drop-down Menu	Description
	Users Connected	View and, if desired, remove users currently connected to the link containing the currently selected device (or devices).
Help		Displays the version number of Device Save/Restore installed.

Here are examples of the drop-down Menus:

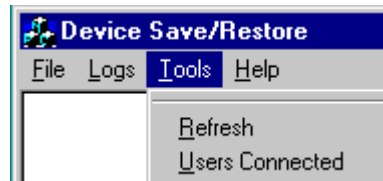
File menu



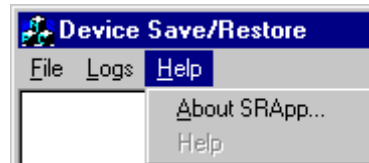
Logs menu



Tools menu



Help menu



Device Save/Restore Synchronization with TPN Checkpoint Files

At the time the Device Save/Restore saves the link or device data, the TPN database should also be checkpointed.

No automatic checkpointing is available for Device Save/Restore at the present time.

Master Checkpoint and Backup Checkpoint

The initial Save of the selected item(s) results in a Master checkpoint for all items at the selected level and below. For example, if the link is saved as a Master checkpoint for the link, its schedule, and all devices on the link are saved. Subsequent saves move the Master to the Backup and the most recent Save becomes the Master.

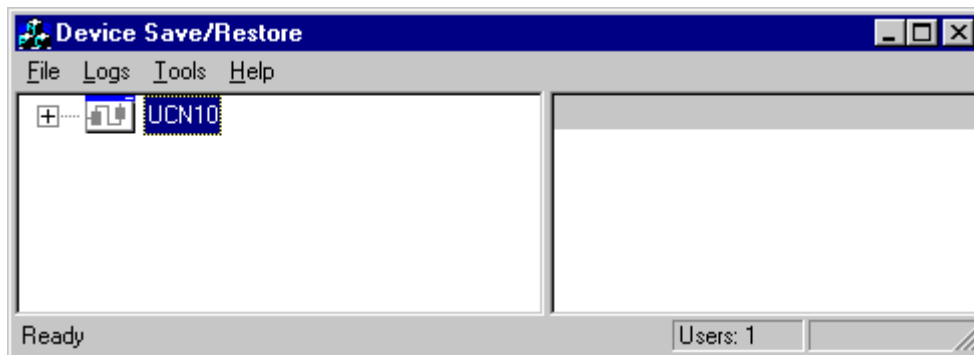
13.3 Starting the Device Save/Restore Application

Launching the Application

The Device Save/Restore application can be launched by Start>Programs>TPS>Device Save/Restore, or if the system has been set up with an icon on the desktop, from that icon.

View from the Open Application

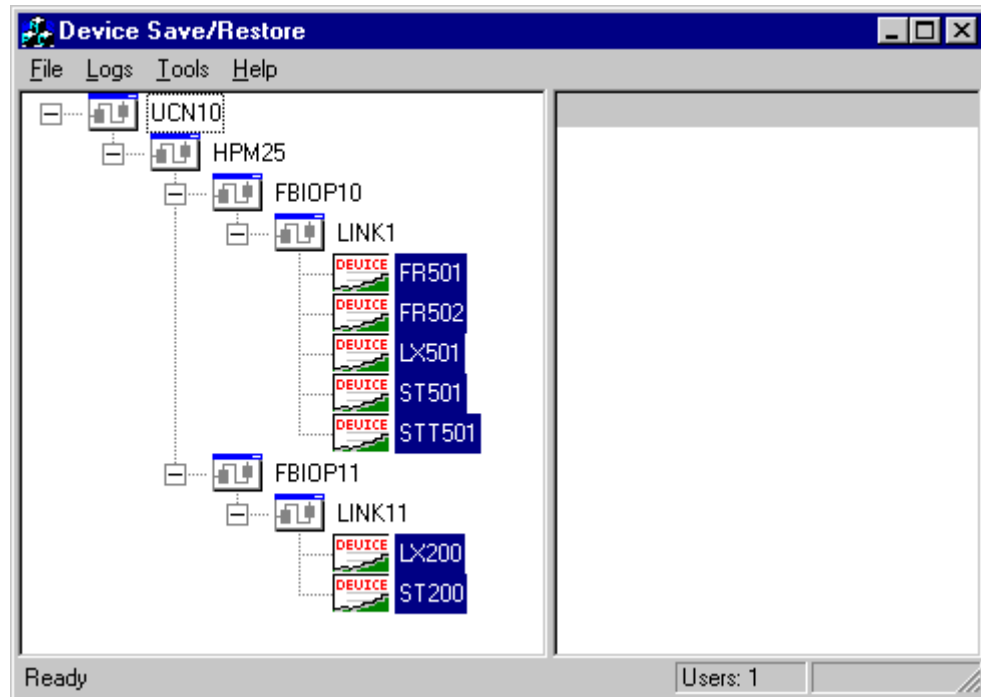
When the Device Save/Restore application opens, the Device Save/Restore window appears. It is divided into 2 panes, A tree view of the device structure at the UCN level is displayed in the left pane. A typical Device Save/Restore application window is shown in the following figure:



13. Device Save/Restore Application

13.3. Starting the Device Save/Restore Application

When expanded, the view displays the FBIOPs, Links, and devices configured for the project. This is shown in the following figure:



13. Device Save/Restore Application

13.4. Saving a Device

13.4 Saving a Device

Required Preparation

Before attempting to perform a save, ensure that the conditions are met; they are the same as a download.

- 1 The Native Window is operating and the Engineering Menu or Builder Commands is displayed.
- 2 File transfer is running.
- 3 The History Module has been mapped as a network drive.

Choosing the Save Level

It is recommended that Save or Restore to a device be performed at the lower level at the device. For example, if the save is performed by selecting UCN 14, all HPMs shown in the project which have FBIOPs configured with links and devices will be saved. In the previous figure, only one HPM, HPM39 is shown; if selected at that level all devices below that will be saved all checkpoints will be updated. This may be appropriate for an initial installation, but as devices are added to links and links are updated, it may be expedient to save only the link, and its accompanying schedule.



ATTENTION

To keep your device data current, a Save must be performed after any configuration change. TPN checkpoints should also be kept current and synchronized with device data.

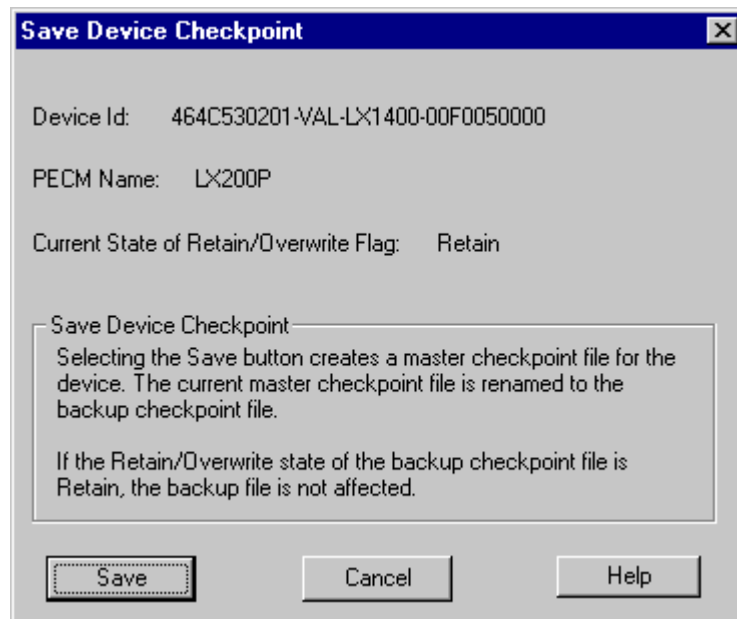
Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the device (or devices) you want to save and then select them.	The selected device (or devices) is (are) highlighted.
3	Select File .	The File menu is displayed.

13. Device Save/Restore Application

13.4. Saving a Device

Step	Action	Result
4	Select Save .	The Link is checked to see if any updating is required. If an unprocessed Link Definition File was found, a prompt is displayed, asking if you want to update the Link before proceeding with the Save. If no unprocessed Link Definition File was found, the Save Device Checkpoint dialog box is displayed. An example follows.



5	Select Save .	The Save procedure begins and a progress bar is displayed.
6	If errors are encountered, for example, the IOP has lost communication with a device,	the Save will not be completed, and an error message will be displayed.
7	If no errors are detected,	the Save will be completed, and a prompt issued.

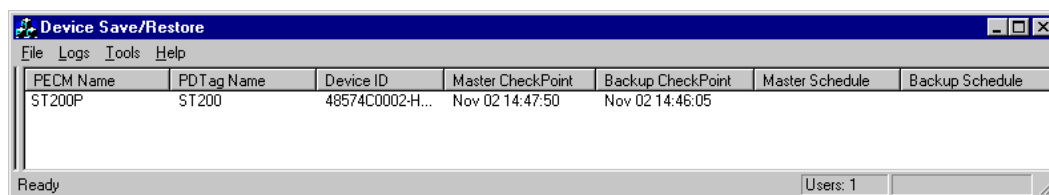
13. Device Save/Restore Application

13.4. Saving a Device

Step	Action	Result
8	Click OK .	The prompt will disappear.
NOTE: If a previous Save has been performed, it is moved from the Master to the Backup, and the most recent Save becomes the Master.		

Update of the Directory

When a save has been completed, the Device Save/Restore directory is updated, as shown in the following figure:

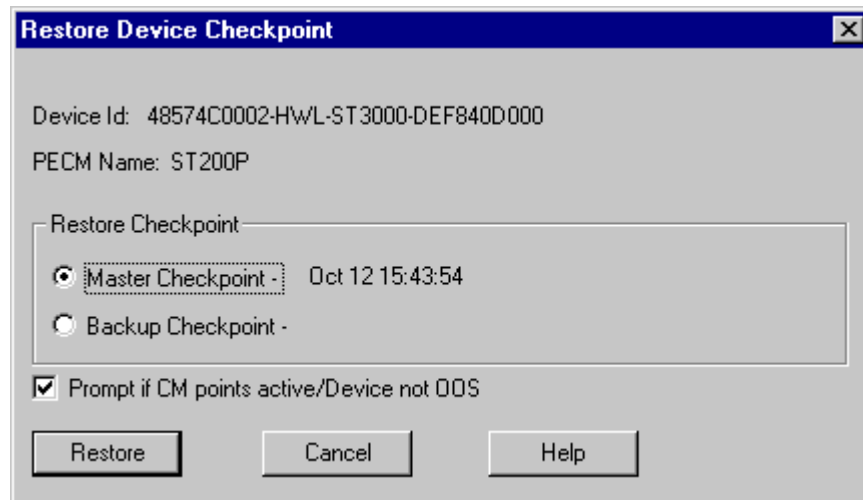


13.5 Restoring a Device

Procedure

The conditions for restoring a device are the same as for the download procedure, except that the device must have been SAVED. Use the following procedure to restore a device.

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the device (or devices) you want to restore and then select them.	The selected device (or devices) is (are) highlighted.
3	Select File . NOTE: If a single device is selected and if it has not yet been checkpointed, the Restore function will not be available.	The File menu is displayed.
4	Select Restore .	The Restore Device Checkpoint dialog box is displayed. An example follows.



13. Device Save/Restore Application

13.5. Restoring a Device

Step	Action	Result
5	Select the desired radio button to choose the either the Master Checkpoint or the Backup Checkpoint. (The device shown in this example has not been backed up.)	The checkpoint designated will be used in the Restore function. The default is to prompt if the points are active and the device is not out of service.
6	Click Restore .	The selected device(s) is (are) restored.

13.6 Device Save/Restore Database

Location

The Device Save/Restore database is maintained in the C:\HWIAC\databases\SaveRestore\data directory. The Device Schedule Files and Checkpoint files are in the C:\HWIAC\databases\SaveRestore\checkpoint directory.

Automatic Database Updating

When the Device Save/Restore application is invoked, it examines the Download Interface Files present in the C:\HWIAC\databases\TPS Builder\download directory. It determines if the files have already been used to update the database by comparing the timestamp of the Download Interface File for a Link against the last update information maintained for the Link.

Device Save/Restore uses any files with a later timestamp than held in the database to update the database.

Database Updating Details

The following table describes in detail the various actions that can take place during a Device Save/Restore database update.

Condition	Action
The Link does not already exist in the database.	The Link and all Device details are added to the database. At this time no checkpoint details are present for the Devices in the Link, but a current schedule file may be present.
The Link already exists in the database	The Link details are updated. The Device details are updated. If the device is already present, the details are updated. If the device is not present, the details are added. If the Device is present on the database, but does not appear in the Link Definition File, the details are deleted from the database. Any associated Device Schedules Files and Checkpoint files are also deleted.

13. Device Save/Restore Application

13.6. Device Save/Restore Database

Updating from the File Menu

After the application is running, you can initiate updating of the Device Save/Restore database from the Main menu. This is described in the next subsection. For example, you might want to trigger an update is if a Link was downloaded after the application was initiated.

13.7 Updating the Database

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the device (or devices) you want to update and then select them.	The selected device (or devices) is (are) highlighted.
3	Select File .	The File menu is displayed.
4	Select Update Database .	The database updating begins. Refer to the Database Updating Details subsection for details.

13. Device Save/Restore Application

13.8. Deleting a Link

13.8 Deleting a Link

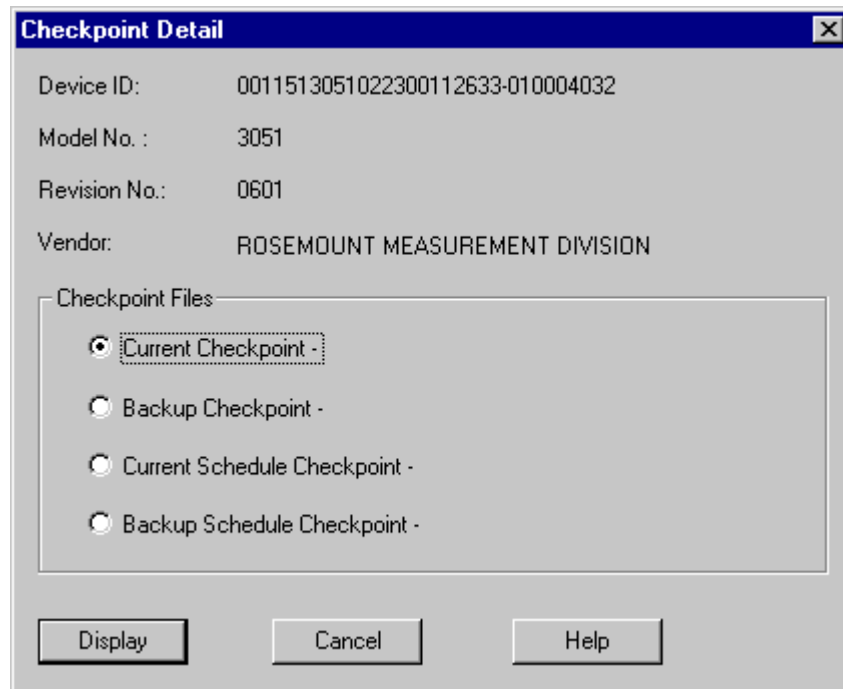
Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the link you want to delete and then select it.	The selected link is highlighted.
3	Select File .	The File menu is displayed.
4	Select Delete Link .	The following items for the selected Link are deleted: • All the Link details in the Device Save/Restore database. • The .cpt and .sch files in the C:\HWIAC\databases\SaveRestore\checkpoint directory.

13.9 Displaying Details

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the device you want to display and then select it.	The selected device is highlighted.
3	Select File .	The File menu is displayed.
4	Select Display .	The Checkpoint Detail dialog box is displayed. An example follows.



13. Device Save/Restore Application

13.9. Displaying Details

Step	Action	Result
5	Select the radio button to indicate the checkpoint file you want to see and then click Display .	The saved data for this point is displayed. An example of the current checkpoint data for a point follows.

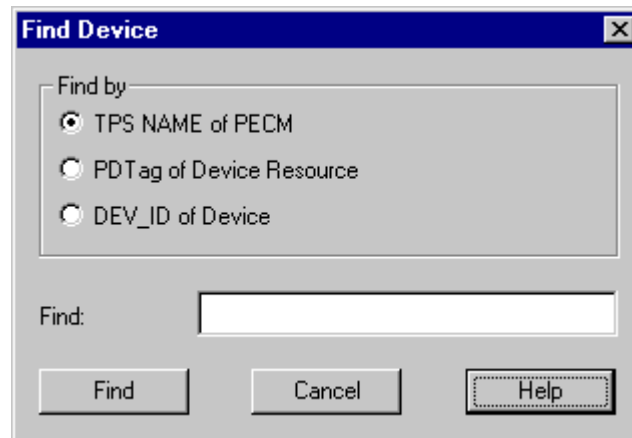
```

Name,Value,Type,Size,0D,Name1,Val1,Name2,Val2,Name3,Val3,Name4,Val4,Name5,Val5,N
FileName, g:\databases\dsr\Checkpoint\FR501P_MasterCPT.txt
DateTime, Wed Nov 15 14:05:14 2000
MajorVersion, 1
MinorVersion, 0
Dev_Id, 0011513051022300112633-010004032
VendorName, ROSEMOUNT MEASUREMENT DIVISION 3051
ModelName, 3051
Revision, 0601
NumberOfMgtParams, 54
NumberOfAppParams, 0
NumberOfBlockParams, 142
Mode_Blk_Resource_00S, 0x01010101,69,4,305,
ScheduleActivationVariable, 0,6,2,650,
UcrStaticEntry(2), #,97,44,853,ArTypeAndRole,0,LocalAddr,0000,RemoteAddr,0000,:
UcrStaticEntry(3), #,97,44,854,ArTypeAndRole,0,LocalAddr,0000,RemoteAddr,0000,:
UcrStaticEntry(4), #,97,44,855,ArTypeAndRole,0,LocalAddr,0000,RemoteAddr,0000,:
UcrStaticEntry(5), #,97,44,856,ArTypeAndRole,0,LocalAddr,0000,RemoteAddr,0000,:
  
```

13.10 Finding a Device

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select File .	The File menu is displayed.
3	Select Find .	The Find Device dialog box is displayed. An example follows.

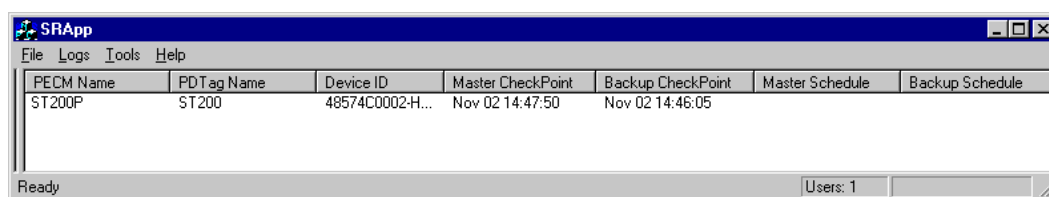


- | | | |
|---|--|--------------------------|
| 4 | Select the desired Find by radio button to indicate the parameter on which the search will be based. | |
| 5 | Enter the value for the search parameter selected in the Find data input port. | |
| 6 | Click Find . | The search is initiated. |

13. Device Save/Restore Application

13.10. Finding a Device

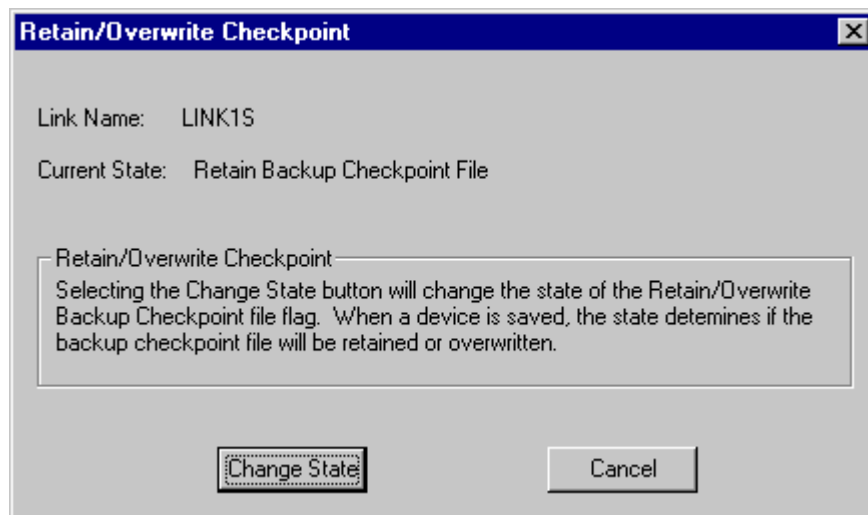
Step	Action	Result
7	If the specified device cannot be found,	An error message will be displayed, indicating that the device could not be found. Click OK to clear the error message and to terminate the Find request.
8	If the specified device is found,	The checkpoint and schedule information for the device is displayed. See the following example.



13.11 Retaining or Overwriting a Checkpoint

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Expand the device structure in the left pane (click on the + sign) until you find the Link you want to retain or overwrite, and then select it.	All the Devices in the link are highlighted.
3	Select File .	The File menu is displayed.
4	Select Retain/Overwrite Checkpoint .	The Retain/Overwrite Checkpoint dialog box is displayed. An example follows.



- | | | |
|---|---|--|
| 5 | Note the Current State information for the selected link. | The Current State: will indicate either "Retain Backup Checkpoint File" or "Overwrite Backup Checkpoint File." |
|---|---|--|

13. Device Save/Restore Application

13.11. Retaining or Overwriting a Checkpoint

Step	Action	Result
6	To change the Current State, click Change State .	The Current State is changed to the alternate value.
7	To leave the Current State unchanged, click Cancel .	The current status is not changed.

13.12 Exiting

Prerequisite

The Device Save/Restore application must be running.

Procedure

Step	Action	Result
1	Select File .	The File menu is displayed.
2	Select Exit .	The Device Save/Restore application is terminated.

13. Device Save/Restore Application

13.13. Logs

13.13 Logs

Overview

When you run the Device Save/Restore application, data will be automatically collected for four different logs. The following table lists the logs and their content.

Log	Contents
Activity Log	Device Save/Restore procedures that have been executed.
Audit Log	Changes made to the Device Save/Restore database.
Save/Restore Log	The separate stages associated with Save and Restore functions.
Error Log	Errors detected during Save and Restore functions.

Activity Log Data

The following table lists the functions that will generate Activity Log data and the actual data items that are recorded.

Function	Data Recorded
Update Link	Update started Update completed
Save Device	Save started Save completed
Clear Device	Clear started Clear completed
Restore Device	Restore started Restore completed
Delete Link	Link deleted
Delete Checkpoint	Checkpoint deleted

Audit Log Data

The following table lists the functions that will generate Audit Log data and the actual data items that are recorded.

Function	Data Recorded
Update Link	Link added Link updated Device added Device updated Device deleted
Delete Link	Link deleted
Delete Checkpoint	Checkpoint deleted

Save/Restore Log Data

The following table lists the functions that will generate Save/Restore Log data and the actual data items that are recorded.

Function	Data Recorded
Clear Device	Device xxx Block Parameters Set OOS (where xxx = Resource, PID, AI, AO, or Transducer) Device management parameters cleared
Restore Device	Device xxx Block Parameters Written (where xxx = Resource, PID, AI, AO, or Transducer) Device management parameters written

13. Device Save/Restore Application

13.13. Logs

Error Log Data

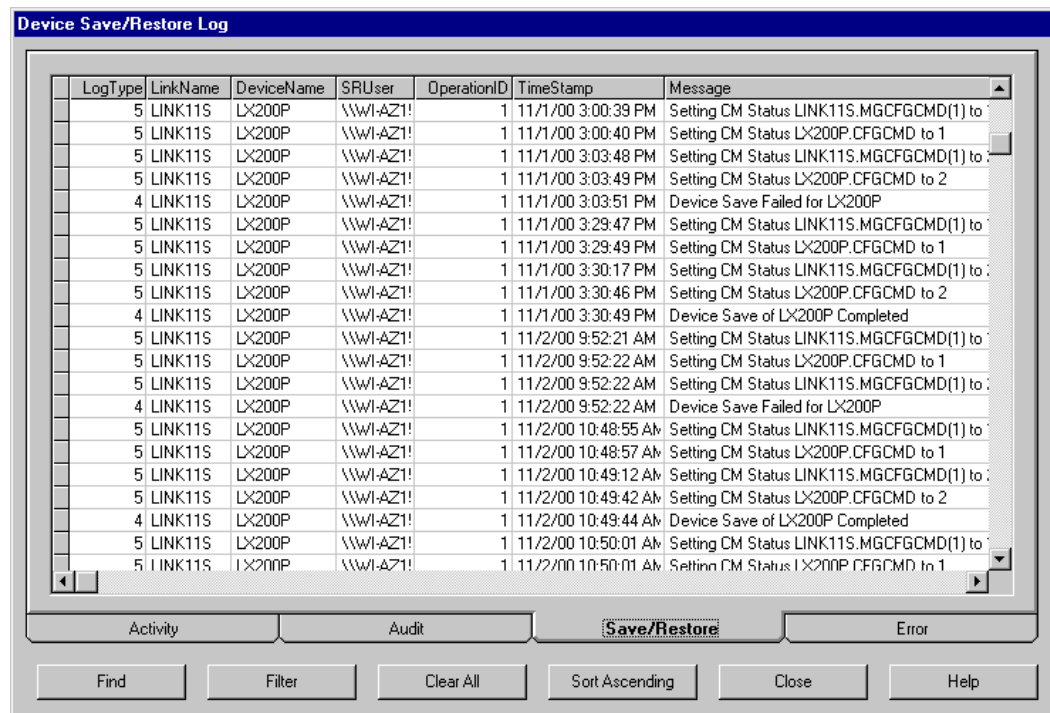
The following table lists the functions that will generate Error Log data and the actual data items that are recorded.

Function	Data Recorded
Save Device	Link status invalid Device status invalid Inconsistent device information Could not set block OOS Could not write parameter information
Clear Device	Link status invalid Device status invalid Inconsistent device information Could not set block OOS Could not write parameter information
Restore Device	Link status invalid CM point status invalid Device status invalid/mismatch Inconsistent device information Could not write parameter information Could not transition CM point status

13.14 Displaying Logs

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select Logs .	The Logs menu is displayed.
3	Select Logs .	The Log window is displayed.
4	Select the desired tab (Activity, Audit, Save/Restore, or Error) to display the desired log.	The first page of the selected log appears. An example of the Log window with the Save/Restore Log tab selected follows.



13.15 Refreshing the Device Tree

Overview

Although a single Device Save/Restore application is allowed to run from one system, more than one system can run the Device Save/Restore application and connect to the Device Save/Restore database. In order to allow multiple users to access the same database, the functions that could change anything in the database are disabled while any change to a Link or to any device in the link is in progress. This is accomplished by disabling the affected menu items. Users will be notified that the Device Save/Restore database is either currently being updated or has already been updated and that a refresh action is required.

Prerequisite

The status of the current device tree, as displayed in the status bar at the bottom of the Device Save/Restore window, must indicate “Refresh Required.”

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select Tools .	The Tools menu is displayed.
3	Select Refresh .	The Device Save/Restore database is updated.

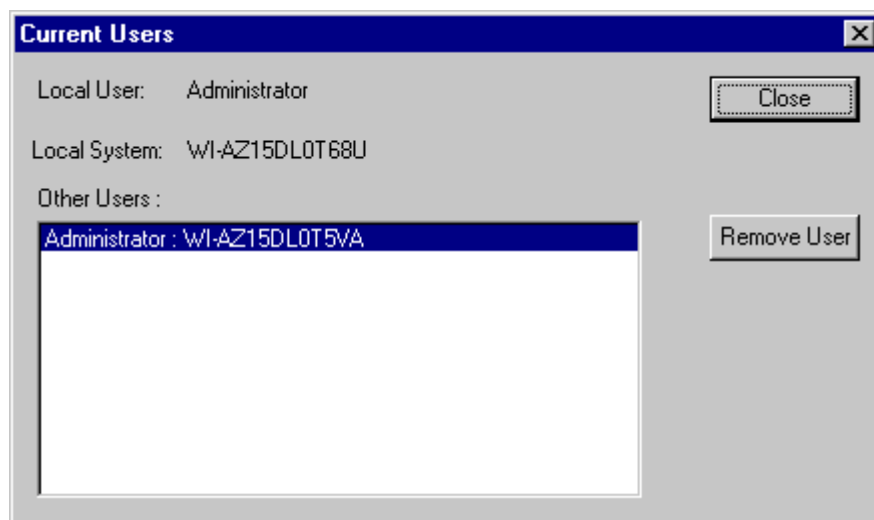
13.16 Showing Users Connected

Overview

Although a single Device Save/Restore application is allowed to run from one system, more than one system can run the Device Save/Restore application and connect to the Device Save/Restore database. In order to allow multiple users to access the same database, the Current Users function identifies all users currently connected to the Device Save/Restore database and provides the ability to remove them and any database entries they may have made.

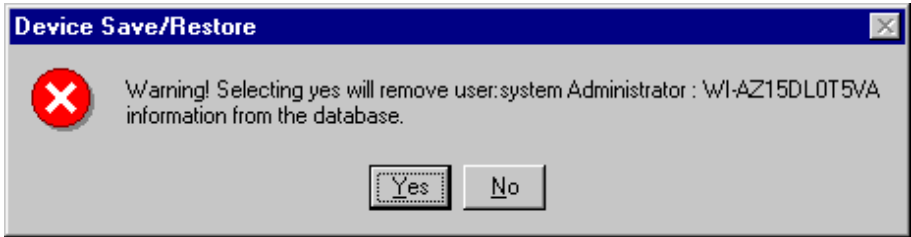
Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select Tools .	The Tools menu is displayed.
3	Select Users Connected .	The Current Users dialog box is displayed. An example follows.



13. Device Save/Restore Application

13.16. Showing Users Connected

Step	Action	Result
4	If desired, select (highlight) the Other User you want to remove and click Remove User .	A Warning dialog box appears asking if you really want to proceed. An example follows.
		
5	To disconnect the selected user, click Yes .	The selected user will be disconnected and any database entries removed
6	To cancel the user removal request, click No .	No action will be taken.

13.17 Displaying Revision Information

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select Help .	The Help menu is displayed.
3	Select About SRApp....	The About Device Save/Restore dialog box is displayed. An example follows.

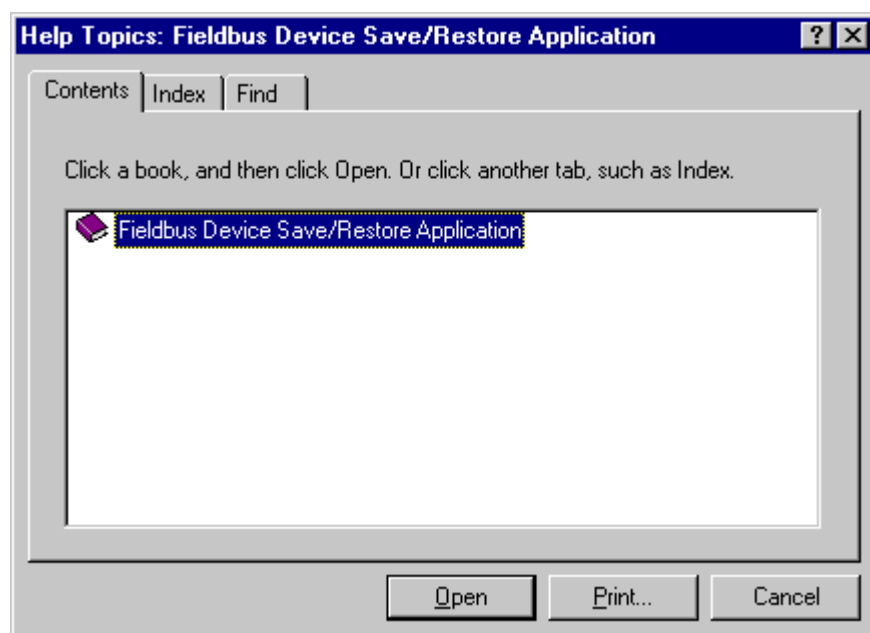


4	Click OK .	The About Device Save/Restore dialog box disappears.
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13.18 Displaying Help

Procedure

Step	Action	Result
1	Launch the Device Save/Restore application.	The Device Save/Restore application window is displayed.
2	Select Help .	The Help menu is displayed.
3	Select Help .	The Fieldbus Device Save/Restore Application Help Topics window appears. An example follows.



- | | |
|---|---|
| 4 | Use the Windows Help system developed for this application. |
|---|---|
-

14. Configuration Examples

14.1 Introduction to the Examples

Description

The examples that are provided here are for building Fieldbus Loops using TPS Builder, using a step-by-step approach. The material here is to be used in conjunction with the *Fieldbus Overview* and TPS Builder training material. These examples assume that the user has no prior experience with TPS Builder, and are therefore elementary.

For both these examples, the following TPS Builder options must be loaded:

- Fieldbus Function Block Library
- Fieldbus Function Block Library
- Fieldbus Link Library
- Fieldbus Device Library
- UCN Library that includes a template for the CM Strategy
- PM/APM/HPM Library

In addition, you must have:

- TPN Server
- Primary Domain Controller
- NI Interface
- File Transfer Program
- History module mapped as a Network Drive

14. Configuration Examples

14.1. Introduction to the Examples

Conventions Used in these Examples

Convention	Use
<i><Tools/Fieldbus/ForwardBuild/Download></i>	Menu options are shown in <i>Italics</i>
<i>Start_Time_Offset</i>	FF parameters are shown in Initial <i>Capital Italics</i>
<i>FB_CITYPE(n)</i>	TPS Builder fieldbus parameters are shown in italicized <i>CAPITAL</i> letters.
CISRC(n)	TPN parameters are shown in bold CAPITAL letters.
Fieldbus Link	Libraries and Project Views are shown in Bold Initial Capitals .

Terminology Used in Configuration

Click — press and quickly release the left mouse button once on an object or menu selection.

Double-click — click the left mouse button twice in rapid succession on an object.

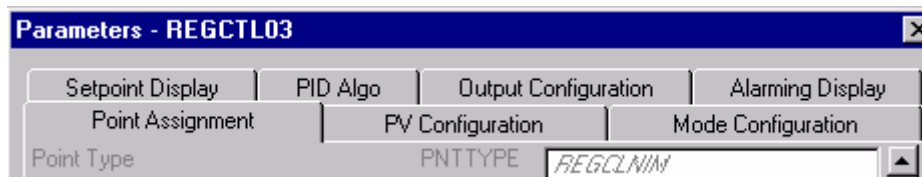
Drag and drop — with the cursor on an object, press and hold the left mouse button, move the mouse and object to a new location, and release the mouse button.

Right-click — same as click, except use the right mouse button.

Using the Parameter Configuration Form

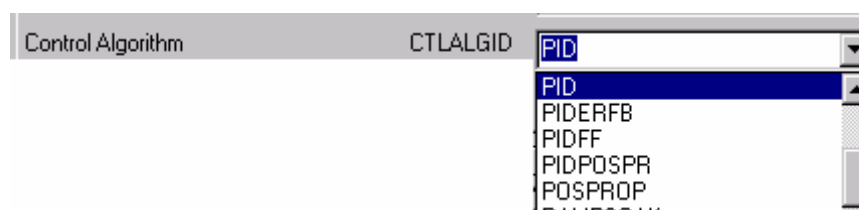
In the Parameter Configuration form, some parameter entries are gray, meaning that they template-defining, and cannot be changed, as shown in the following figure.

Example of Template-Defining Parameter



You may encounter a Combo box, , that when selected (by a click) will invoke a menu from which to choose, as shown in below.

Example of Combo Box



14.2 Configuration Example 1

Before You Begin

Synchronize the time as shown in “Date and Time” in this document.

Get your hardware definitions as shown in “Hardware Definitions” in this document. In addition, get the 32-character device identifications supplied by the manufacturer.

Requirements

Check the SCN/Customer Release guide for the correct list and versions of the software.

Hardware Definitions

You will need to know the HPM configuration and the IOP number for which the fieldbus segment is configured.

In this exercise we have used UCN = 1, HPM = 19, Module Number 5, FTA = 1 (segment position 1, address/slot number 120). This is the type of information needed.

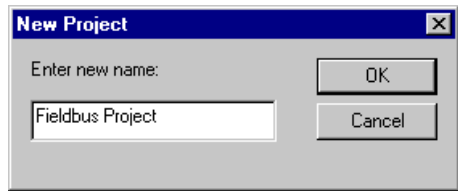
Configuring a Fieldbus Link with Devices

Example 1 uses functions blocks that are resident in the devices; no inputs, outputs or control points reside outside the FBIOP.

The control scheme is a simple loop with a pressure input, a PID controller in the input device, and output to a control valve. The PID from the output is recommend for configuration of loops, but the purpose of this exercise is to familiarize the user with TPS Builder and the configuration of fieldbus segments using this tool. Configuration of a loop using the PID in the output block is similar to the exercise shown here. See “Performance Considerations in Configuring Loops” in this document.

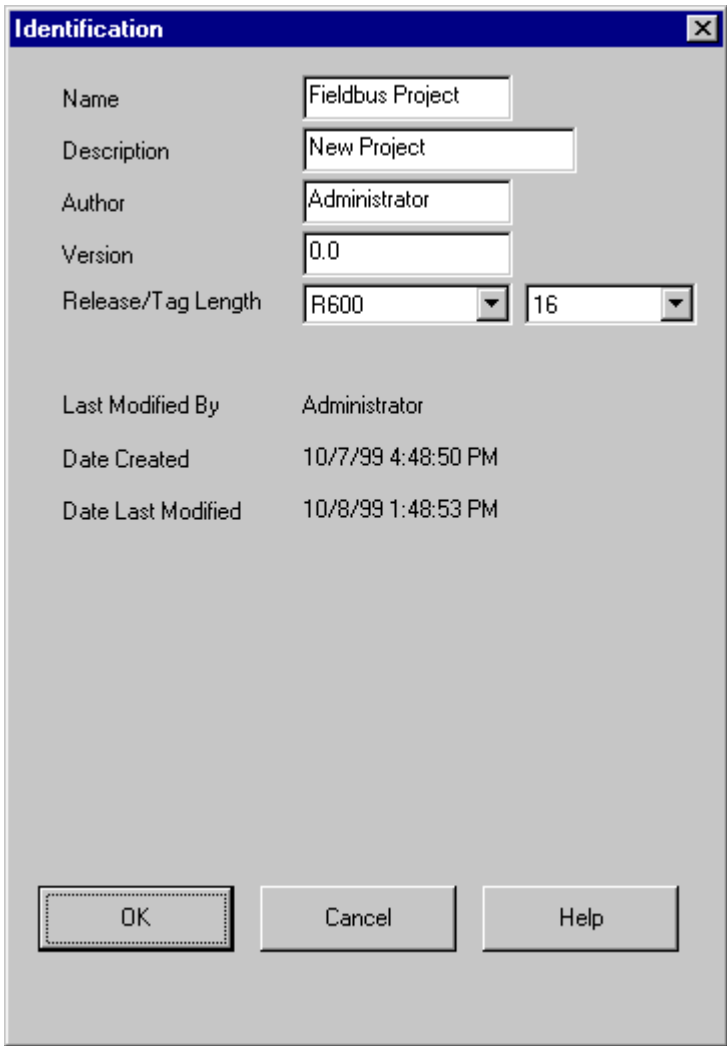
The devices used are a Honeywell ST3000 Smart Pressure Transmitter, and a Logix 1400 Valve.

Step 1 — Creating a Project

Step	Action	Result/Notes
1	Open TPS Builder (double click on icon.).	The TPS Builder program opens with an empty screen.
2	<File>New>Project>	The New Project dialog box opens. 
3	Enter name: Fieldbus Project and click OK.	A new project is created. (It will open in the view selected in the bottom half of the window.)
4	Select the Fieldbus Project and select Hardware View (they will turn color when selected).	The fieldbus project opens in the hardware view.
5	Right click and select <Identification>.	The Identification dialog page opens.

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
		
6	From the combo boxes for Release/Tag length, select R600 and 16, respectively, and click OK.	The release is set to R600 before the fieldbus can be configured. Prior releases do not support fieldbus.

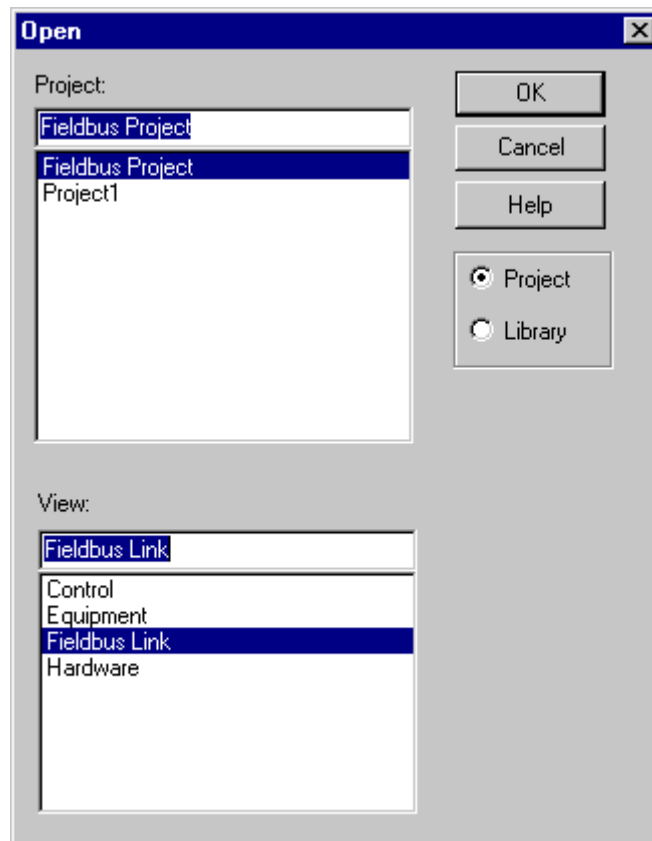
Step 2 — Opening Project Views and Libraries



TIP

If it is bothersome to you, after opening the views, you may choose *<Window>Tile>*. This tool will resize and arrange the displays to prevent overlap (or you can drag the displays to prevent this overlap).

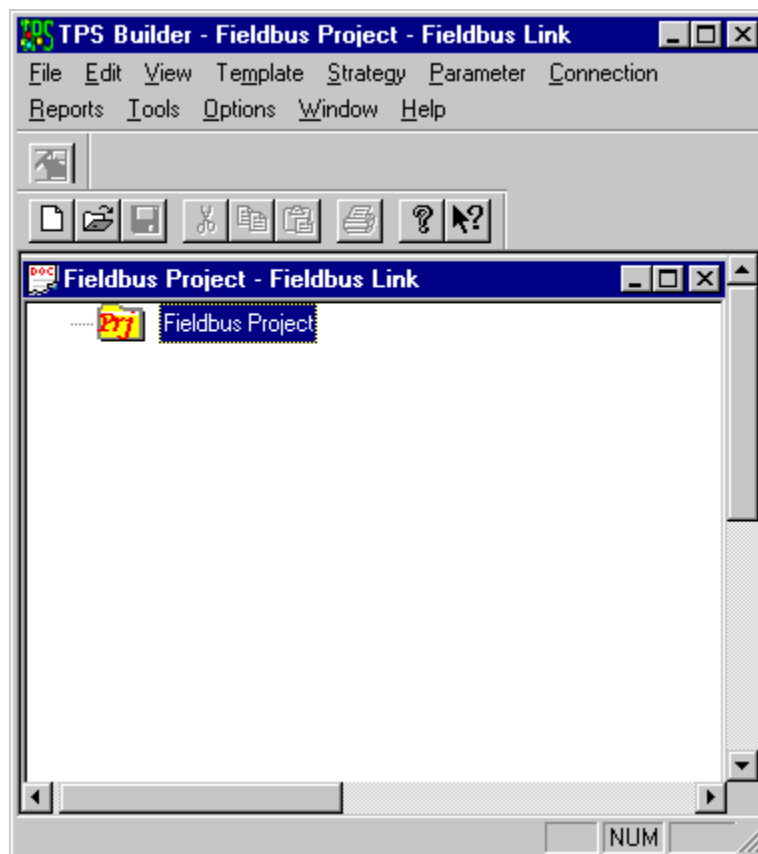
Step	Action	Result/Notes
1	<i>File>Open>Project.</i>	The project dialog box appears.



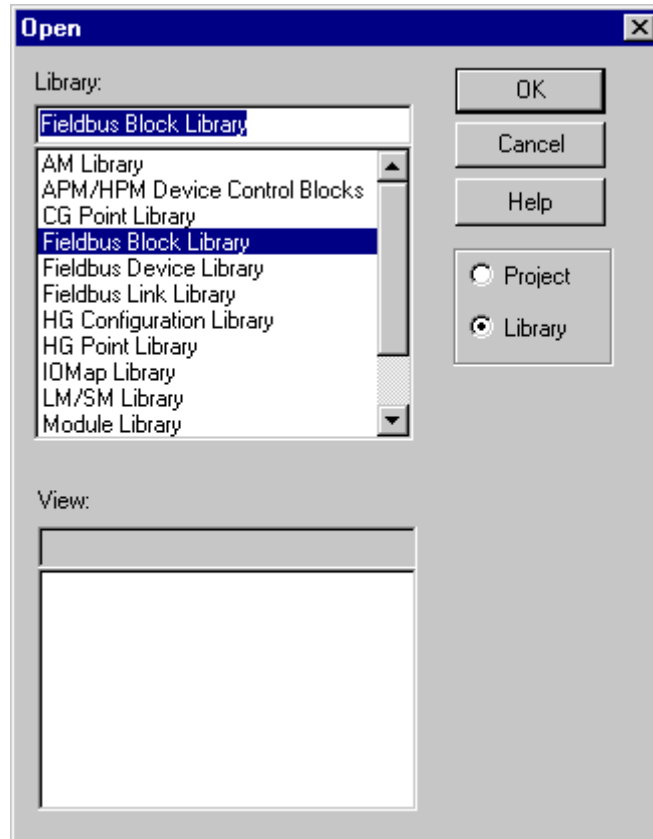
14. Configuration Examples

14.2. Configuration Example 1

- 2 Choose **Fieldbus Project** in the upper box and **Fieldbus Link View** in the Lower box. The **Fieldbus Project** is opened in the **Fieldbus Link View**.



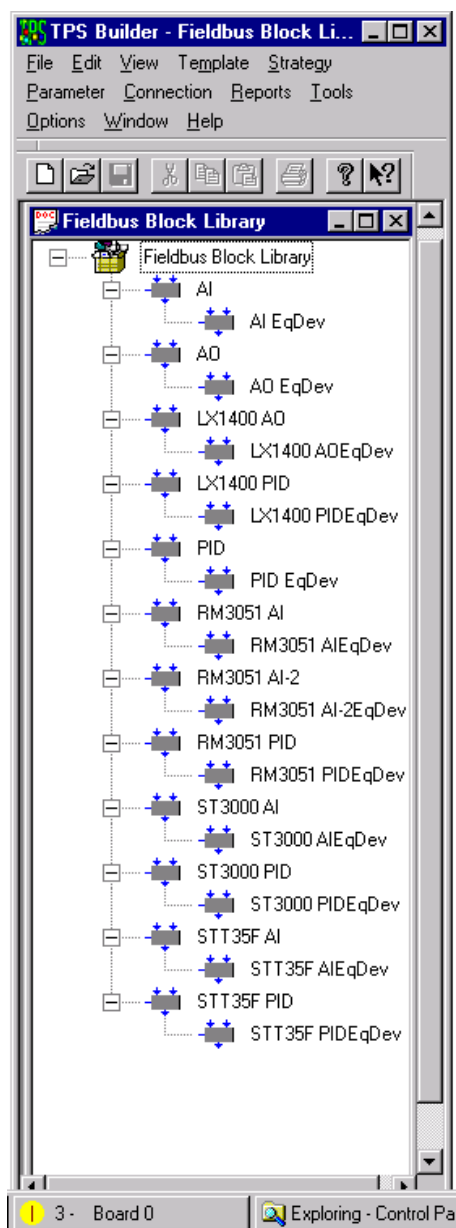
- 3 <File>Open> **Fieldbus Project, Equipment View** and click OK. The **Fieldbus Project Equipment View** is displayed.
- 4 <File>Open>Library>. Choose **Fieldbus Block Library** and click OK. The **Fieldbus Block Library** opens.



- 5 Expand the library by clicking on the  in front of the icon. The **Fieldbus Block Library** expands to show the blocks.

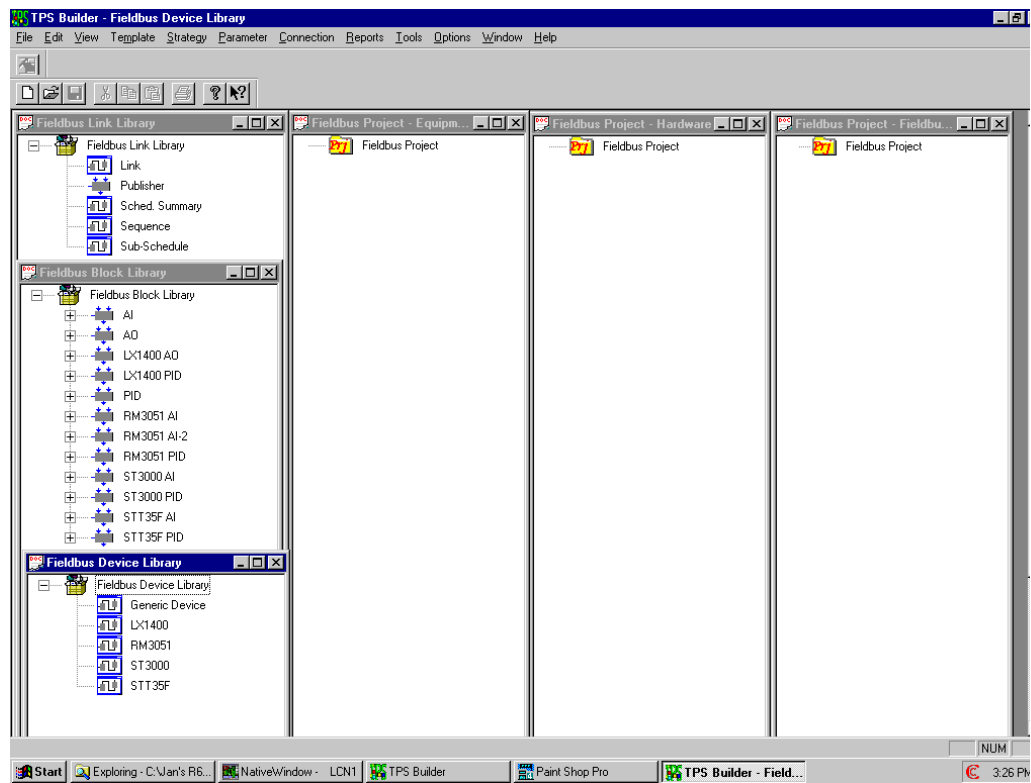
14. Configuration Examples

14.2. Configuration Example 1



- | | | |
|---|--|--|
| 6 | <File>Open>. Choose Fieldbus Device Library and click OK | The Fieldbus Device Library is displayed. |
| 7 | <File>Open>Library>. Choose Fieldbus Link Library and click OK. | The Fieldbus Link Library is displayed. |

Fieldbus Project Views and Libraries



14. Configuration Examples

14.2. Configuration Example 1

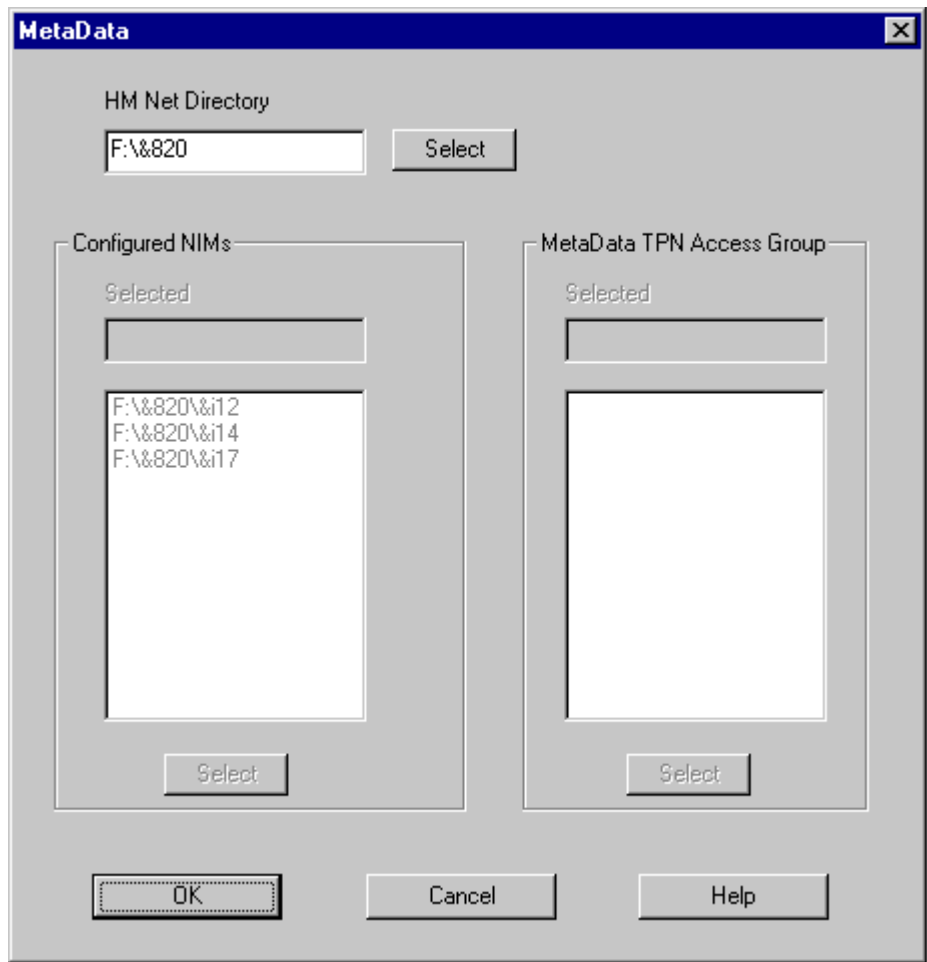
Step 3 — Create MetaData

The Detail Display MetaData file describes the structure of the fieldbus device data to a NIM. Each NIM receives a complete definition of the Fieldbus Function Block Library.

After the MetaData file is created, and the data is forward built and downloaded, the Operating/Engineering displays are built. The Operator/Engineer can then view, and operate the points, using these displays on a Universal or a GUS station.

Step	Action	Result/Notes
1	From the <Tools> Menu, select MetaData and Create MetaData.	The MetaData creation sets up the necessary TPN display database to build CM Points and their allied displays. See the figure following.
2	Navigate to the directory that points to the correct History Module on which the MetaData file is stored. Click OK.	The MetaData is created.

MetaData Dialog Box



Step 4 — Defining the Hardware

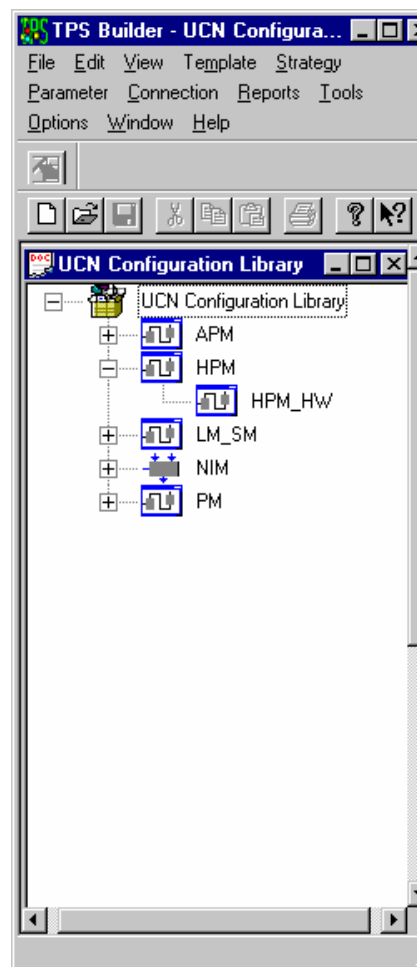
This step creates the physical layout of the TPS Network Node.

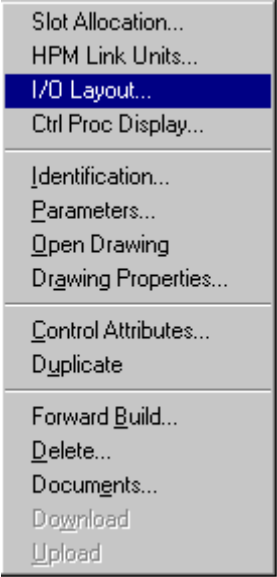
Step	Action	Result/Notes
1	Open the Fieldbus Project Hardware View .	The empty project appears.

14. Configuration Examples

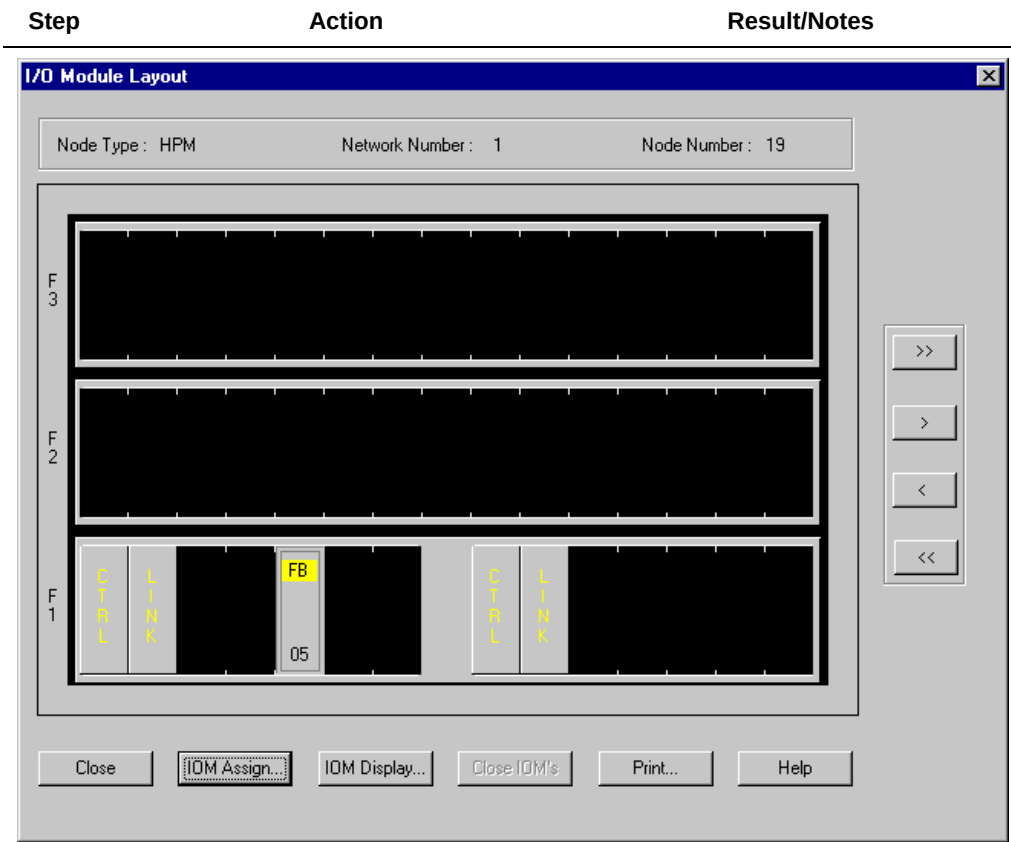
14.2. Configuration Example 1

Step	Action	Result/Notes
2	Open the UCN Configuration Library .	The UCN Configuration Library appears.
3	Click on the  in front of the HPM hardware to expose the HPM_HW.	The HPM_HW template is exposed.



Step	Action	Result/Notes
4	Drag and drop the HPM_HW template into the Fieldbus Project Hardware View .	A port opens, New Strategy.
5	In the port, type the name: Enter HPM0119.	The hardware icon appears with the name, HPM0119.
6	Important Note: In this example, we reconstituted the point \$NM0`B19 and backbuilt the HPM to show the exact configuration before adding the Fieldbus IOP. Note that this particular configuration is a Split 7. The backbuilding will not only configure the I/O Module layout, but will also reflect the current configuration such as NCTLSLOT, NPVSLOT, etc. If your HPM has not been configured, it will need to be configured using the Parameter Configuration Form.	
7	After the backbuilding is complete, select HPM0119 and right-click.	A menu appears. 
8	From the menu select <I/O Layout>.	The I/O Layout appears.

14. Configuration Examples
14.2. Configuration Example 1



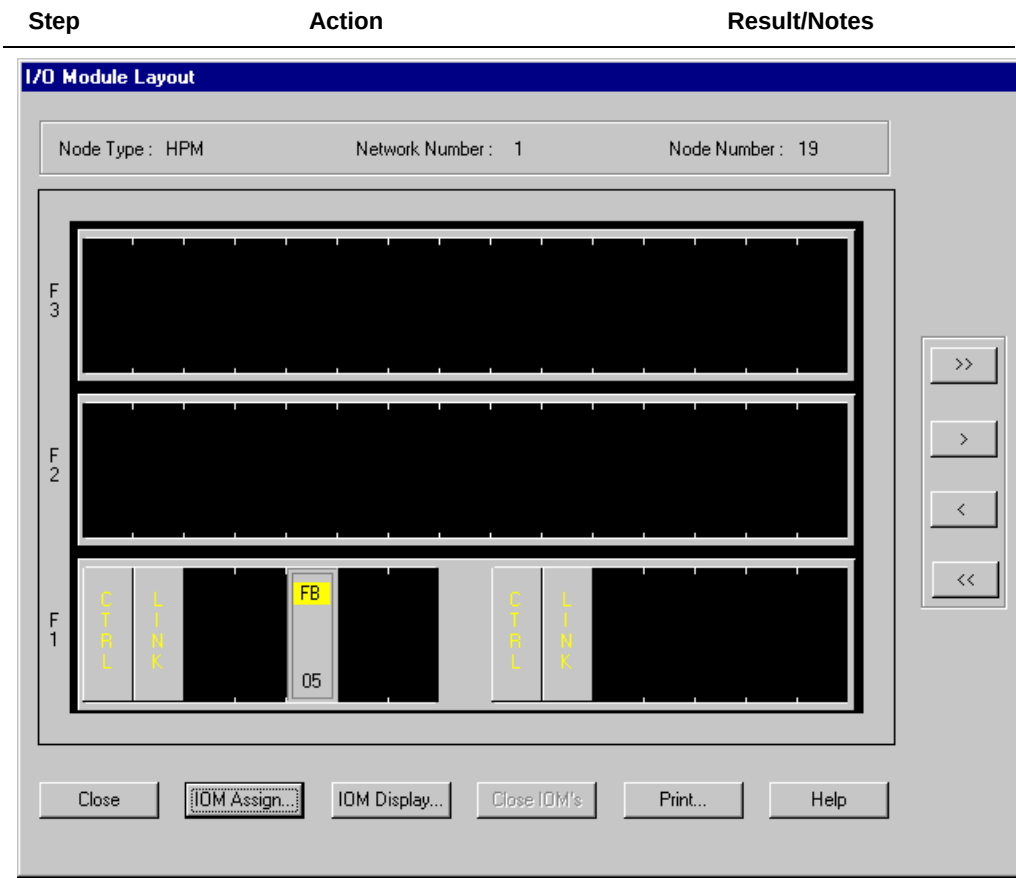
- | | | |
|---|---|---------------------------------------|
| 9 | Select IOM Card 5 in the configuration and right-click. | I/O Module Assignment dialog appears. |
|---|---|---------------------------------------|

Step	Action	Result/Notes

- | | | |
|------------------|---|--|
| <p>10</p> | <p>Select FBUS and click OK.</p> | <p>The fieldbus IO Module is assigned to slot 5.</p> |
|------------------|---|--|

14. Configuration Examples

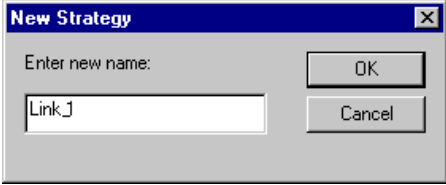
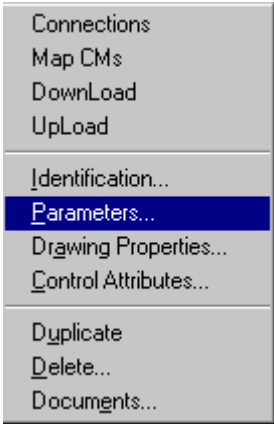
14.2. Configuration Example 1



- 11 Close the file and return to the Project Link View.
-

Step 5 — Adding Fieldbus Link

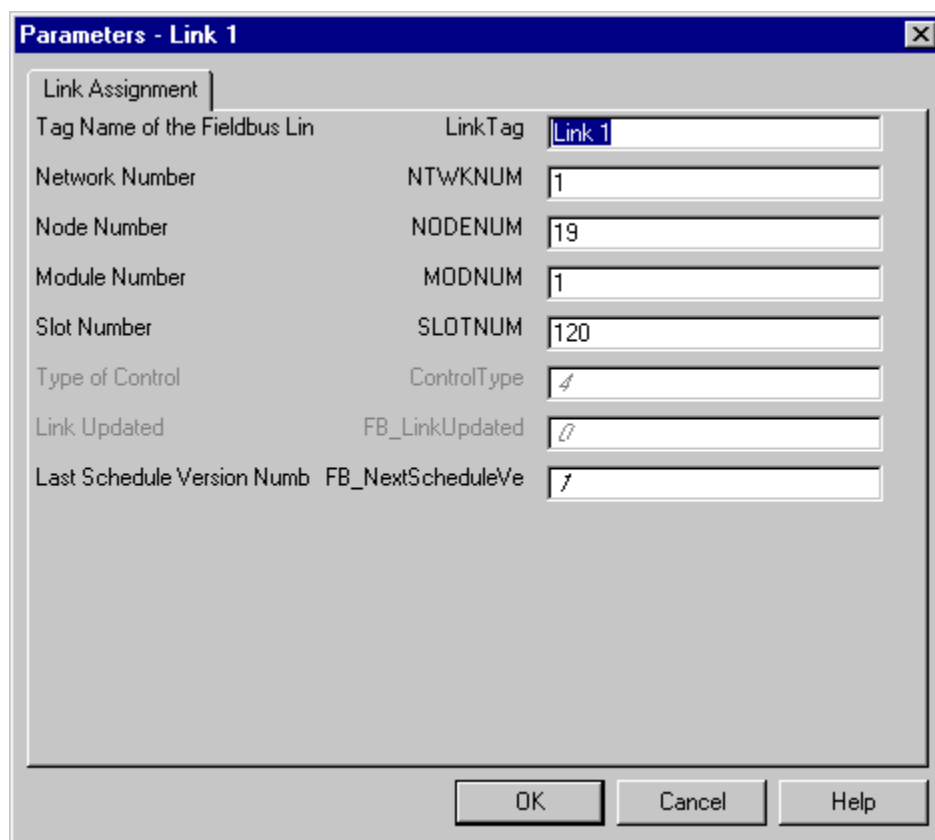
This step creates the Fieldbus Link. This strategy creates the fieldbus link (also called segment or network) and puts it on the first FTA position.

Step	Action	Result/Notes
1	Click on the  in front of the Fieldbus Link Library to expand the view to expose the Link element.	The Link elements are shown.
2	With the Fieldbus Project Fieldbus Link View open, drag the Link from the Fieldbus Link Library and drop onto the Fieldbus Link View .	A port opens: 
3	Enter: Link_1 and click OK.	The Link and its schedule element are displayed in the Fieldbus Project Fieldbus Link View .
4	Assign the Link to a slot Number on the IOP where the Fieldbus Link resides. In the Fieldbus Link View , select the template for Link_1 (it will turn color) and right-click.	The menu of choices appears: 

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
5	From the menu select <Parameters>.	The Parameter Dialog Box appears.



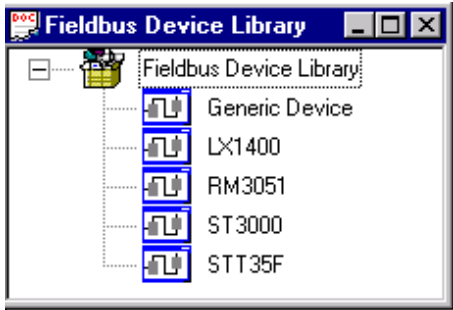
The image shows a dialog box titled "Parameters - Link 1". It has a tab labeled "Link Assignment". Inside the dialog, there are several input fields arranged in two columns. The first column contains labels: "Tag Name of the Fieldbus Lin", "Network Number", "Node Number", "Module Number", "Slot Number", "Type of Control", "Link Updated", and "Last Schedule Version Numb". The second column contains corresponding labels: "LinkTag", "NTWKNUM", "NODENUM", "MODNUM", "SLOTNUM", "ControlType", "FB_LinkUpdated", and "FB_NextScheduleVe". Each label is followed by a text input field. The values entered in the fields are: "Link 1", "1", "19", "1", "120", "4", "0", and "7". At the bottom of the dialog, there are three buttons: "OK", "Cancel", and "Help".

Label	Value
Tag Name of the Fieldbus Lin	Link 1
Network Number	1
Node Number	19
Module Number	1
Slot Number	120
Type of Control	4
Link Updated	0
Last Schedule Version Numb	7

Step	Action	Result/Notes
6	Assign: - NETWKNUM = 1 (UCN Number) - NODNUM = 19 (HPM Node Number) - MODNUM = 5 (I/O Module Number) - SLOT NUMBER = 120 (Software slots 120-123 are reserved for Fieldbus Links) - The control type is defaulted to 4. Click OK.	Comment: The slot number is the software slot for the SECM in position 1 of the 4 positions on the FTA. Note: If you are configuring an actual link, you need to supply your own parameters for the example above; however Link 1 will always occupy SLOTNUM 120.

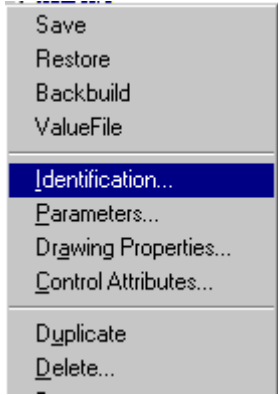
Step 6A— Add Fieldbus Devices and Function Blocks for the Input Device

This step adds and names the ST3000 to Link_1. The function blocks for the input device is configured and named.

Step	Action	Result/Notes
1	Click on the  in the Fieldbus Device Library to expand the devices.	The devices are shown in the library. 
2	Select the ST3000 Device Library template and drag it onto the link icon in the project's Link View .	A port opens, New Strategy.

14. Configuration Examples

14.2. Configuration Example 1

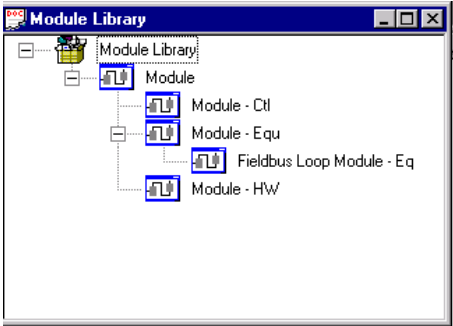
Step	Action	Result/Notes
3	In the port, enter the strategy: ST-101.	ST-101 and its accompanying function block and transducer block templates appear below the Link Schedule in the Link View.
4	In the Link View , ST-101, right-click and select Parameters .	The Parameters menu appears. It may be in tabbed format or in Alphabetical.
5	In the tabbed format, the parameter for the Device Identification is under the tab, Device Header .	In the Device identification port, enter the 32-character device id and click OK.
6	In the Link View , select the ST3000AI function block template (below the VF) and right click.	A menu appears. 
7	Right mouse click and select <i><Identification></i> .	The Identification Dialog Box appears.
8	Enter the in the NAME port: PI-101 and click OK.	The default name for the analog input function block changes to PI-101.
9	Select the PID function block (the default name starts with PI100. . .) and right mouse click.	A list menu appears.
10	Select <i><Identification></i> .	The Identification dialog box appears.
11	Enter the name: PC-102 and click OK.	The name, PC-102, appears below ST-101 and replaces the default name for the PID function block.

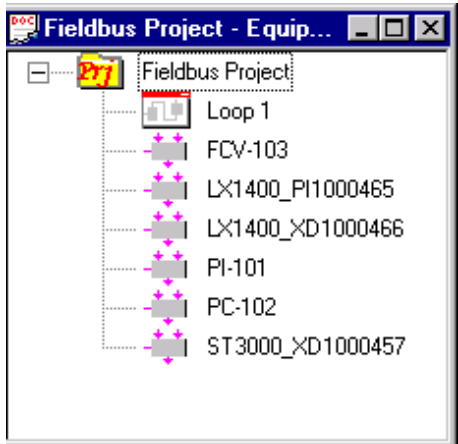
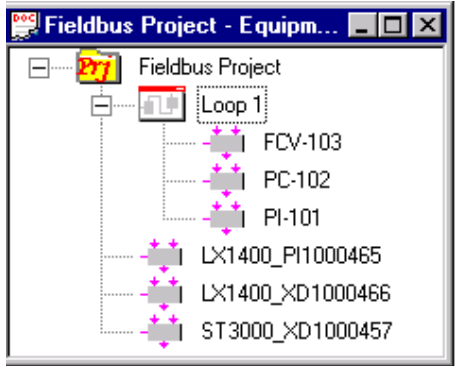
Step 6B— Add Fieldbus Devices and Function Blocks for the Output Device

Step	Action	Result/Notes
1	From the Fieldbus Device Library , drag and drop the LX1400 template onto Link 1 in the Fieldbus Link View .	A port opens, New Strategy.
2	Name the strategy: LX-101 and click OK.	The LX1400 device is named LX-101. Its accompanying function and transducer block templates appear below the Link in the Link View of the project
3	In the Link View , select the device LX-101 and right-click.	The menu appears.
4	Select Parameters .	The tabbed form or the alphabetical listing of the parameters appears.
5	In the tabbed format, the parameter for the Device Identification is under the tab, Device Header .	In the Device Identification port, enter the 32-character device id and click OK.
6	Select the AO function block template below the LX-101 template and right mouse click.	The list of choices appears.
7	Select <i><Identification></i> from the list.	The identification dialog box appears.
8	Name the AO function block: FCV-103 and click OK.	The AO function block is renamed FCV103.

Step 7 — Configuring the Loop Strategy

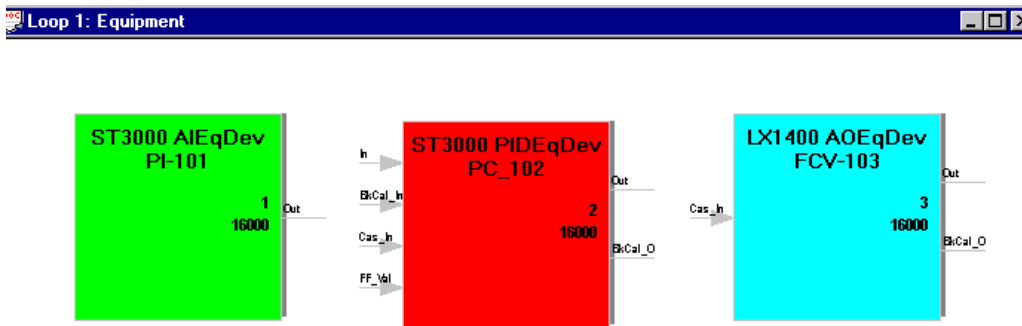
This step creates a loop in which the blocks will reside. From the loop drawing, the blocks can be displayed, rearranged, and connected.

Step	Action	Result/Notes
1	<File/Open/ Module Library>.	The Module Library is displayed.
		
2	Click on the  in front of the Module Library to expose the different modules.	The module templates are displayed.
3	Click on the  in front of Module – Equ to expose the Fieldbus Loop Module -Equ.	
4	Open the project's Equipment View . (<File/Open/Project/Equipment View/)	The project's Equipment view is displayed.
5	Drag the Fieldbus Loop Module – Equ onto the project's Equipment View .	A port opens, New Strategy.

Step	Action	Result/Notes
6	Give the New Strategy the name Loop 1 and click OK.	The loop strategy is added to the view.  The transducer blocks are associated with their respective analog input and analog output blocks when the project is forward built/downloaded.
7	Drag and drop the following function block strategies onto Loop 1: PI-101, PC-102 and FCV-103.	The corresponding function block strategies appear below Loop 1 the Equipment View. 

14. Configuration Examples
14.2. Configuration Example 1

Step	Action	Result/Notes
8	Double click on Loop 1 to open the drawing.	The drawing opens, exposing the blocks in their colors: green for the AI, red for the PID, and blue for the AO block.



9	Place the blocks into this order: PI-101,TC-102,FCV-103.	The blocks move into position.
10	The next step is performed with the loop drawing open, so don't close the drawing.	

Step 8 — Assigning the Function Blocks' Execution Sequence and Execution Periods

The blocks will execute in the following sequence:

Sequence	Block
1	PI-101
2	PC-102
3	FCV-103

The execution period for the loop must be multiples of ¼ second; the execution period for Loop 1 will be ½ second (16000 clock units).

Step	Action	Result/Notes
1	Select the AI Block PI-101 (handles will appear) and right mouse click.	A menu appears.
2	From the menu, select <Parameters>.	The Parameters tabs appear.
3	Select the BlockAssign tab.	The BlockAssign dialog box appears.

Parameters - PI-101

BlockAssign | BlockHead | Parameters-AI | Mfr Parameters | AI Mode-ODS | AI Mode-Set

Physical Device Tag PDTag -- --

VFD Relative Index Vfd 2

Function Block Number in VF BlockNo 1

Block OD Index for Addressin BLOCK_INDEX 257

Type of Control ControlType 1

Block Name BlockName ST3000 AI Block

Block Type BLOCK_TYPE AI

Sequence of Block Execution BlockExecutionSequ 1

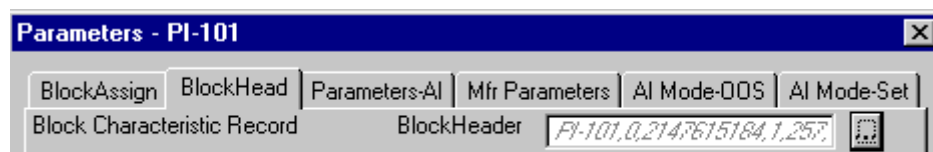
OK Cancel Help

- | | | |
|---|-------------------------------------|---|
| 4 | Enter 1 for the BlockExecutionSequ. | This block will execute first in the ½ second sequence. |
|---|-------------------------------------|---|

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
5	Select the BlockHead tab.	The BlockHead dialog box appears.

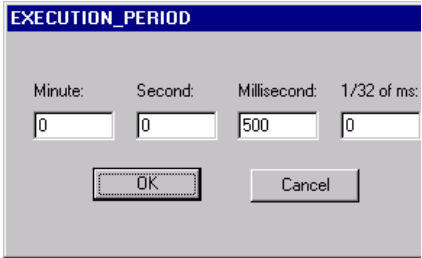


6	Click on the ellipsis  .	The Attributes of BlockHeader dialog box appears.
---	---	---

Step	Action	Result/Notes																																							
Attributes of BlockHeader <table> <tr> <td>Block_Tag</td> <td>BLOCK_TAG</td> <td>PI-101</td> </tr> <tr> <td>DD_MemberID</td> <td>DD_MEMBER</td> <td>0</td> </tr> <tr> <td>DD_Item</td> <td>DD_ITEM</td> <td>0x800201D0</td> </tr> <tr> <td>DD_Revis</td> <td>DD_REVIS</td> <td>0x0001</td> </tr> <tr> <td>Profile</td> <td>PROFILE</td> <td>0x0101</td> </tr> <tr> <td>Profile_Revision</td> <td>PROFILE_REVISID</td> <td>0x0401</td> </tr> <tr> <td>Execution Time</td> <td>EXECUTION_TIME</td> <td>736 ...</td> </tr> <tr> <td>Execution Period</td> <td>EXECUTION_PERI</td> <td>0 ...</td> </tr> <tr> <td>Number of Parameters in Bloc</td> <td>NUM_OF_PARAMS</td> <td>39</td> </tr> <tr> <td>Next Function Block to Exec</td> <td>NEXT_FB_TO_EXE</td> <td>0</td> </tr> <tr> <td>Views DD Index</td> <td>VIEWS_INDEX</td> <td>485</td> </tr> <tr> <td>Number_View_3</td> <td>NUMBER_VIEW_3</td> <td>1</td> </tr> <tr> <td>Number_View_4</td> <td>NUMBER_VIEW_4</td> <td>1</td> </tr> </table> OK Cancel Help			Block_Tag	BLOCK_TAG	PI-101	DD_MemberID	DD_MEMBER	0	DD_Item	DD_ITEM	0x800201D0	DD_Revis	DD_REVIS	0x0001	Profile	PROFILE	0x0101	Profile_Revision	PROFILE_REVISID	0x0401	Execution Time	EXECUTION_TIME	736 ...	Execution Period	EXECUTION_PERI	0 ...	Number of Parameters in Bloc	NUM_OF_PARAMS	39	Next Function Block to Exec	NEXT_FB_TO_EXE	0	Views DD Index	VIEWS_INDEX	485	Number_View_3	NUMBER_VIEW_3	1	Number_View_4	NUMBER_VIEW_4	1
Block_Tag	BLOCK_TAG	PI-101																																							
DD_MemberID	DD_MEMBER	0																																							
DD_Item	DD_ITEM	0x800201D0																																							
DD_Revis	DD_REVIS	0x0001																																							
Profile	PROFILE	0x0101																																							
Profile_Revision	PROFILE_REVISID	0x0401																																							
Execution Time	EXECUTION_TIME	736 ...																																							
Execution Period	EXECUTION_PERI	0 ...																																							
Number of Parameters in Bloc	NUM_OF_PARAMS	39																																							
Next Function Block to Exec	NEXT_FB_TO_EXE	0																																							
Views DD Index	VIEWS_INDEX	485																																							
Number_View_3	NUMBER_VIEW_3	1																																							
Number_View_4	NUMBER_VIEW_4	1																																							

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
7	Click on the ellipsis for Execution Period.	The Execution Period dialog box appears. 
8	Enter 500 in the millisecond port and click OK.	The Execution Period dialog box appears again with 16000 in the Execution Period, which is ½ second for the macrocycle.

Step	Action	Result/Notes

- 9 Click OK on the Attributes of BlockHeader tab and close the dialog box. The dialog box closes.

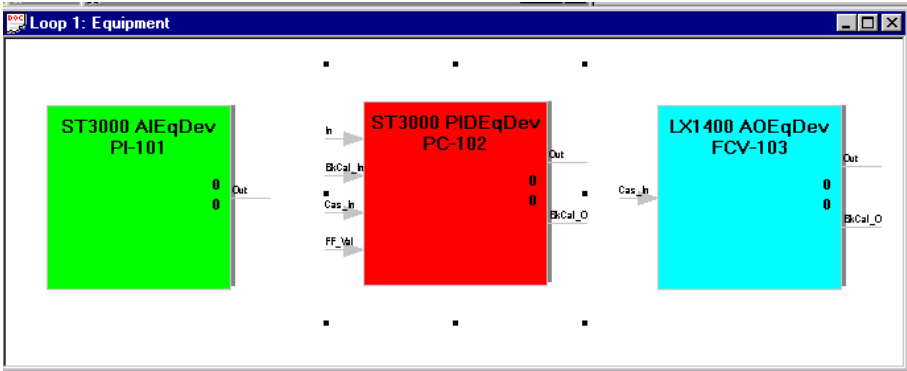
14. Configuration Examples
14.2. Configuration Example 1

Step	Action	Result/Notes									
10	Repeat steps 1 through 9, entering the sequences as follows for the following function blocks. <table><tr><td>Block</td><td>Seq.</td><td>EXECUTION_PERIOD</td></tr><tr><td>PC-102</td><td>2</td><td>500</td></tr><tr><td>FCV-103</td><td>3</td><td>500</td></tr></table>	Block	Seq.	EXECUTION_PERIOD	PC-102	2	500	FCV-103	3	500	The macrocycle for Loop 1 is set to ½ second and the sequence of the execution of the function blocks is set. When the CM Mapping is performed (later in this exercise), the actual time for the execution of each block will be visible.
Block	Seq.	EXECUTION_PERIOD									
PC-102	2	500									
FCV-103	3	500									
11	Leave Loop 1 drawing open for the next step, which is connecting the blocks.										

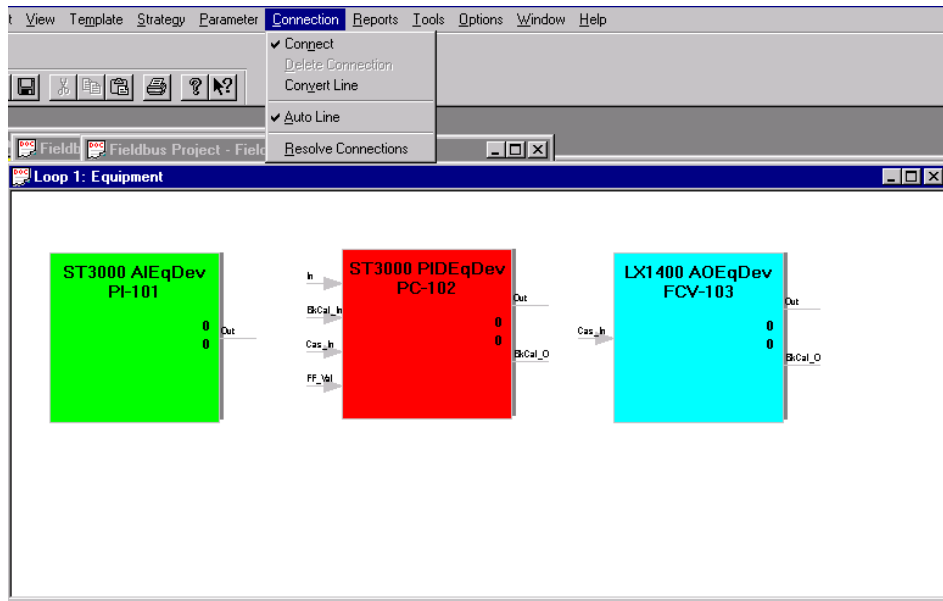
Step 9 — Connecting the Function Blocks in Loop 1

At this point the function blocks in Loop 1 are not connected as shown in the following figure.

Exposing the Functions Blocks in the Loop Strategy



Connection Tool



Step	Action	Result/Notes
1	On the Toolbar under Connections, select <Connect>.	A checkmark appears by the Connect tool.
2	Place the cursor on the <i>Out</i> connection of PI-101, depress the left mouse button, and drag the cursor to the <i>In</i> connection on PC-102.	A line is drawn from <i>Out</i> of the AI function block PI-101 to the <i>In</i> connection of the PID block PC102.
3	Connect the <i>Out</i> of PC-102 to the <i>Cas_In</i> of FCV-103.	A line validates the connection. The <i>BkCal_Out</i> to <i>BkCal_In</i> is made.
4	Save the drawing and close it.	If you forget to save the drawing as you try to close it you will get a prompt asking if you want to save.

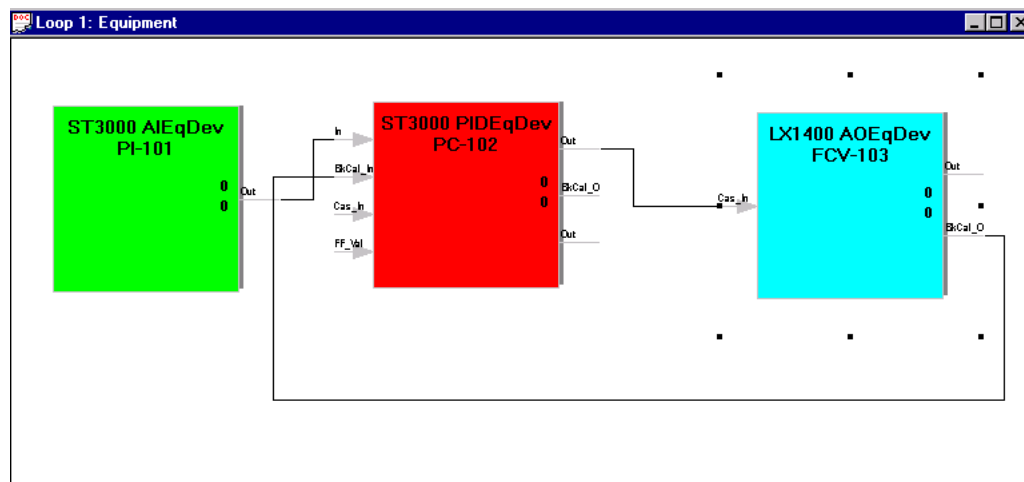
The next step is to perform the CM Mapping.

14. Configuration Examples

14.2. Configuration Example 1

The completed drawing is shown below.

Loop 1 Drawing with Connections



Step 10 — Performing the CM Mapping

Step	Action	Result/Notes
1	In the project's Link View , position the cursor on the Link strategy and right click.	The menu of choices appears.
2	Select <MapCMs>.	Link_1 is mapped to LINK_1S ST-101 is mapped to ST_101P LX-101 is mapped to LX_101P PI-101F is mapped to PI101F PC-102 is mapped to PC102F FCV-103 is mapped to FCV103F LINK_1S is assigned software slot 120. The devices (ST_101P and LX_101P) are assigned software slots 119 downward. The function blocks are assigned software slots 1 upward.

CM points are SECM (segments), PECM (physical devices), or FBCM (function block strategy points).

The **S** following the **LINK_1** designates that this is a SECM point. During the mapping, all link/segments will have an **S** following the configured name.

All device points, such as ST-101 will have all illegal characters deleted for substituted (a hyphen is an illegal character in the TPN System). Note that with the devices the hyphen is replaced with the underscore character, which is a legal character. TPS Builder also appends a P to the configured point name to designate that this is a PECM point, which represents a physical device in the system.

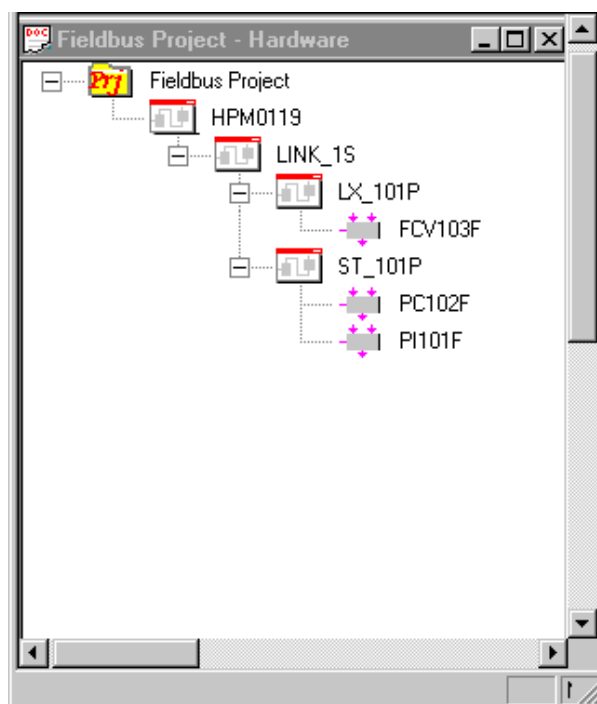
During the CM Mapping the names of the function blocks had their illegal hyphen characters deleted and an **F** appended to their names to designate Function blocks.

14. Configuration Examples

14.2. Configuration Example 1

The hardware view of the project (after CM Mapping has been performed) is shown in the following figure.

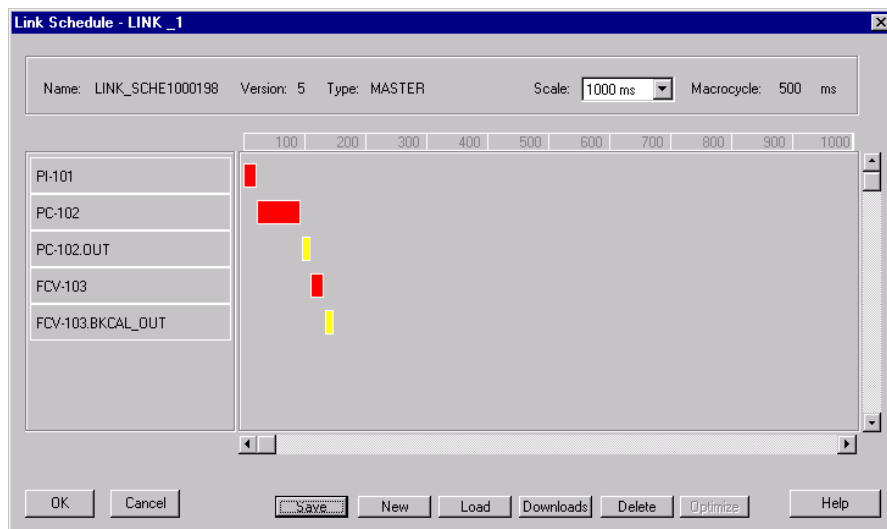
Hardware View after CM Mapping



Step 11 — Creating and Saving the Link Schedule

Step	Action	Result/Notes
1	In the project's Link View , position the cursor on the schedule element (it is just below the link name) and right mouse click.	A menu of choices appears.
2	Select <i><Schedule></i> .	The schedule creates (this may take a few seconds). A graphic view of the schedule is presented, as shown in the next figure.
3	Examine the schedule, and click OK if it is correct. The schedule is NOT saved until you click on OK.	The schedule is saved and the window closes.

Example of TPS Builder Generated Fieldbus Schedule

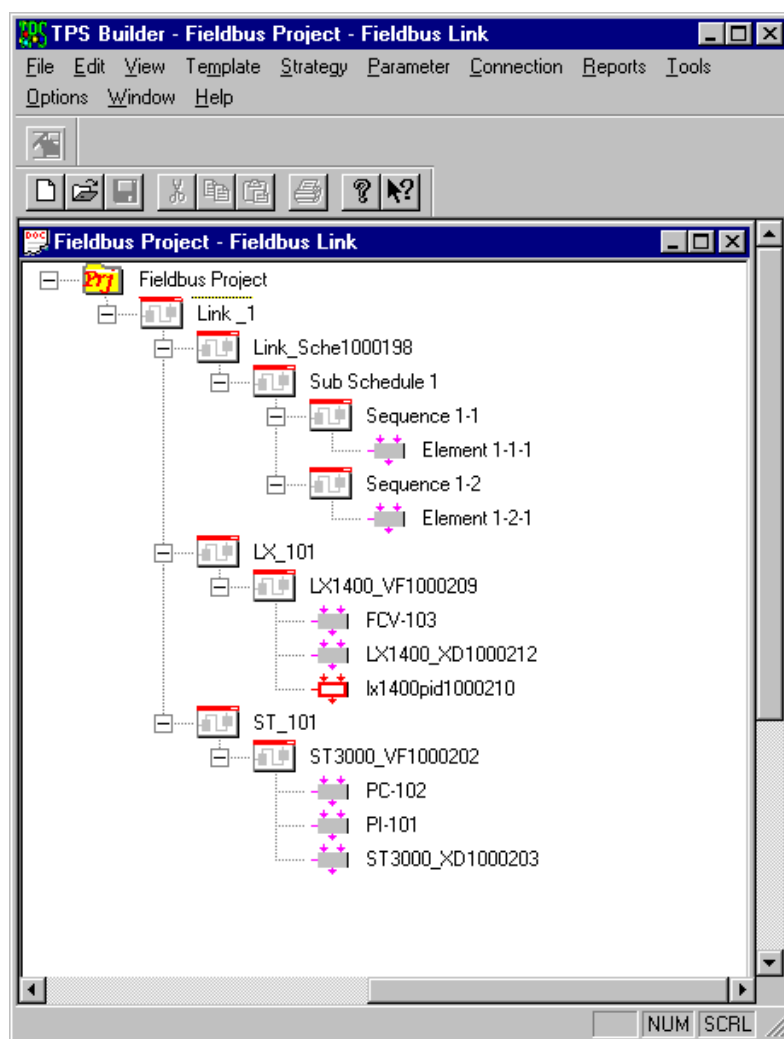


The following figure shows the Link View after the schedule has been created and saved. The elements have been expanded to show the sub-schedule elements.

14. Configuration Examples

14.2. Configuration Example 1

Link View of Project after Schedule Created and Saved



Step 12 — Entering a Valid Unit Number

Valid unit numbers **must** be entered for the following points, which will be built on the TPS network.

Step	Action	Result/Notes
1	In the project's Hardware View , position the cursor LINK_1S and right mouse click.	A menu of choices appears.
2	Select <Parameters>.	The parameters dialog box appears.

Parameters - LINK_1S

Link Objects	Devices	IOP Address	LAS Download
Point Assignment	IOP Stack	FB Schedules	Link Schedule
Point Type - CM	PNTTYPE	CM/IM	
Tag Name	NAME	LINK_1S	
Node Type	NODETYP	HFM	
Point Form	PNTFORM	FULL	
Point Descriptor	PTDESC		
Point Keyword	KEYWORD		
Associated Display	ASSOCDSP		
Unit Id	UNIT	A1	
Network Number	NTWKNUM	1	
Node Number	NODENUM	19	
Module Number	MODNUM	5	
Slot Number	SLOTNUM	120	
Primary Module Point Id	PRIMMOD		

OK Cancel Help

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
3	Enter a valid unit. We entered A1 , which is a valid unit for our area in TPN. Click OK.	The unit is accepted.
4	Repeat steps 1 through 3 for the following points: LX_101P, FCV103F, ST_101P, PC102F, and PI101F.	Points will not download correctly without a valid unit.

Step 13 — Configuring the ACK_Options for the Blocks

After CM Mapping has been performed, enable alarming for the devices and AI and PID function blocks. Using the Parameter Configuration Form, configure the ACK_Option, and Reports.

Step	Action	Result/Notes
1	In the Equipment View of the project, position the cursor on PI102F. Right click, and select <Parameters>.	The Parameter Configuration Form appears.

Step	Action	Result/Notes

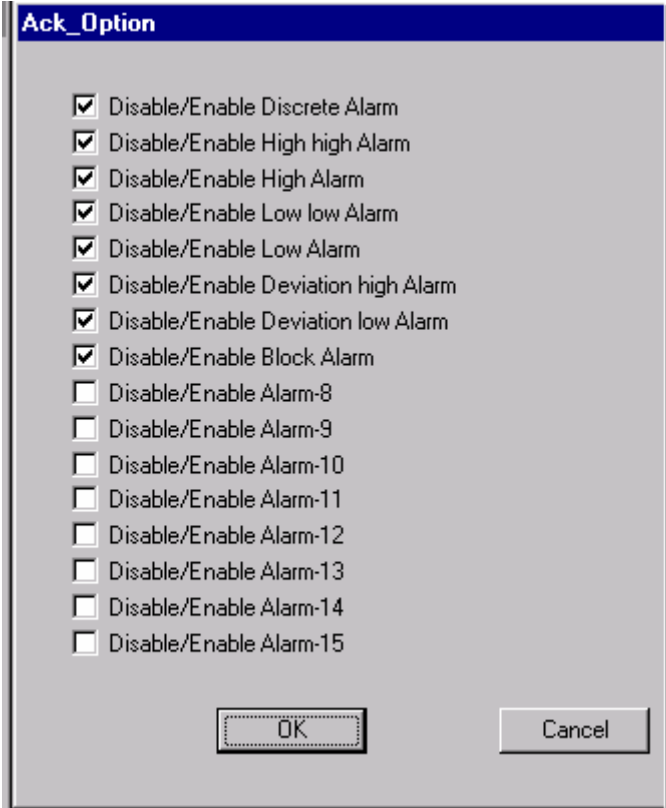
- | | | |
|----------|--|--|
| <p>2</p> | <p>If you are working from the tabbed dialog, select the Parameters-AI tab, and scroll until you find the Ack_Option. If you are working from an alphabetical listing, scroll to it.</p> | <p>The Parameters AI dialog box appears.</p> |
|----------|--|--|

14. Configuration Examples

14.2. Configuration Example 1

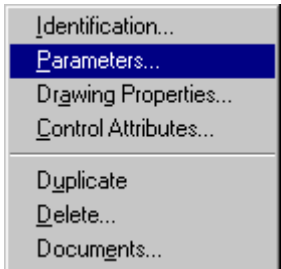
Step	Action	Result/Notes
The screenshot shows the 'Parameters-Alarm' tab of a configuration window. The 'Ack_Option' field is set to '0xFF00' and has an ellipsis button to its right. An arrow points to this ellipsis button. Other fields include 'L_Type' (Direct), 'Low_Cut' (0.000000), 'PV_Ftime' (0.000000), 'Field_Val' (Bad: OutOfService: Constant), 'Update_Evt' (Acknowledged, Undefined, 1), 'Block_Alm' (Acknowledged, Uninitialized), 'Alarm_Sum' (0,0,0,0), 'Alarm_Hys' (0.500000), 'Hi_Hi_Pri' (0), 'Hi_Hi_Lim' (+INF), 'Hi_Pri' (0), and 'Hi_Lim' (+INF). Buttons for OK, Cancel, and Help are at the bottom.		

- Click on the ellipsis. The ACK_Option dialog box appears.

Step	Action	Result/Notes
		
4	If the Disable/Enable alarms are checked as shown in the preceding figure, click OK and close the dialog. Otherwise position the cursor in the unchecked boxes and enable/disable the AI alarms.	These alarms must be checked for the alarms to be reported to the TPS system network.
5	Repeat steps 1,2 and 3 for PC102F.	The ACK_Option is configured.

Step 14-A — Configure the Resource Block's Features and Features Select

The device's resource blocks Features and Features Select must be configured to get events and alarms reported.

Step	Action	Result/Notes
1	In the Link View of the project, position the cursor on the icon for the VFD for ST_101 and right click.	The list of menu items appears: 
2	The Parameter Configuration Form appears. Choose <Resource Parameters> tab.	The Resource parameters appear.

Step	Action	Result/Notes
Mfr Parameters Resource Mode-00S Resource Mode-Set VFD Assign FBAP Directory FBAP Link Object VFD Assign Device Clear (2) BlockAssign BlockHead Resource Parameters Allow manual restart to be initi Restart Run Supported resource block opt Features 0x4800 Selected resource block opti Feature_Sel 0x0000 Block Execution Methods av Cycle_Type 0x8000 Block Execution Method sele Cycle_Sel 0x0000 Time duration of resource's mi Min_Cycle_T 4000 Available Configuration Memo Memory_Size 0 Interval between writing NV p NV_Cycle_T 28800000 Pct of memory available for fu Free_Space 0.000000 Pct of Block Proccessing Tim Free_Time 68.493000 Time duration -> failed to writ Shed_Rcas 96000 Time duration -> failed to writ Shed_Rout 8000 Condition set by loss of comm Fault_State Clear OK Cancel Help		

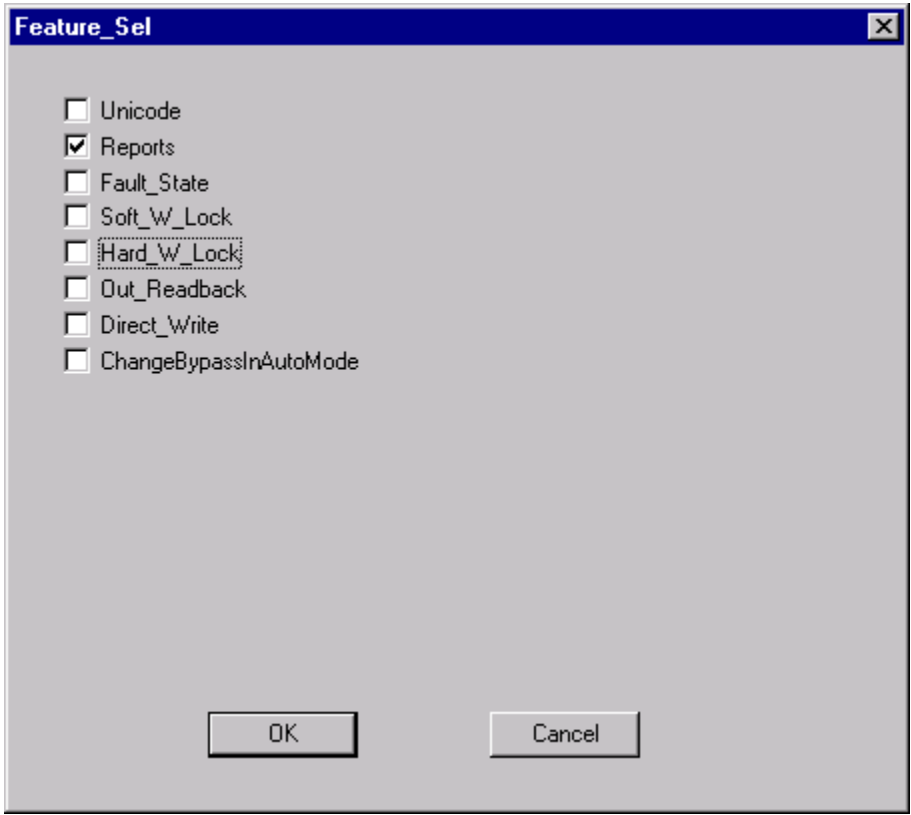
- 3

Scroll to **Features_Sel** and click on the ellipsis.

The Features dialog box appears:

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
		
4	Check the reports and click OK.	The Feature for resource block for ST_101 is configured.
5	Repeat steps 1 through 4 for LX_101P.	The Feature_Sel for the resource blocks is configured.

Step 14-B (Optional) — Configuring the Process Parameters, Including Alarms

This is an optional step. Setpoints and alarm parameters may be configured from the point's detail display after they are downloaded. When the TPS database (as opposed to the device database) is checkpointed (or saved), these parameters will be saved at their current value, not the configured value.

If you want to save the process configuration parameters in the TPS Builder repository, do the configuration in TPS Builder.

Step	Action	Result/Notes
1	In the project's Equipment View , position the cursor on PI101F and right mouse click.	A menu of choices appears.
2	Select <Parameters> and click on the Parameters-AI tab.	The Alarm Parameters dialog box appears.

14. Configuration Examples

14.2. Configuration Example 1

Step	Action	Result/Notes
Parameters - PI-101 BlockAssign BlockHead Parameters-Al Mfr Parameters AI Mode-00S AI Mode-Set Amt PV must return within Alm Alarm_Hys 0.500000 High High Alarm Priority Hi_Hi_Pri 0 High High Alarm setting (in E Hi_Hi_Lim +INF High Alarm Priority Hi_Pri 0 High Alarm setting (in EU) Hi_Lim +INF Low Alarm Priority Lo_Pri 0 Low Alarm Setting (in EU) Lo_Lim -INF Low Low Alarm Priority Lo_Lo_Pri 0 Low Low Alarm Setting (in EU Lo_Lo_Lim -INF Status of the Hi_Hi_Alarm an Hi_Hi_Alm Acknowledged, Uninitialized, ... Status of the HI Alarm and its Hi_Alm Acknowledged, Uninitialized, ... Status of the Low Alarm and i Lo_Alm Acknowledged, Uninitialized, ... Status of the Low Low Alarm Lo_Lo_Alm Acknowledged, Uninitialized, ... OK Cancel Help		

Step	Action	Result/Notes
3	Enter process information on the AI tab, such as setpoints, alarm limits, and so forth as it applies to your particular process. If you are not familiar with the parameter, refer to the <i>Fieldbus Parameter Reference</i>. Click OK. The parameters on this tab are similar to the parameters shown on a Data Entry Display. The alarm parameters are displayed on the various operational displays for the FBCM points.	Process alarms are configured for PI101F.
4	Repeat steps 1 and 2 for FCV103F and PC102F.	The process parameters are configured.

Step 15 — Preparing to Download

Refer to in the “Checklist before Download Strategies” in this document to ensure that conditions are correct before performing the download.

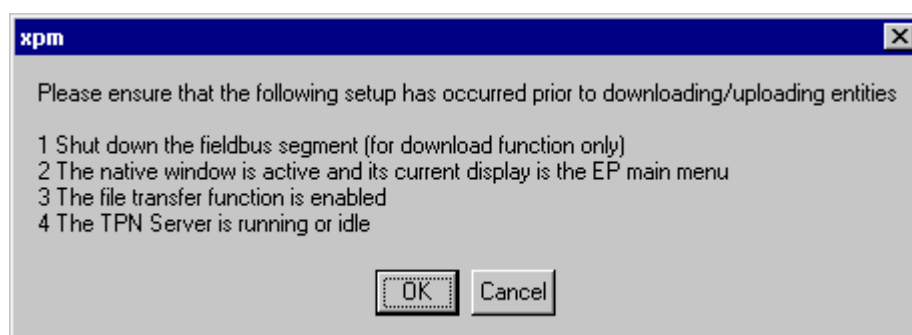
Step 16 — Downloading

Step	Action	Result/Notes
1	In the project's Link View , position the cursor on the link object and right mouse button click. Choose Download from the menu.	The prompt for download appears with a four part warning.

14. Configuration Examples

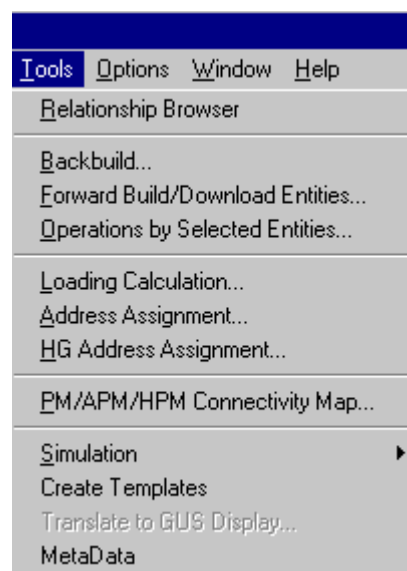
14.2. Configuration Example 1

Step	Action	Result/Notes
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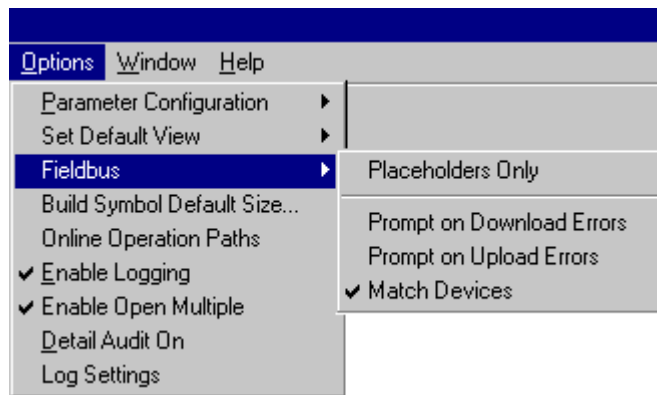


- 1a** Optionally, you may choose from the <Tools> menu.

The choices appear.



Step	Action	Result/Notes
1b	Select <Forward Build/Download Entities>. Select the option to Match Devices (unless you have some good reason not to select this option). Enable logging options.	A progress bar is presented to show the progress on the download. The Link is assigned node address 20. Devices will match their existing addresses. If no address has been assigned, devices with Link Master capability are assigned address in the range of 21, 22, 23 . . . Basic devices (those without the Link Master capability) are assigned address 247 downward.



- 3 Set the TPS Network path to the drive to which your History Module is mapped.
- Set the EB File Directory to the path with your project's name.
- Enable Logging on TPS Builder <Options/  Enable Logging>. (If TPS Builder database is networked; that is, if the database resides on a server or if users are using database located on a single station, only one user should enable logging).



ATTENTION

Do not use the Data Entity Builder (DEB) to delete fieldbus points. Use the Delete function in TPS Builder's Operations by Selected Entities.

Step 17 — Starting up the Loop

The Loading from the Builder should have loaded the points without errors. For a simple PID loop, the following points will be created:

Step	Action
1	One SECM—Link_1S.
2	Two PECMs—one for each device—ST_101P and LX_101P.
3	Three FBCMs—one for each function Block (AI, PID and AO) — PI101F, PC102F, and FC103F.

The Loop should be started in the following order:

SECM

Call up the SECM

Step	Action
1	Press [DETAIL] and LINK_1S.
2	Set the SECM execution state = ACTIVE.
3	Activate the schedule, SECM device detail, page 2.

PECMs

One at a time, call up the PECMs (LX_101P and ST_101P).

Step	Action
1	Press [DETAIL] and LX_101P/ST_101P.
2	Set the PECMs' execution state = ACTIVE and set both to target-mode = AUTO.

FBCMs

Call up each FBCM point.

Step	Action
1	Call up the FBCMs— Press [DETAIL] and PI101F, PC102F, FCV103F.
2	Set the FBCMs' execution state = ACTIVE.
3	Set the Mode on the FBCMs (if they are not set as follows: • AI, PI101F = AUTO • PID, PC103F = AUTO • AO, FCV103F = CAS

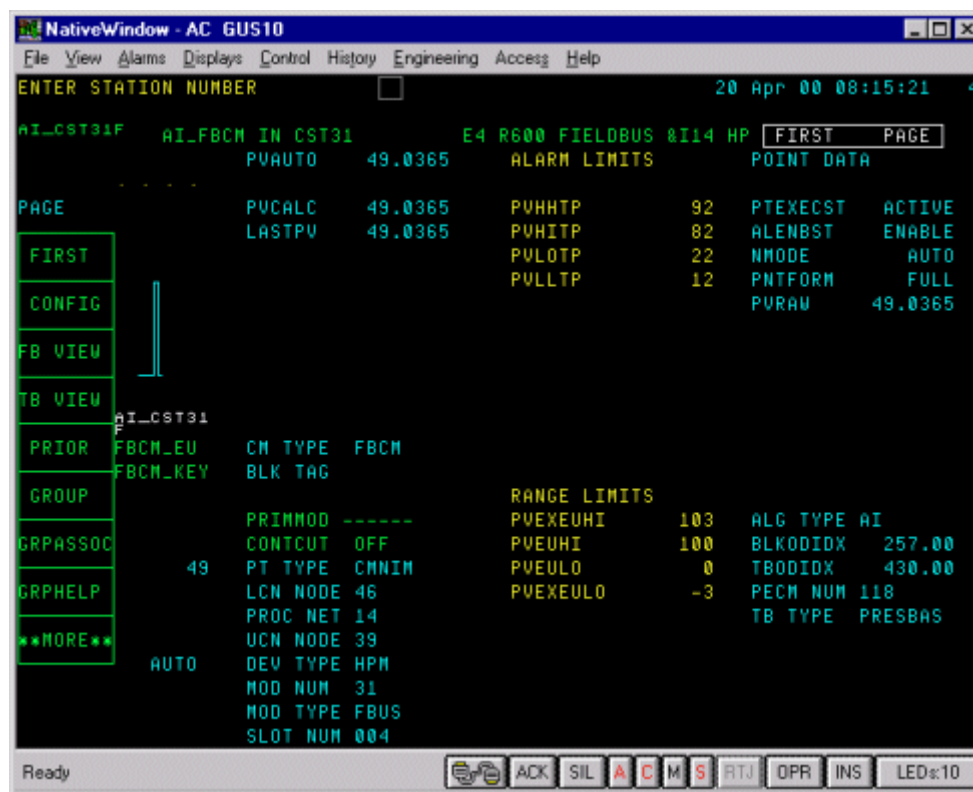
Note: The transducer blocks for the AI and the AO should be set automatically from the Builder download to AUTO. This target mode can be verified (and set if required) from the AI and AO FBCM detail display in TB View Page 1, subpage 1.

Note2: The FBCM detail display will page through pages for the selected VIEWS. You can page through the **FB VIEW** target by the PG FWD function, or by selecting the **FIRST PAGE** target and selecting a view from the menu on the left, as shown in the figure below. Each VIEW may have several pages; these pages may have subpages.

14. Configuration Examples

14.2. Configuration Example 1

FIRST PAGE with Target Selected and Menu



Tune the Loop

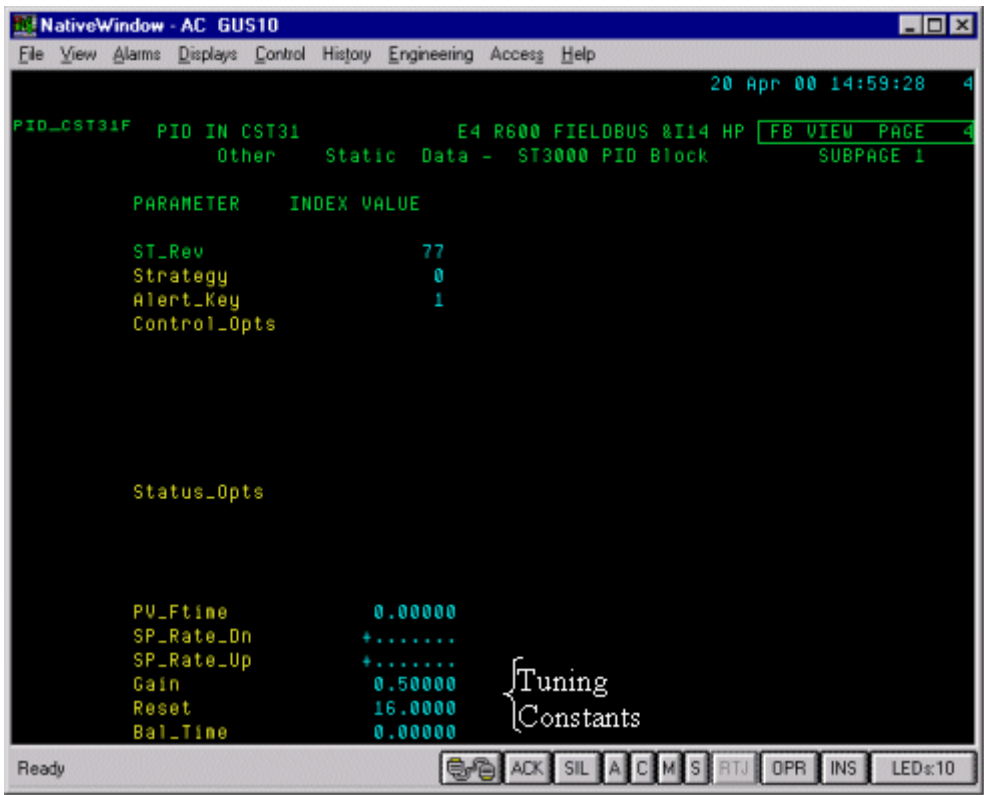
The Tuning Constants are located on the PID's Page 4, subpage 1, as shown in the following display example. Enter the tuning constants (we suggest no values, as each configuration may require a different value).

Step	Action
1	To call up this display, press [DETAIL] and FCV103F.
2	As shown in the preceding figure, select the FIRST PAGE target. The menu as shown on the left of the page appears. Select the FB VIEW target. Page forward to Page 4, subpage 1. The tuning constants are shown in the following figure.
3	Enter the desired constants.

14. Configuration Examples

14.2. Configuration Example 1

Fieldbus PID Tuning Constants



Set the Alarm Limits

Alarm limits depend upon the operating ranges of the process. The process alarms, even if not configured in TPS Builder, can be entered from the FBCM's detail page.

Step	Action
1	Call up the FBCMs— Press [DETAIL] and PI101F, PC102F, FCV103F.
2	Refer to figure in “Set the Alarm Limits” in this document , FIRST PAGE with Target Selected and Menu. The only alarms configured are PVEUHI and PVEULO.

Step	Action
3	All alarm values are absolute values, with the exception of the DEVHITP and DEVLOTP. See the note and the following figure.
4	Alarm entries may be configured on-line by selecting the parameter and entering the values.



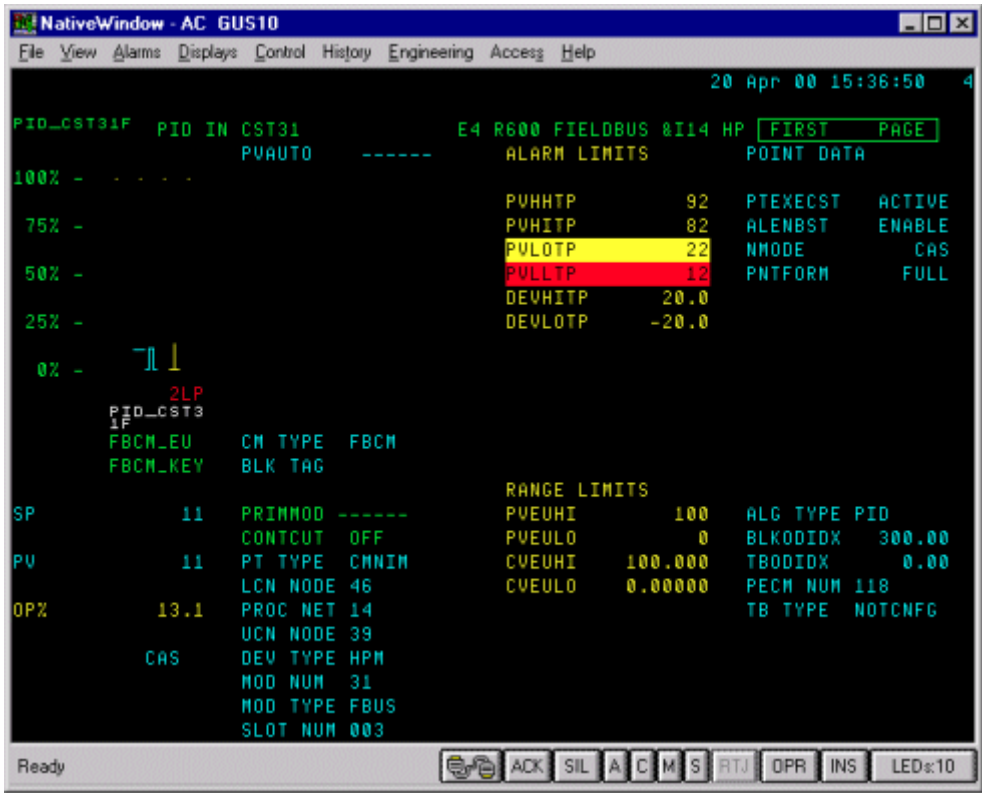
ATTENTION

The deviation high and low trip points (DEVHITP and DEVLOTP) do not operate in fieldbus the same as they do in other TPN based points. In other TPN points, these values are entered as absolute values.

For fieldbus points, the DEVLOTP is a negative value as shown in the preceding figure. The (DEVLOTP) alarm will trip when the PV deviates more than 20 points below the SP. Likewise, if the PV deviates more than 20 points above the SP, the (DEVHITP) alarm will trip.

14. Configuration Examples
14.2. Configuration Example 1

PID First Page with Alarms



Checkpointing

Checkpoint the FBIOP/HPM if you did not get a warning that your data did not download correctly. To checkpoint the FBIOP:

Refer to the procedure for Save Database in the *Process Operations* or the *Operator's Digest*.

This does NOT save the configuration data that has been downloaded to the device. The Save/Restore function saves the device data. To save the device data, refer to the procedure "Sav" in this document.

14.3 Configuration Example 2— Create a Control Scheme for Fieldbus

Introduction

In addition to the libraries used in Example 1, this example requires the **PM/APM/HPM Library**.

In this example, a loop is created with two function blocks and a Regulatory Control Point in the HPM.

Procedures and examples presented in detail in Example 1 are less detailed in Example 2.



TIP

Remember that a fieldbus point can neither push its input nor pull its output to a point outside the fieldbus.

Step 1 — Create Fieldbus Link

The HPM Hardware module has been defined in Exercise 1. Use the same hardware addressing as in Exercise 1.

Step	Action
1	Open the Fieldbus Project in the Fieldbus Link View .
2	Open the Fieldbus Link Library. Click on the  in front of the Fieldbus Link Library to expand the view to expose the Link element.
3	Drag Link template from the Fieldbus Link Library into the Fieldbus Project (link view).
4	When the port opens for the New Strategy, assign the name: Link2 .
5	Assign the Link to a slot Number on the IOP where the Fieldbus Link resides. In the Fieldbus Link View , select the template for Link2 (it will turn color) and right-click.

14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step	Action
6	From the menu select <i><Parameters></i>. NTWKNUM = 1 (This is the UCN number) NODENUM = 19 (This is the HPM number) MODNUM = 5 (This is the I/O Module number) SLOTNUM = 121 (Software slots 120-123 are reserved for Fieldbus Links) The control type is defaulted to 4. Click OK.

Step 2 — Create Fieldbus Devices

Step	Action
1	Open the Fieldbus Device Library. (<i><File>Open>Library></i> .)
2	Click on the + to expand the library to show the device templates
3	Select the STT35F device template, drag and drop onto Link2. Assign this strategy STT_100.
4	The STT35F device appears in the project Link View. - Expand the device by clicking on the  in front of the device. - In the Equipment View, select the AI Block under the STT35F device and give it the Identifier FF_AI100. - Delete the PI Block. (Right-mouse click and <i><Delete></i>).
5	Select the LX1400 device template; drag and drop onto Link2 in the Fieldbus Project Fieldbus Link View. Assign this Strategy LX_100. - The LX1400 device appears in the project in the Link View. - In the Equipment View, expand the device by clicking on the  in front of the device. - The blocks are displayed for AO, PID and Transducer blocks. - Select the AO function block and right-click. Choose <i><Identifier></i> and name the function block FF_AO100. - In the In the Equipment View, delete the PID block under the LX1400 device, delete the PID block under the LX1400 device.

Step 3 — Create Loop Module

Step	Action
1	Open the Module Library . <File>Open>Library>Module Library>.
2	Expand the Library to show the Fieldbus Loop Module -Equ template.
3	Drag and drop the Fieldbus Loop Module -Equ template onto the project in the Fieldbus Equipment View.
4	When the New Strategy port opens, enter the name Loop100.

Step 4 — Add Function Blocks to Loop Module

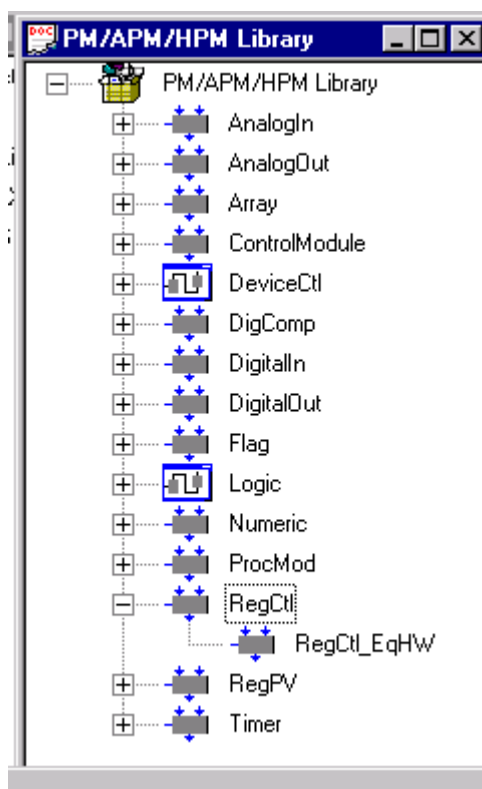
Step	Action
1	Drag AI function FF_AI100 onto Loop100.
2	Drag AO function block FF_AO100 onto Loop100 module

14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step 5 — Adding the Regulatory Control block to the loop

Step	Action
1	Open the PM/APM/HPM Library .
2	Click on the  in front of RegCtl to expose the RegCtl_EqHW template.



- | | |
|---|--|
| 3 | Drag and drop the RegCtl template onto the Loop100 module. |
| 4 | Name the RegCtl Strategy PID_100. |
-

Step 6 — Assigning the Function Blocks' Execution Sequence and Execution Periods

The RegCtl point is not included in the execution sequence.

Step	Action
1	Open Loop100 by double clicking on the icon.

2	Select the FF_AI100 block and right mouse click
3	Select the BlockAssign tab.
4	Enter 1 for the <i>BlockExecutionSequence</i> .
5	Select the BlockHead tab. Click on the ellipsis.
6	The dialog box for the execution period appears. Enter 1 in the <i>Second</i> box. Click OK.
7	32000 clock units appear in the <i>EXECUTION_PERIOD</i> .
8	Click OK and close the Parameter Configuration Form.
9	Select the FF_AO100 block and right mouse click.
10	Enter 2 for the <i>BlockExecutionSequence</i> and 1 second for the <i>EXECUTION_PERIOD</i> .
11	Click OK and close the Parameter Configuration Form

14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step 7 — Configuring the Parameters for the HPM Regulatory Control Strategy

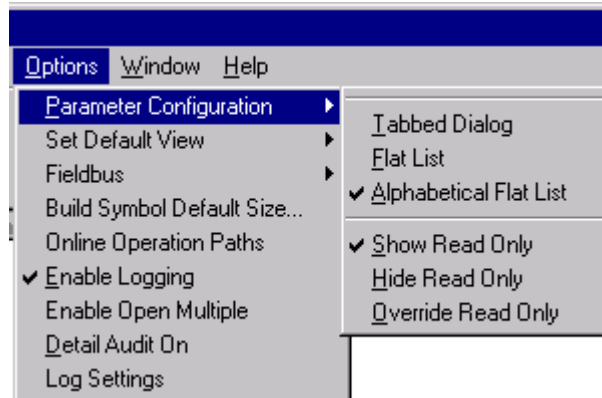
- | Step | Action |
|------|---|
| 1 | Select the RegCtl block PID_100 and right-click. From the menu choose <Parameters>. |
| 2 | The parameter configuration form appears. You can choose to configure parameters in two ways:
a) By selecting the PV Configuration tab, as shown below, or |

Parameters - PID_100

Setpoint Display	PID Algo	Output Configuration	Alarming Display
Point Assignment	PV Configuration	Mode Configuration	
PV Range High	PVEUHI	100.000000	
PV Range Low	PVEULO	0.000000	
PV Decimal Format	PVFORMAT	D1	
PV Source Option	PVSRCOPT	ONLY AUTO	
Overview Value	OVERVAL	25	

OK Cancel Help

Step	Action
b)	By selecting <Options/Parameter Configuration/Alphabetical Flat List>.



14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step

Action

This will generate an alphabetical list of parameters. An example follows.

The screenshot shows a dialog box titled "Parameters - PID_100". It contains a list of parameters for a PID controller, each with a name, a tag, and a value. The parameters are:

Parameter Name	Tag	Value
Auxiliary Unit	\$AUXUNIT	1
OP Tolerance in %	\$OPTOL	0.000000
Set Point Tolerance	\$SPTOL	0.000000
Alarm Enable State	ALENBST	ENABLE
Associated Display	ASSOCDSP	
Bad PV Mode Shed Option	BADCTLOP	NO_SHED
Bad Op Conn Alarm Option	BADOCOPT	OFF
Bad PV Alarm Priority	BADPVPR	LOW
Control Input Source	CISRC(1)	FF_AI100.Out
Destination Point_ID.paramet	CODSTN(1)	FF_AO100.Cas_In
Control Action	CTLACTN	REVERSE
Control Algorithm	CTLALGID	PID
Algorithm Equation Type	CTLEQN	EQ4

At the bottom of the dialog box are three buttons: OK, Cancel, and Help.

Step 8 — Configuring the tabs

Step	Action
1	Select the Point Assignment tab, or if you are using the alphabetical list, go to the next step. In order to keep from getting error messages, you must first configure NODETYP; choose HPM. Scroll down to the bottom of the list; for CTLALGID, choose PID. Scroll back to top of tab. Enter the values for the parameters as follows: PNTFORM = FULL PTDESC = DEG F KEYWORD = TEMP UNIT = A1 NTWKNUM = 1 NODENUM = 19 MODNUM = 0 SLOTNUM = 10
2	Select the PV Configuration tab. PVEUHI = 150 PVEULO = 0 PVFORMAT = D1 PVSRCIOT = ONLYAUTO OVERVAL = 25
3	Select the Mode Configuration tab BADCTLOP = NO_SHED RCASOPT = NONE NMODE = AUTO MODEPERM = PERMIT EXTSWOPT = NONE
4	Select the Setpoint Display tab. SPHILM = 150 SPLOLM = 0 \$SPTOL = 0 SP = (leave blank) SPOPT = NONE RBOPT = NORATBI

14. Configuration Examples

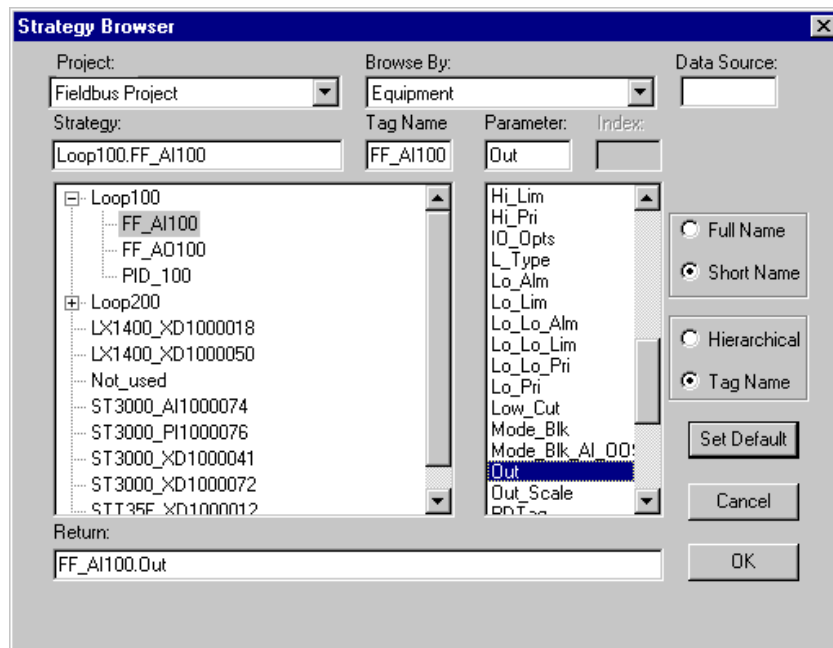
14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step	Action
------	--------

5	Select the PID Algo tab
---	--------------------------------

PIDFORM = INTERACT
CTLEQN = EQA
PVTRACK = NOTRACK
CTLACTN = REVERSE
GAINOPT = LIN
K = .5
T1 = .2
T2 = 0
NOCINPTS = 1

For the CISRC input, click on the ellipsis. Click on the ellipsis for Control Input Source **CIRSC(1)**. The strategy browser appears. Expand the **+** on the loop so that **FF_AI100** is visible. Scroll down the second column and pick the **Out**. Click OK. The parameter accepts the input tagname.parameter.



Step	Action
6	Select the Output Configuration tab. NOCOPTS = 1 For CODSTN(1), click on the ellipsis and navigate to the Strategy browser as shown in the preceding step: From Column 1, select <i>FF_AO100</i> and from the second column select <i>Cas_In</i>. Click OK to configure the parameter as shown below. CODSTN(1) = <i>FF_AO100.Cas_In</i> OPHILM = 105 OPLOLM = 0 SAFEOP (leave blank) OPMCHLM (accept default) OPROCLM (leave blank) \$OPTOL (accept default)
7	Select the Alarming Display tab. BADOCOPT = OFF OPHITP (leave blank) OPLOTP (leave blank) PVALDB = ONE (same as default) PVHITP = 83 PVLOTP = 20 PVHHTP = 98 PVLLTP = 15 PVROCTP = 2 PVROCPPR = LOW PVROCNTP = 22 BADPVPR = LOW

14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

Step 9 — Perform CM Mapping

In the **Fieldbus Link View**, position the cursor on Link2 and right mouse click. Select *<Map CMs>*.

The following points are created (**Hardware View**):

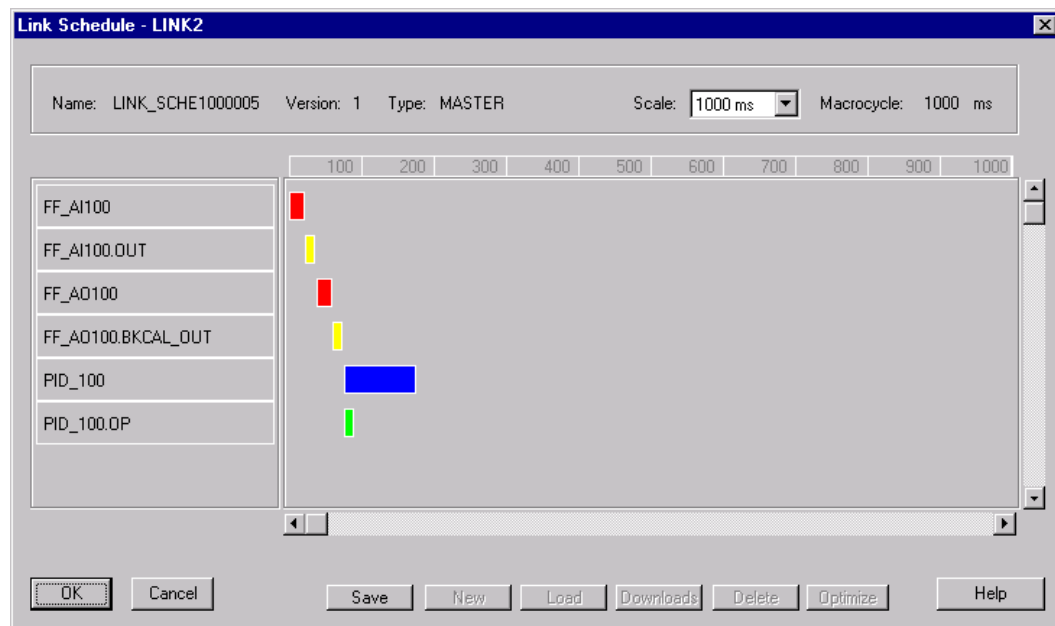
<i>Strategy</i>	<i>Point Name</i>	<i>Explanation</i>
Link2	LINK2S	This is the SECM point for the Link
STT_100	STT_100P	This is the PECM point for the STT35F device
LX_100	LX_100P	This is the PECM for the LX1400 Device
FF_AI100	FF_AI100F	This is the FBCM point for the Analog Input
PID_100	PID_100	This is the Regulatory Control Point in the HPM
FF_AO100	FF_AI100F	`This is the FBCM point for the Analog Output

Step 10 — Create a Schedule

The schedule creation determines the sequence and the duration of the function blocks' execution.

In the **Fieldbus Link View**, position the cursor on the Schedule element under the Link. Right mouse click and select <Schedule>.

The schedule appears.

Schedule for Loop 100**Step 11 — Prepare to Download and Download**

Follow the instructions in “Step 15 — Preparing to Download” and “Step 16 — Downloading” in the Configuration Example 1 section in this document.

14. Configuration Examples

14.3. Configuration Example 2— Create a Control Scheme for Fieldbus

15. Appendix A — VCR List

15.1 VcrList (VCRL)

The VcrList is a list of VCR Definitions. Each entry in the list may contain a defined VCR or it may be empty. After the entire list is loaded a host device must check a capabilities record to determine if an individual VCR may be updated or a new VCR defined by using the FMS Write Service. The VCR List is accessible through an OD Index that exists in the NM Directory. The VcrList is composed of the following attributes:

#	Attribute	M/C	R/W	Description
1	VcrListOdIndex	M	Key	The OD Index of the first object in the VCR List composite object, VcrListControlVariable. The OD Index is in the directory entry for the NM directory.
2	VcrListControlVariable	M	R-W	This attribute is used to control the download of the whole VcrList. The stack capability of the NM agent should be checked to determine if this variable is supported. If supported, writing a 0x01 to this variable causes NMA to abort every established VCR except for the NM and SM standard VCRs. The entries in the VCRL are returned to their initial state as define by the device manufacturer. The NMgr may now reload the entire list.
3	VcrListCharacteristicsRecord	M	R	This attribute describes the characteristics of the whole VCRL. The attribute is often called the VCRL header. The record contains the major and minor version of the VCRL, the maximum number of static VCRs that can be defined, the number of permanent non-volatile entries into the VCRL, the number of static entries configured, a dynamics supported flag, and a statistics supported flag.

15. Appendix A — VCR List

15.1. VcrList (VCRL)

#	Attribute	M/C	R/W	Description
4	ListOfVcrStaticRecords	C	R-W	This attribute contains an array of VcrStaticEntry Records. The VcrStaticEntry record represents the VCR as a combination of its layer protocol components. There are 17 protocol components that are to be defined.
5	ListofVcrDynamicRecords	C	R	This attribute contains a list of VcrDynamicEntry records. The VcrDynamicEntry record defines a management object for the dynamic portion of the VCR.
6	ListOfVcrStatisticsRecords	C	R-W	This attribute contains a list of VcrStatisticsEntry records. A VcrStatisticsEntry record contains a writable OD Index to the VcrStaticEntry record for which statistics are being gathered. Writing True to the ClearVcrStatistics flag clears the statistics. VCR statistics include the number of aborts, timeouts, and the data protocol units sent and received.

In this and the following table, **M** indicates a mandatory attribute and **C** indicates a conditional attribute. **R** indicates that the attribute is readable and **W** indicates that the attribute is writeable.

VcrStaticEntry

The VcrStaticEntry record contains the static part of a VCR that must be configured by a Host Interface device. A single VCR or the whole static VCR list may be written sequential by FMS. Writing to a single VCR aborts the connection. The following attributes are contained in the VcrStaticEntry:

#	Attribute	M/C	R/W	Description
1	FasArTypeAndRole	M	R-W	This attribute contains the FAS Application Relationship type, role, and FMS bypass flag. Types include QUB, QUU, and BNU. Roles include peer initiator, peer responder, client, server, publisher, subscriber, source, sink.
2	FasDIILocalAddr	M	R-W	This attribute is a DLCEP, DLSAP, or a FREE relationship that is indicated by a zero.
3	FasDIConfiguredRemoteAddr	M	R-W	This attribute is either the remote QUB DLCEP address or the remote QUU DLSAP address. A zero indicates the relationship is FREE.
4	FasDIISDAP	M	R-W	This attribute describes the AP's Data Delivery Features, Authentication Type, and Delivery Priority. It also indicates whether the AREP is LAS scheduled.
5	FasDIIMaxConfirmDelayOnConnect	M	R-W	A QUB and QUU AREP's use this attribute. It defines the maximum amount of time that the data link layer to complete the DL Connection establishment process.

15. Appendix A — VCR List

15.1. VcrList (VCRL)

#	Attribute	M/C	R/W	Description
6	FasDIIMaxConfirmDelayOnData	M	R-W	A QUB and QUU AREP's use this attribute. It defines the amount of time that the data link layer has to complete individual connectionless and connection-oriented data transfers.
7	FasDIIMaxDlsduSize	M	R-W	This is the maximum DLSDU size that can be sent on this VCR.
8	FasDIIMaxResidualActivitySupported	M	R-W	This attribute is for QUB VCRs only. If true, residual activity monitoring is performed by the DLL in each direction of the transfer.
9	FasDIITimelinessClass	M	R-W	This attribute is for BNU VCRs only. This attribute defines the publisher and the subscriber Timeliness Classes.
10	FasDIIPublisherTimeWindowSize	M	R-W	This attribute is for BNU VCRs only. This optional parameter provides the time window of a Publisher provided by a DLL.
11	FasDIIPublisherSynchronizingDLcep	M	R-W	This attribute is for BNU VCRs only and then only if the publisher timeliness class is update or synchronous. This is the DLCEP address used for synchronization.
12	FasDIISubscriberTimeWindowSize	M	R-W	This attribute is for BNU VCRs only. This optional parameter provides the time window of a subscriber provided by a DLL.
13	FasDIISubscriberSynchronizingDLcep	M	R-W	This attribute is for BNU VCRs only and then only if the subscriber timeliness class is update or synchronous. This is the DLCEP address used for synchronization.

15. Appendix A — VCR List**15.1. VcrList (VCRL)**

#	Attribute	M/C	R/W	Description
14	FmsVfdId	M	R-W	This attribute references the VFD associated with this VCR.
15	FmsMaxOutstandingServicesCalling	M	R-W	This attribute is for QUB VCRs only. This corresponds to the number of sending state machines.
16	FmsMaxOutstandingServicesCalled	M	R-W	This attribute is for QUB VCRs only. This corresponds to the number of receiving state machines.
17	FmsFeaturesSupported	M	R-W	This attribute indicates the services that are supported by FMS for this VCR. For example, Read, Write, InitiateDownloadSequence, DownloadSegment, TerminateDownLoadSequence might be supported. This is a two BitStrings with a maximum length of 8 octets. Each bit indicates a service. The first bitstring represents the client and the second bitstring represents the server.

15. Appendix A — VCR List

15.1. VcrList (VCRL)

16. Appendix B — VCR Entry

16.1 VCR Entry

Field Value Table

Field	Value
VCR Type	QUU
VCR Role	Server
Local Address	0x08
Remote Address	Node (IOP) + Group

16. Appendix B — VCR Entry

16.1. VCR Entry

17. Appendix C — Valid I/O Connections

17.1 Types of Connection Links

The table below shows the valid input and output parameters on each of the standard FF function blocks. The general rule is that only output parameters can be linked to input parameters through link objects. Contained parameters may not be specified in link objects. One exception to the general rule is that an input parameter may be linked to another input parameter in the same resource. Another exception is the treatment of the input parameters in the Analog Human Interface and Discrete Human Interface function blocks. For example, the IV_1 and IV_D1 are classified as input parameters yet may be linked to the input of other blocks.

Parameters that are linked must be of the same type. I/O parameters may be of the types: DS-65 Value & Status — Floating Point, DS-66 Value & Status — Discrete and DS-67 Value & Status — Bit String.

Valid Connection Parameters

The following table shows the valid connection parameters available and the Fieldbus Function Blocks¹ on which they appear:

Parameter	Use	Algorithms	Data Type
Binary_CL	IN	CmplxAO, CmplxDO	DS-66
Binary_OP	IN	CmplxAO, CmplxDO	DS-66
BkCal_In	IN	ManLoad, BiasGain, PD, PID, Ratio, SPRamp, CtrlSel	DS-65
BkCal_In1	IN	Splitter	DS-65
BkCal_In2	IN	Splitter	DS-65
BkCal_In_D	IN	DevCtrl	DS-66
Cas_In	IN	BiasGain, PD, PID, RATIO, AO, CmplxAO, StepCtrl, Splitter	DS-65
Cas_In_D	IN	DO, CmplxDO, DevCtrl	DS-66
Dead_Time	IN	DeadTime	DS-65

17. Appendix C — Valid I/O Connections

17.1. Types of Connection Links

Parameter	Use	Algorithms	Data Type
Disable_1	IN	SnglSel	DS-66
Disable_2	IN	SnglSel	DS-66
Disable_3	IN	SnglSel	DS-66
Disable_4	IN	SnglSel	DS-66
En_Close	IN	CmplxAO, CmplxDO	DS-66
En_Open	IN	CmplxAO, CmplxDO	DS-66
FF_Val	IN	PD, PID	DS-65
Follow	IN	LeadLag, DeadTime	DS-66
In	IN	ManLoad, PD, PID, Ratio, StepCtrl, LeadLag, DeadTime, ArithMet, AnlgAlm	DS-65
Interlock_D	IN	DevCtrl	DS-66
In_1	IN	BiasGain, Ratio, SnglSel, CharCtzt, ArithMet, CalcAnlg, CalcDisc, Integrat	DS-65
In_2	IN	SnglSel, CharCtzt, ArithMet, CalcAnlg, CalcDisc, Integrat	DS-65
In_3	IN	SnglSel, ArithMet, CalcAnlg	DS-65
In_4	IN	SnglSel, CalcAnlg	DS-65
In_D	IN	DevCtrl	DS-66
In_D1	IN	CalcAnlg, CalcDisc, Timer, DiscAlm	DS-66
In_D10	IN	CalcDisc	DS-66
In_D11	IN	CalcDisc	DS-66
In_D12	IN	CalcDisc	DS-66
In_D13	IN	CalcDisc	DS-66
In_D14	IN	CalcDisc	DS-66
In_D15	IN	CalcDisc	DS-66
In_D16	IN	CalcDisc	DS-66

17. Appendix C — Valid I/O Connections

17.1. Types of Connection Links

Parameter	Use	Algorithms	Data Type
In_D2	IN	CalcAnlg, CalcDisc, Timer, DiscAlm	DS-66
In_D3	IN	CalcAnlg, CalcDisc, Timer, DiscAlm	DS-66
In_D4	IN	CalcAnlg, CalcDisc, Timer, DiscAlm	DS-66
In_D5	IN	CalcDisc	DS-66
In_D6	IN	CalcDisc	DS-66
In_D7	IN	CalcDisc	DS-66
In_D8	IN	CalcDisc	DS-66
In_D9	IN	CalcDisc	DS-66
In_Lo	IN	ArithMet	DS-65
IV_1	IN	AnlgHum	DS-65
IV_2	IN	AnlgHum	DS-65
IV_3	IN	AnlgHum	DS-65
IV_4	IN	AnlgHum	DS-65
IV_D1	IN	DiscHum	DS-66
IV_D2	IN	DiscHum	DS-66
IV_D3	IN	DiscHum	DS-66
IV_D4	IN	DiscHum	DS-66
Lag_Time	IN	LeadLag	DS-65
Lead_Time	IN	LeadLag	DS-65
OP_Select	IN	SnglSel	DS-66
Pause	IN	SPRamp	DS-66
Permissive_D	IN	DevCtrl	DS-66
PSP	IN	AnlgAlm	DS-65
Reset_In	IN	SPRamp, Integrat, Timer	DS-66

17. Appendix C — Valid I/O Connections

17.1. Types of Connection Links

Parameter	Use	Algorithms	Data Type
Rev_Flow1	IN	Integrat	DS-66
Rev_Flow2	IN	Integrat	DS-66
SafeGuard_C L	IN	CmplxAO, CmplxDO	DS-66
SafeGuard_O P	IN	CmplxAO, CmplxDO	DS-66
Sel_1	IN	CtrlSel	DS-65
Sel_2	IN	CtrlSel	DS-65
Sel_3	IN	CtrlSel	DS-65
Shutdown_D	IN	DevCtrl	DS-66
Start	IN	SPRamp	DS-66
Trk_In_D	IN	ManLoad, BiasGain, PD, PID, RATIO, DevCtrl	DS-66
Trk_Val	IN	ManLoad, BiasGain, PD, PID, RATIO	DS-65
BkCal_Out	OUT	BiasGain, PD, PID, Ratio, AO, CmplxAO, StepCtrl, Splitter	DS-65
BkCal_Out_D	OUT	DO, CmplxDO, DevCtrl	DS-66
BkCal_Sel1	OUT	CtrlSel	DS-65
BkCal_Sel2	OUT	CtrlSel	DS-65
BkCal_Sel3	OUT	CtrlSel	DS-65
N_Reset	OUT	Integrat	DS-65
Out	OUT	AI, ManLoad, BiasGain, CtrlSel, PD, PID, Ratio, AO, PulseIn, CmplxAO, SPRamp, SnglSel, LeadLag, DeadTime, ArithMet, Integrat, AnlgAlm	DS-65
Out_1	OUT	Splitter, CharCtzt, CalcAnlg, CalcDisc	DS-65
Out_2	OUT	Splitter, CharCtzt, CalcAnlg	DS-65
Out_Accum	OUT	PulseIn	DS-65
Out_Alm	OUT	AnlgAlm, DiscAlm	DS-65

17. Appendix C — Valid I/O Connections

17.1. Types of Connection Links

Parameter	Use	Algorithms	Data Type
Out_D	OUT	DI, DO, CmplxDIO, StepCtrl, DevCtrl, Timer, DiscAlm	DS-66
Out_D1	OUT	CalcAnlg, CalcDisc	DS-66
Out_D2	OUT	CalcAnlg, CalcDisc	DS-66
Out_D3	OUT	CalcDisc	DS-66
Out_D4	OUT	CalcDisc	DS-66
Out_Exp	OUT	Timer	DS-65
Out_Ptrip	OUT	Integrat	DS-66
Out_Rem	OUT	Timer	DS-65
Out_Trip	OUT	Integrat	DS-66
Selected	OUT	SnglSel	DS66
Step_Posn	OUT	SPRamp	DS-66
Time_Posn	OUT	SPRamp	DS-65
Time_Posn_T	OUT	SPRamp	DS-65

17. Appendix C — Valid I/O Connections

17.1. Types of Connection Links

18. Appendix D Alert Link Objects

18.1 Alert Link Object

The Alert Link Object is the first object created in the device.

Field Name	Value
Local Index	0
VCR Number remote communication	OD Index of the associated VCR.
Remote Index	0
Service Operation	ALERT
Stale Count Limit	N/A

18. Appendix D Alert Link Objects

18.1. Alert Link Object

19. Appendix E — VCR Definitions

19.1 Purpose

Appendix D describes the definition of VCRs. VCRs are created as part of the scheduling process.

19.2 VCR Types

The following table lists the attributes in a VCR and the type of VCR that may have data configured for that attribute. The type **BNU** is normally associated with a **Publisher-Subscriber** role. The type **QUB** is normally associated with the **Client-Server** role. The type **QUU** is normally associated with the **Source-Sink** role of the Alarm-Trend reporter-receiver.

No.	Attribute	BNU	QUB	QUU
1	FasArTypeAndRole	X	X	X
2	FasDIILocalAddr	X	X	X
3	FasDIILConfiguredRemoteAddr	X - Subscriber	X	X
4	FasDIISDAP	X	X	X
5	FasDIIMaxConfirmDelayOnConnect	X	X	
6	FasDIIMaxConfirmDelayOnData		X	X
7	FasDIIMaxDisduSize	X	X	X
8	FasDIILResidualActivitySupported		X	
9	FasDIILTimelinessClass	X		
10	FasDIIPublisherTimeWindowSize	X		
11	FasDIIPublisherSynchronizingDlcep	X		
12	FasDIISubscriberTimeWindowSize	X		
13	FasDIISubscriberSynchronizingDlcep	X		
14	FmsVfdId	X - Publisher	X - Server	X – Sink

Appendix E — VCR Definitions

VCR Types

No.	Attribute	BNU	QUB	QUU
15	FmsMaxOutstandingServicesCalling		X	
16	FmsMaxOutstandingServicesCalled		X	
17	FmsFeaturesSupported	X	X	X

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